

Abstract Linear Algebra

Prof. Lanier

jlancier8@uchicago.edu

Please pick up a copy of the handout.

Try those warm-up problems.

LIN-ee-er

Abstract Linear Algebra

Prof. Lanier Luh-NEER
jlanier8@uchicago.edu

Please pick up a copy of the handout.

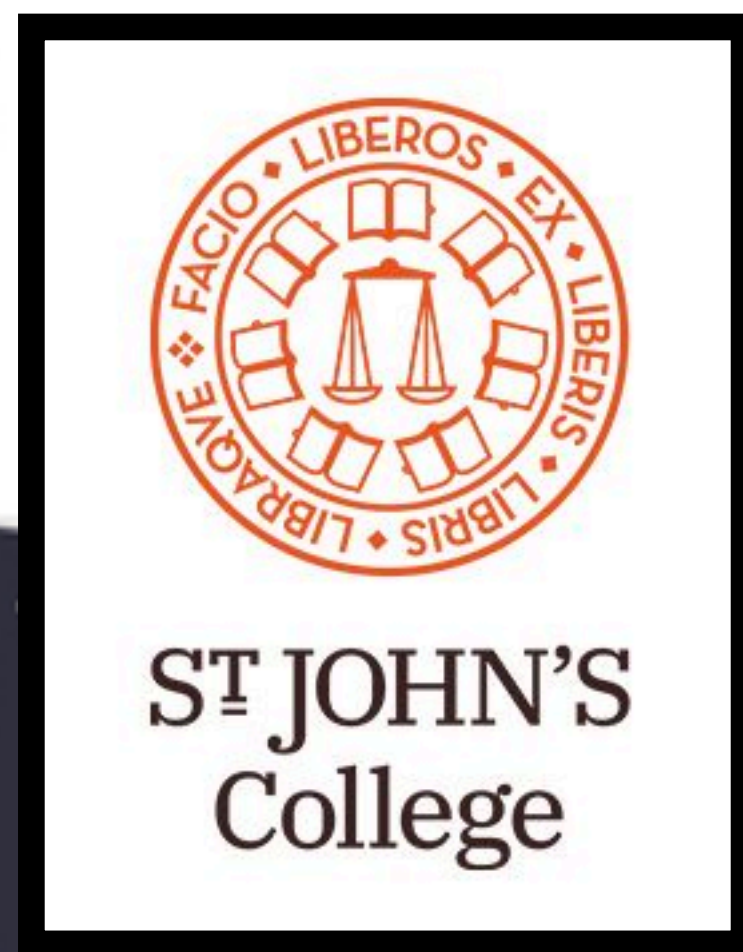
Try those warm-up problems.

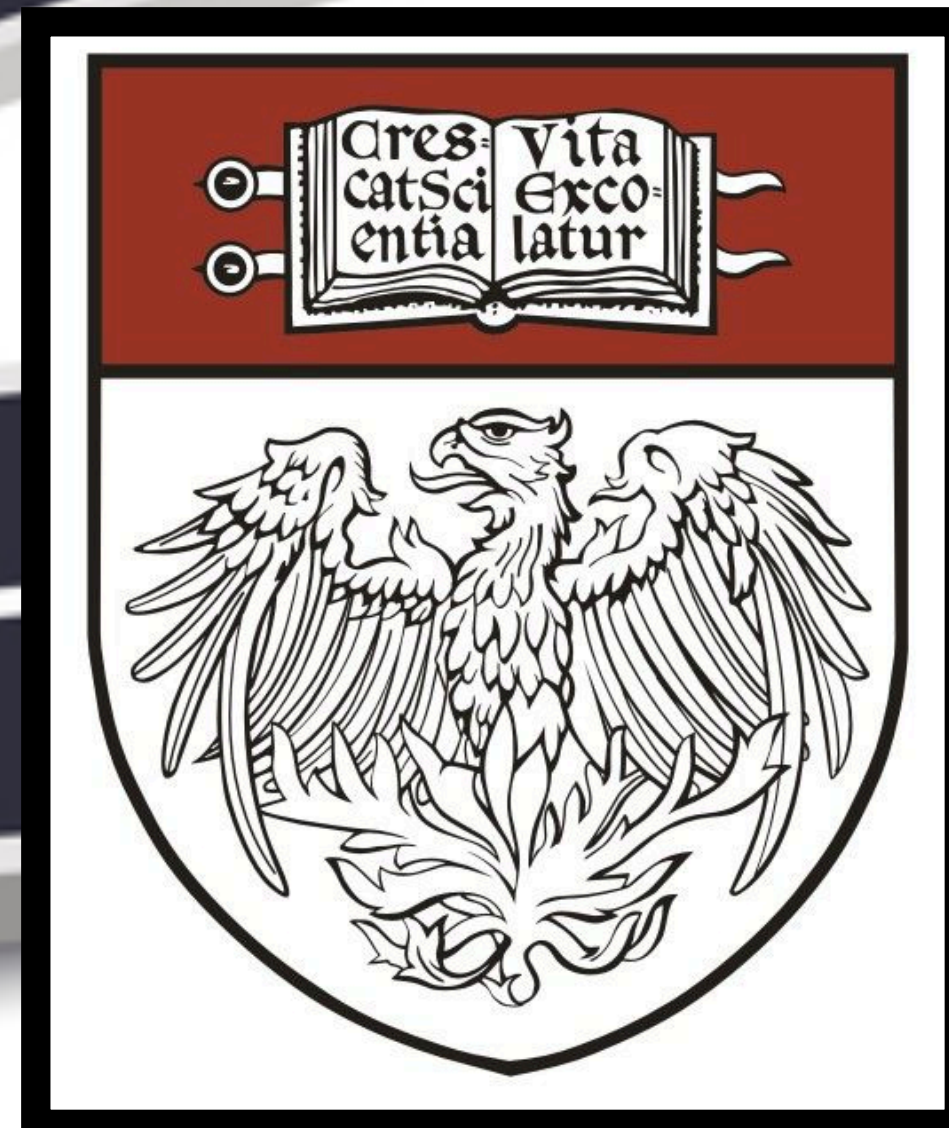
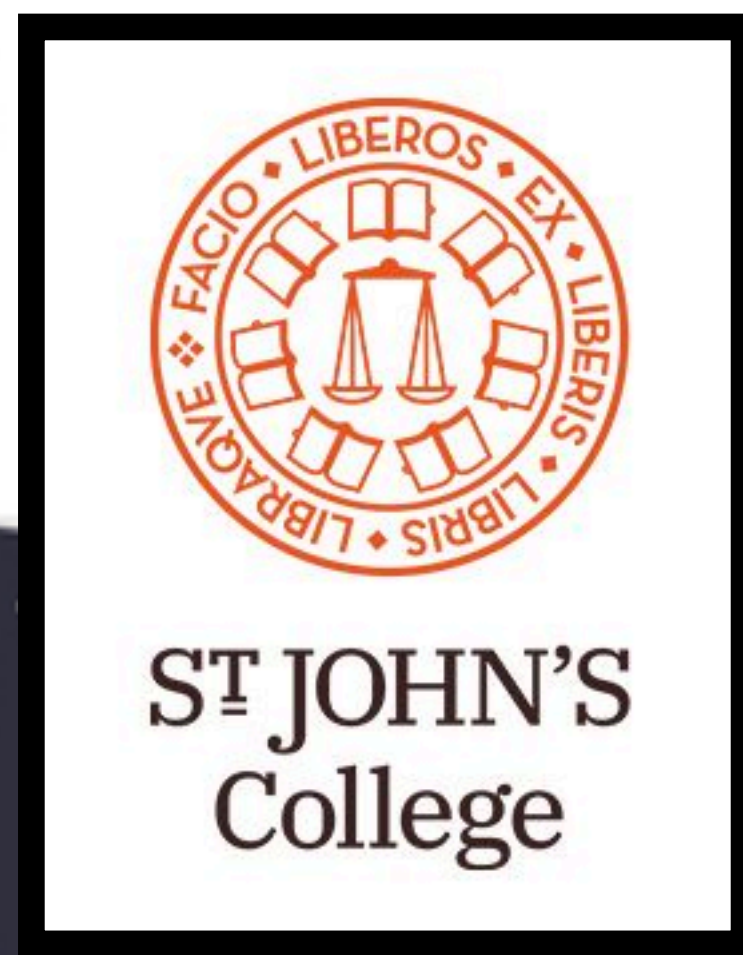
For you to do ASAP:

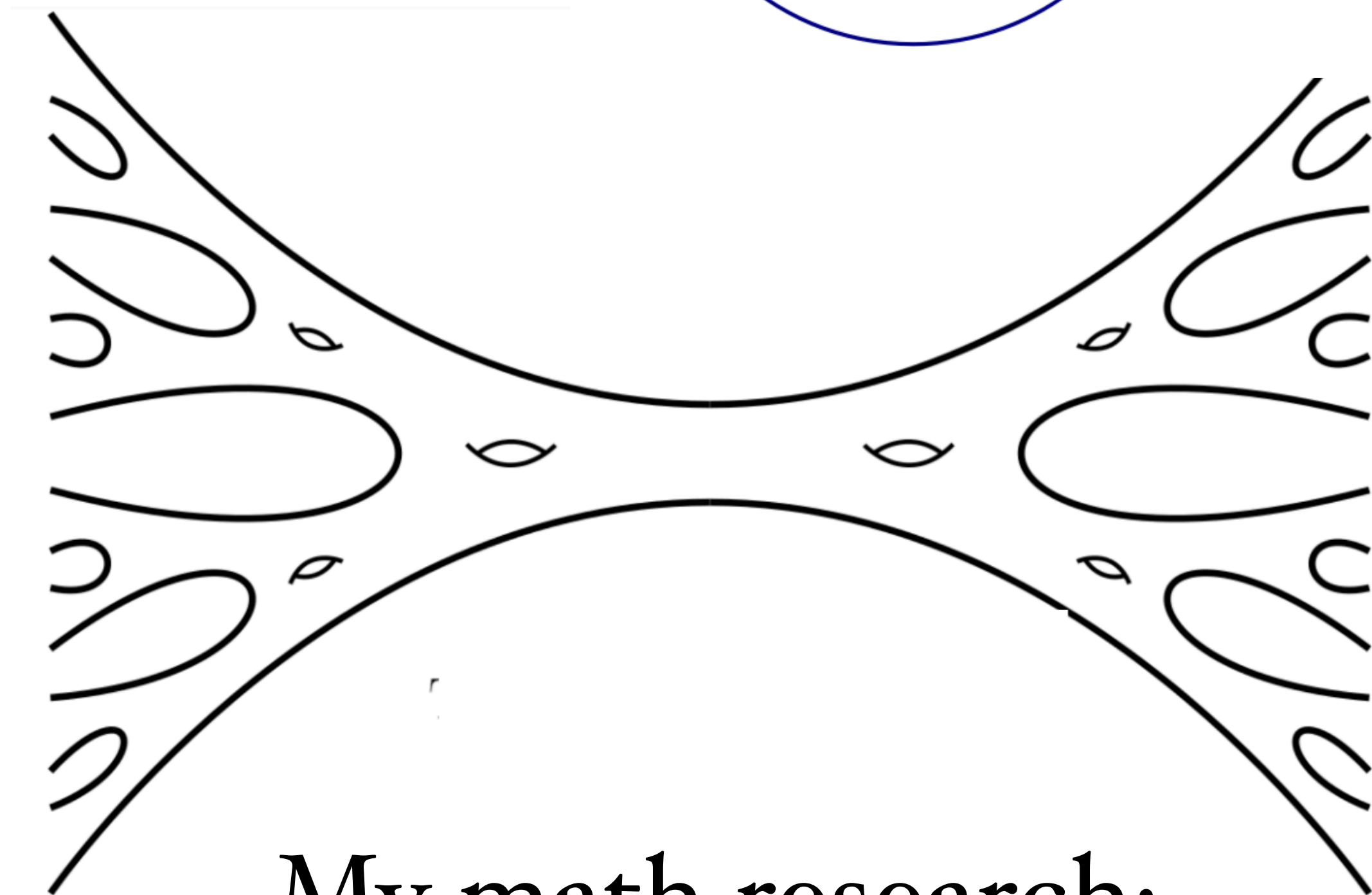
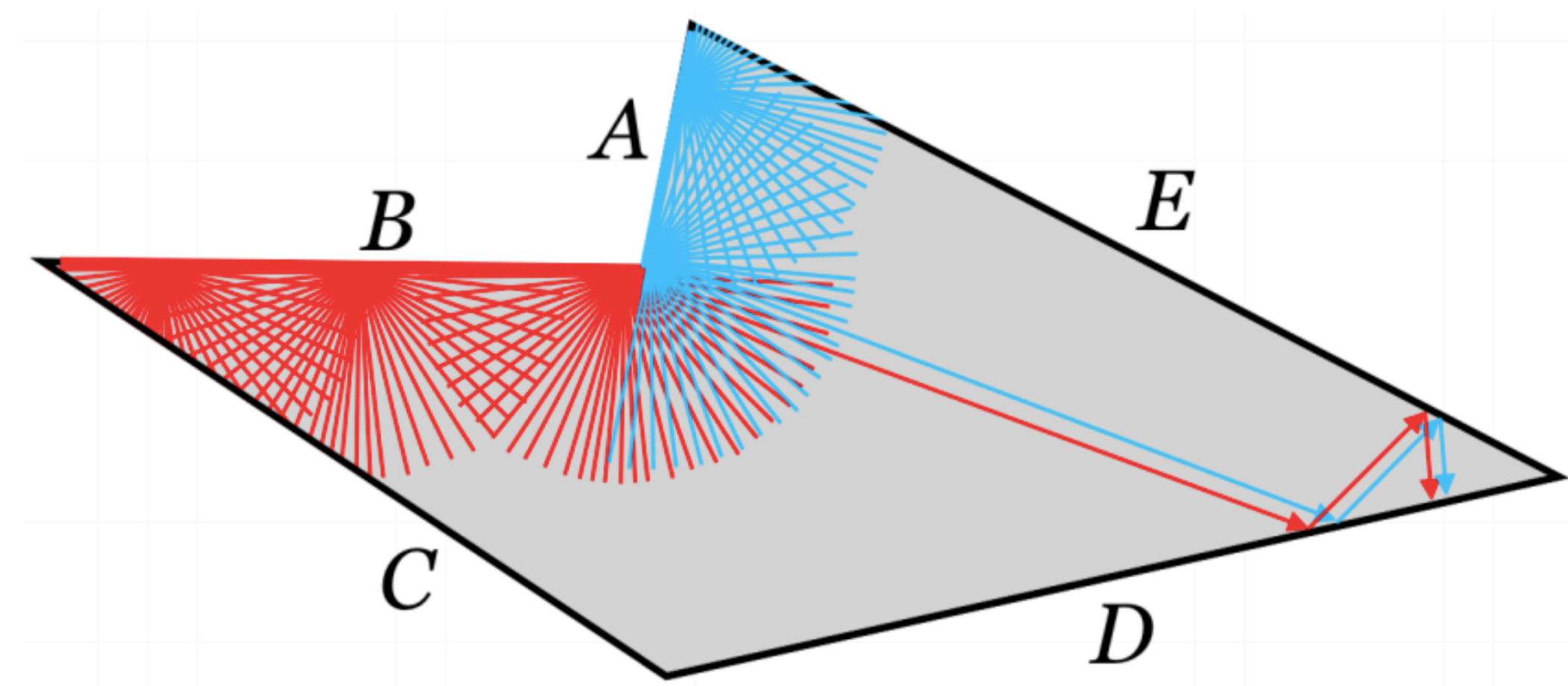
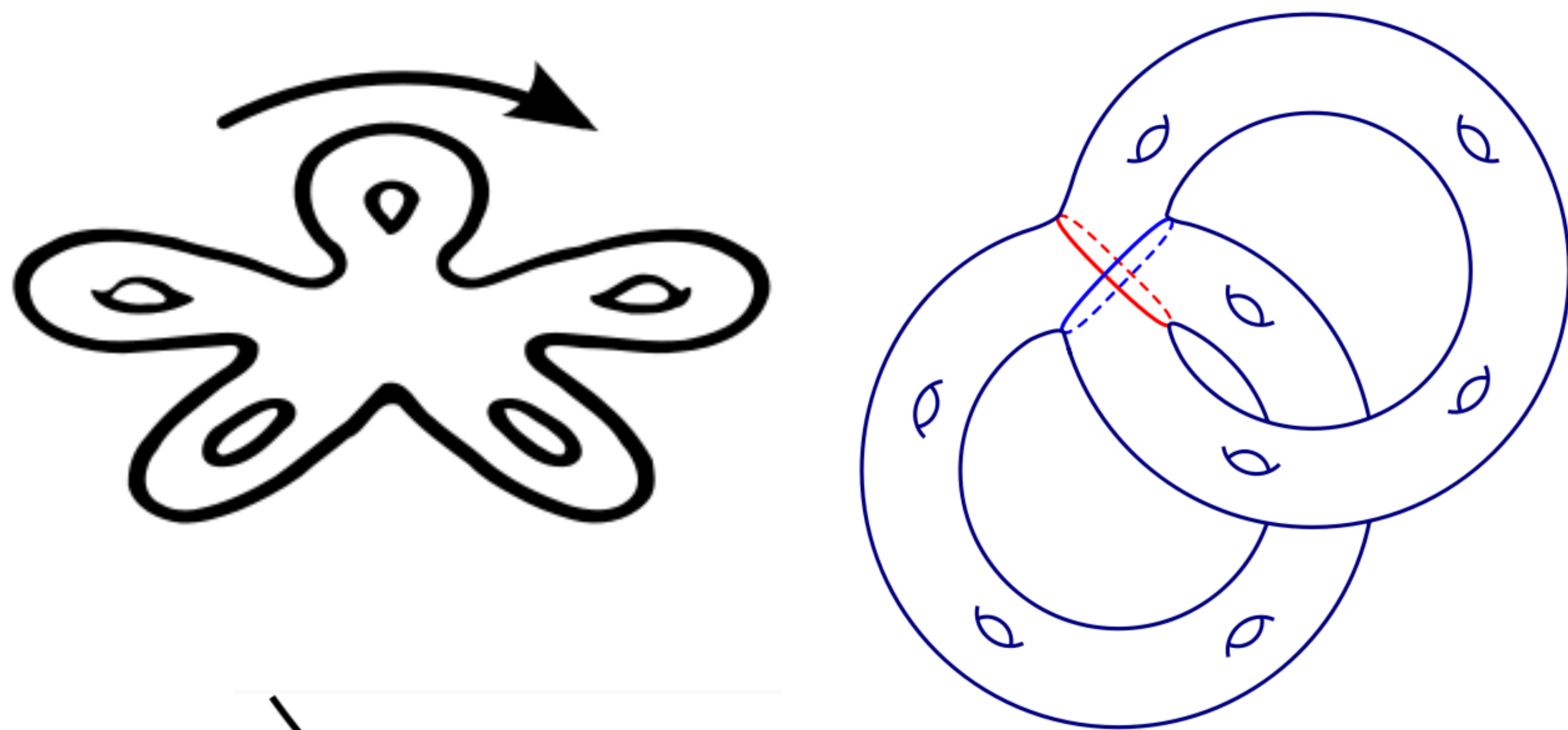
- Read through the syllabus and course docs
- Fill out the start-of-course survey
- Read sections 1-4 of Chapter 1 of *LADW*
- Form a study group
- Explore the Canvas site some
- Come to office hours!

Today in class:

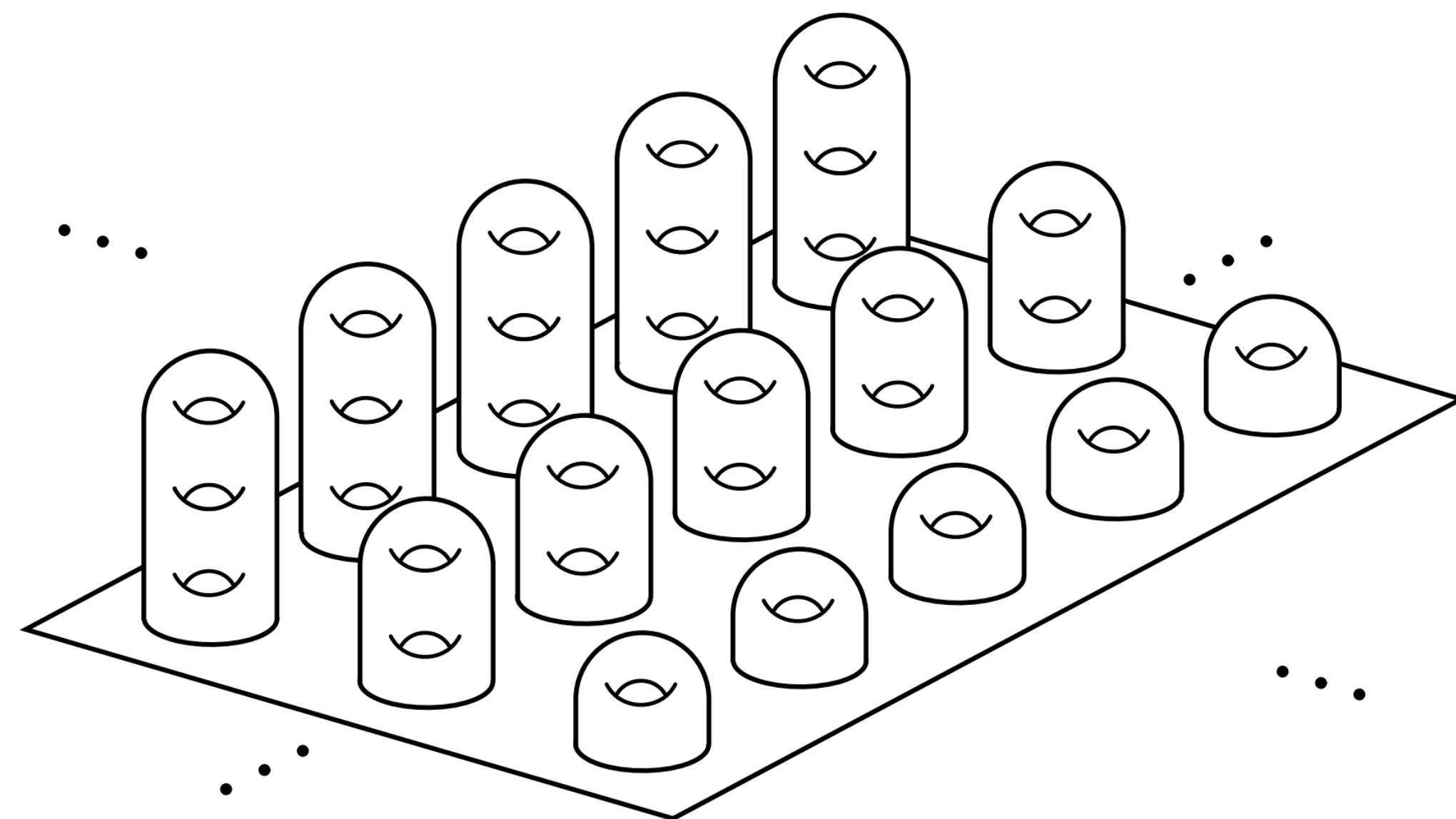
- Go over the basics of the syllabus
- Begin to give an overview of the main ideas of the course (Ch 1-2)







My math research:
surfaces and their symmetries



The group $\mathrm{Sp}(2g, \mathbb{Z})$ contains $M = (\mathbb{Z}/3\mathbb{Z})^g$ as a subgroup, with generators m_1 to m_g of the form

$$m_i = \begin{bmatrix} I_{2i-2} & 0 & 0 \\ 0 & A & 0 \\ 0 & 0 & I_{2g-2i} \end{bmatrix}$$

Here A is the 2×2 matrix

$$\begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix}$$

of order 3 and I_n denotes the $n \times n$ identity matrix. Every nontrivial element in M is a

Proof. Fix some $g \geq 1$ and consider again the mapping classes T_A and T_B as above. Much more than proving $T_A^{-1}T_B$ is pseudo-Anosov, Thurston proved that there is a positive number μ and a homomorphism

$$\langle T_A, T_B \rangle \rightarrow \mathrm{PSL}_2(\mathbb{R})$$

with

$$T_A \mapsto \begin{bmatrix} 1 & -\mu^{1/2} \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad T_B \mapsto \begin{bmatrix} 1 & 0 \\ \mu^{1/2} & 1 \end{bmatrix},$$

and so that $f \in \langle T_A, T_B \rangle$ is pseudo-Anosov if and only if its image in $\mathrm{PSL}_2(\mathbb{R})$ (or, rather, a representative in $\mathrm{SL}_2(\mathbb{R})$) has an eigenvalue $\lambda > 1$; moreover in this case λ is the stretch factor of f (the number μ is the Perron–Frobenius eigenvalue of NN^T , where N is the intersection matrix between the a_i and b_i ; see [16]).

Consider then the mapping class $f_n = T_A^{-1}T_B^n$. Its image in $\mathrm{PSL}_2(\mathbb{R})$ is

$$\begin{bmatrix} 1 & \mu^{1/2} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ n\mu^{1/2} & 1 \end{bmatrix} = \begin{bmatrix} 1 + n\mu & \mu^{1/2} \\ n\mu^{1/2} & 1 \end{bmatrix}.$$

The trace of this matrix is larger than 2, and so it has two real eigenvalues. The larger

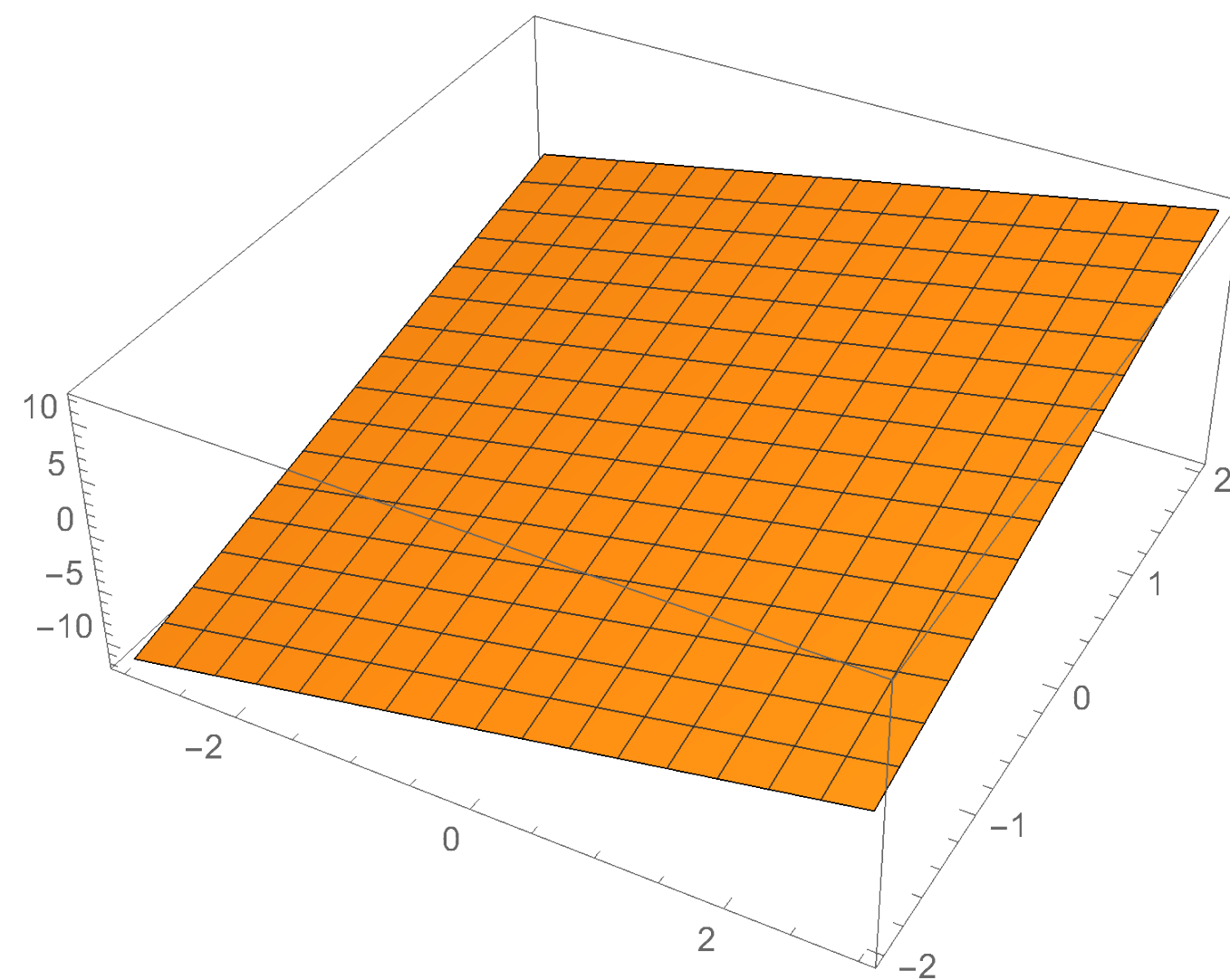
The function L in the proof of Lemma 3.8 is linear, and the fixed point of L is an eigenvector with eigenvalue 1. So the fixed point can be computed explicitly.

Abstract Linear Algebra

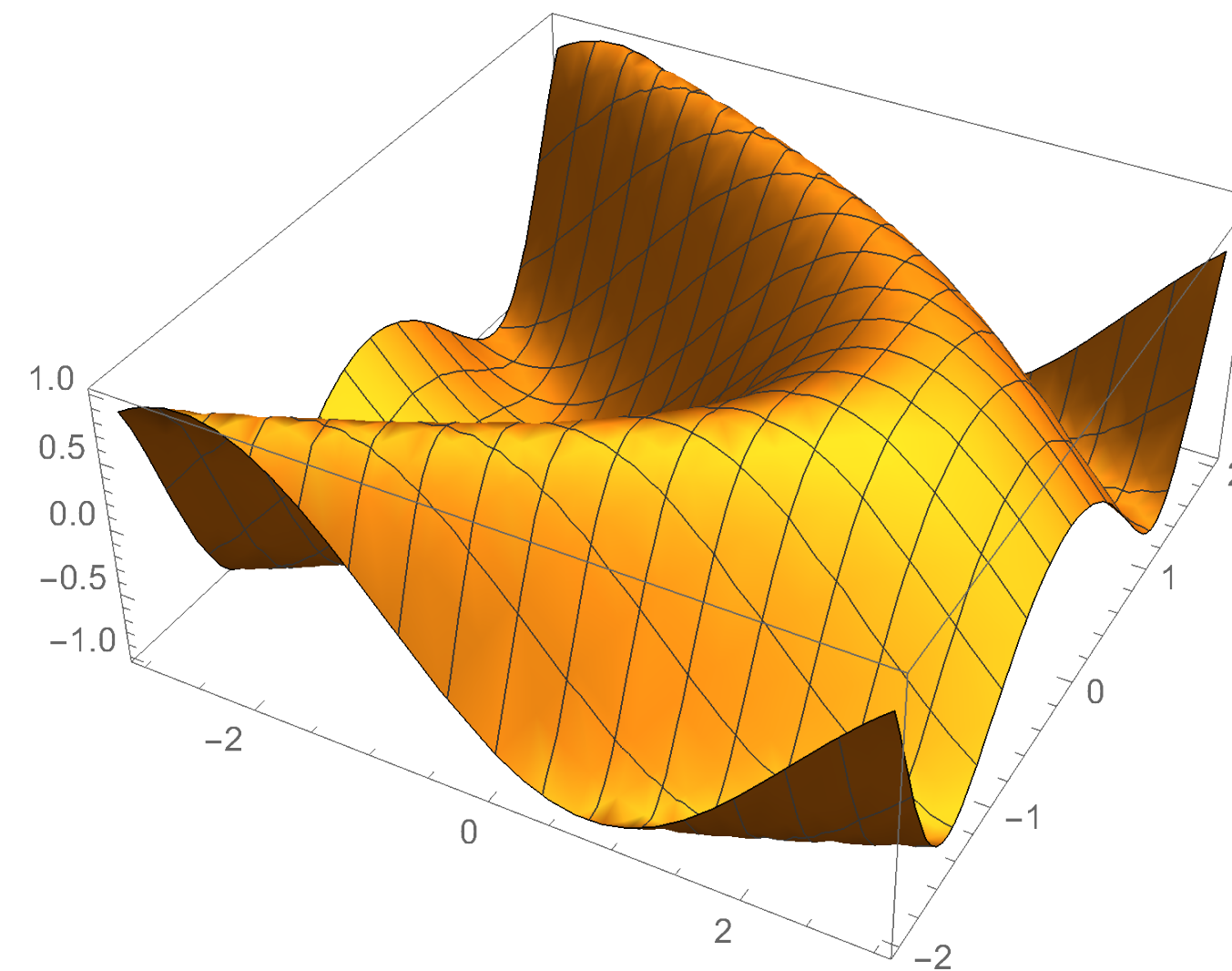
Abstract Linear Algebra

Abstract Linear Algebra

$$z = 2x + 3y$$



$$z = \sin(x + y^2)$$



$$A(u + v) = Au + Av$$

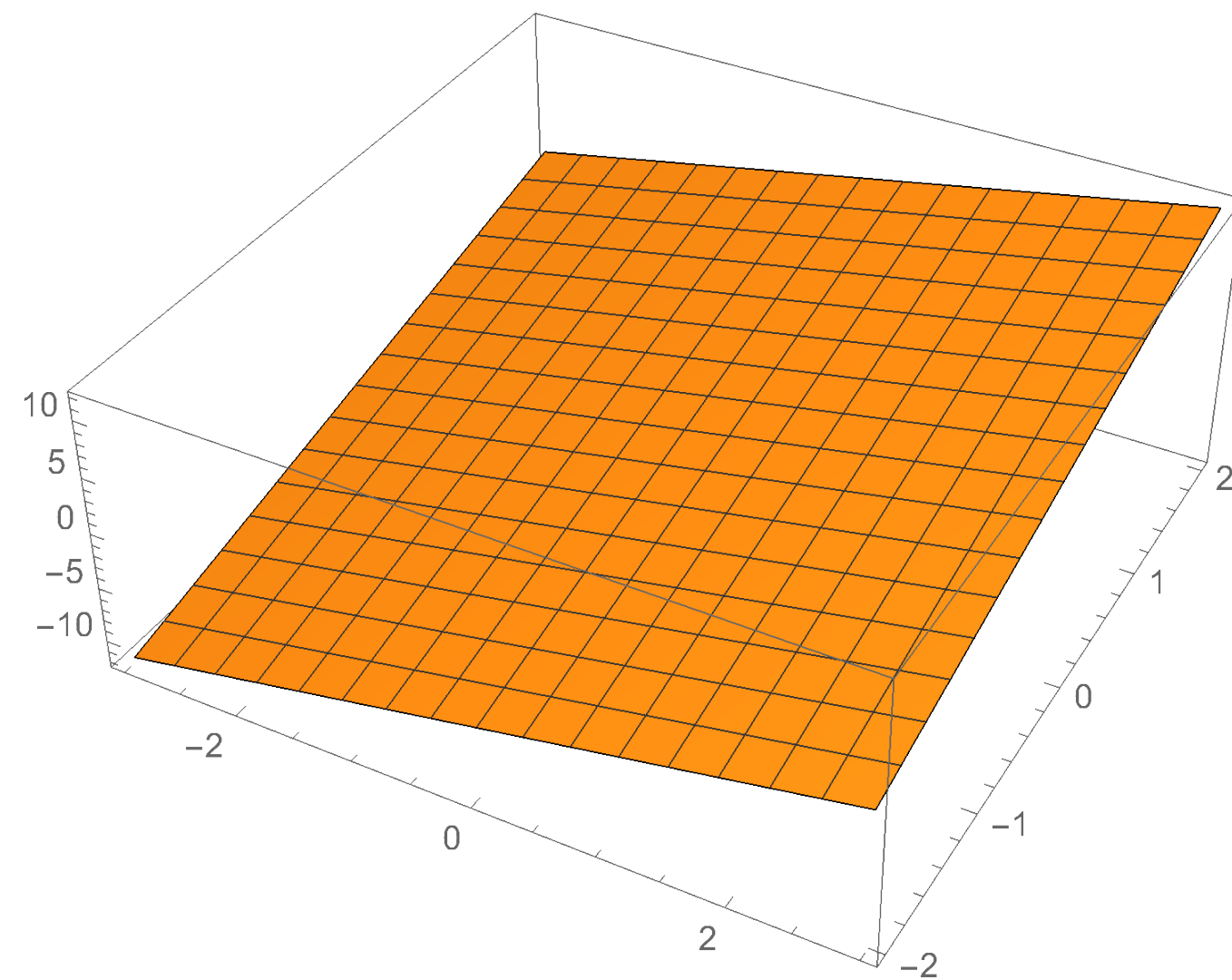
linear

$$\sqrt{a + b} \neq \sqrt{a} + \sqrt{b}$$

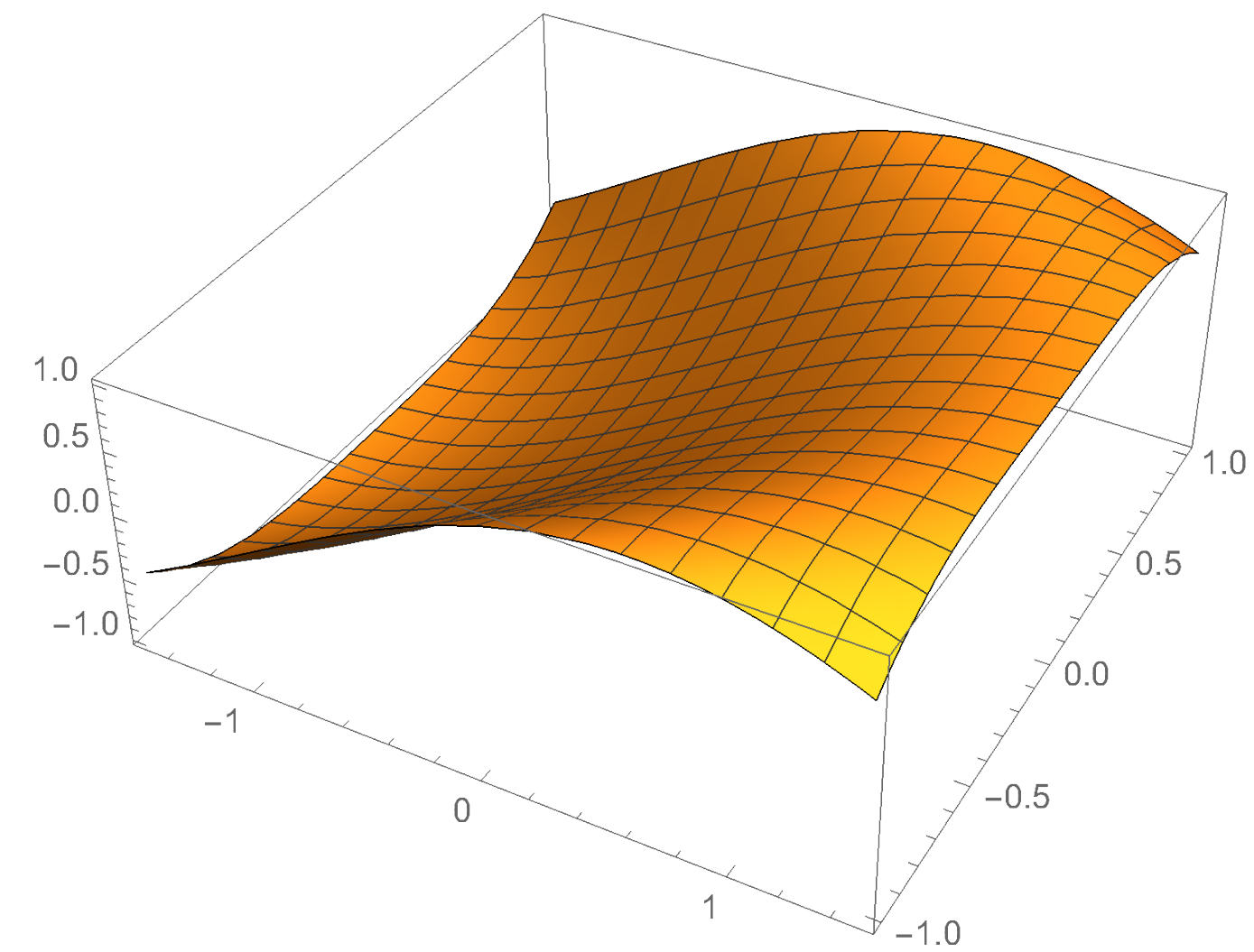
not linear

Abstract Linear Algebra

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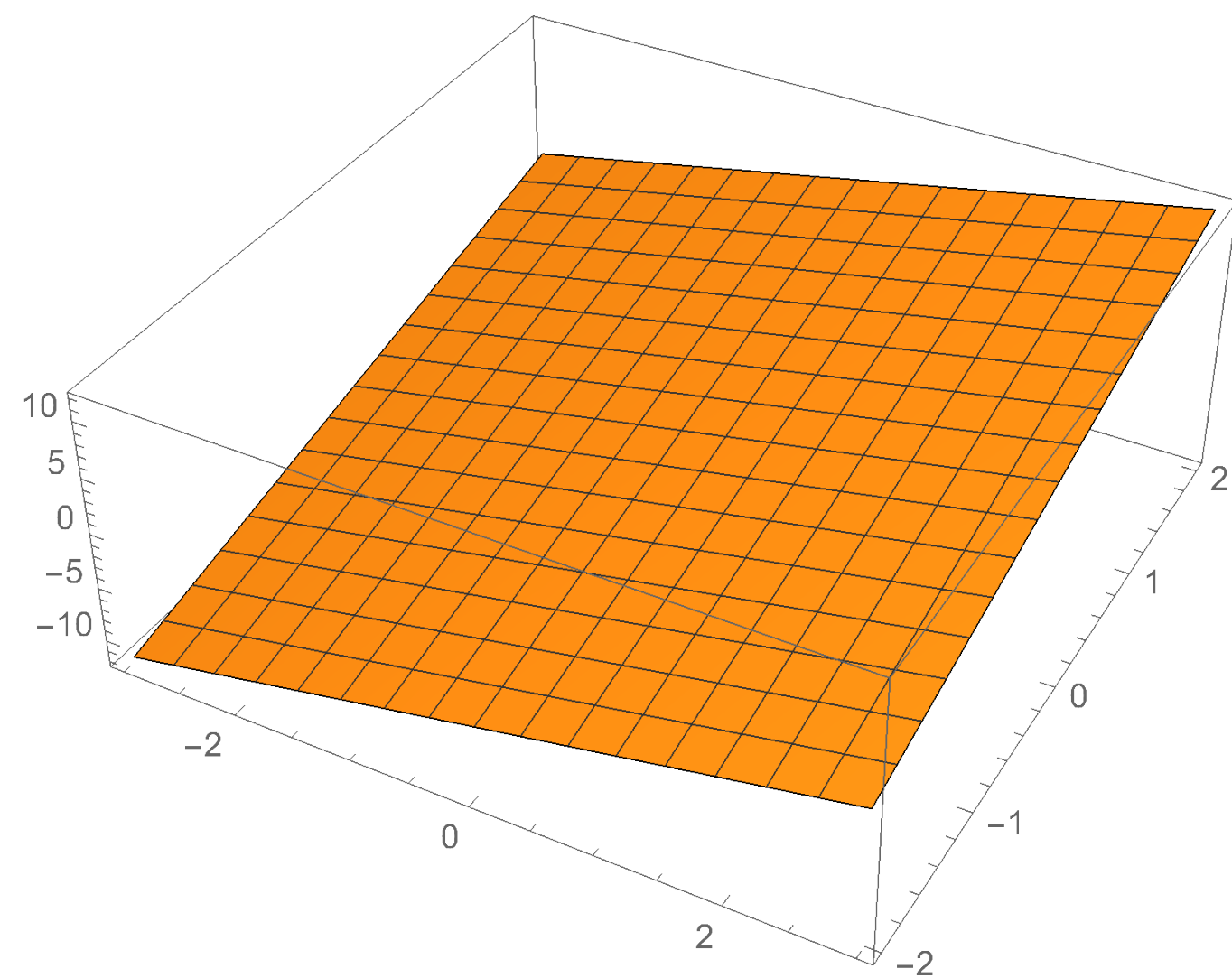


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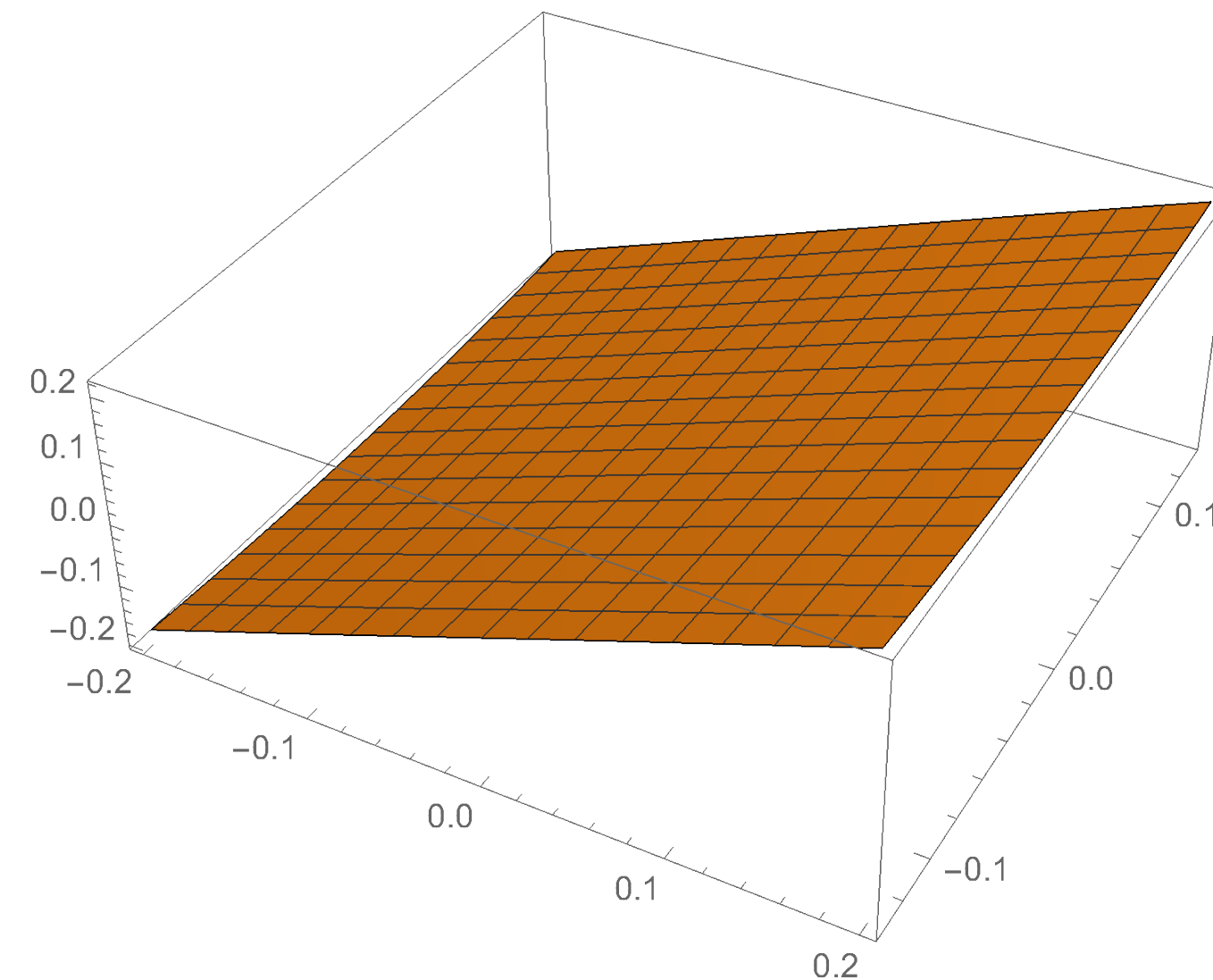
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$$2x + 3y = 0$$

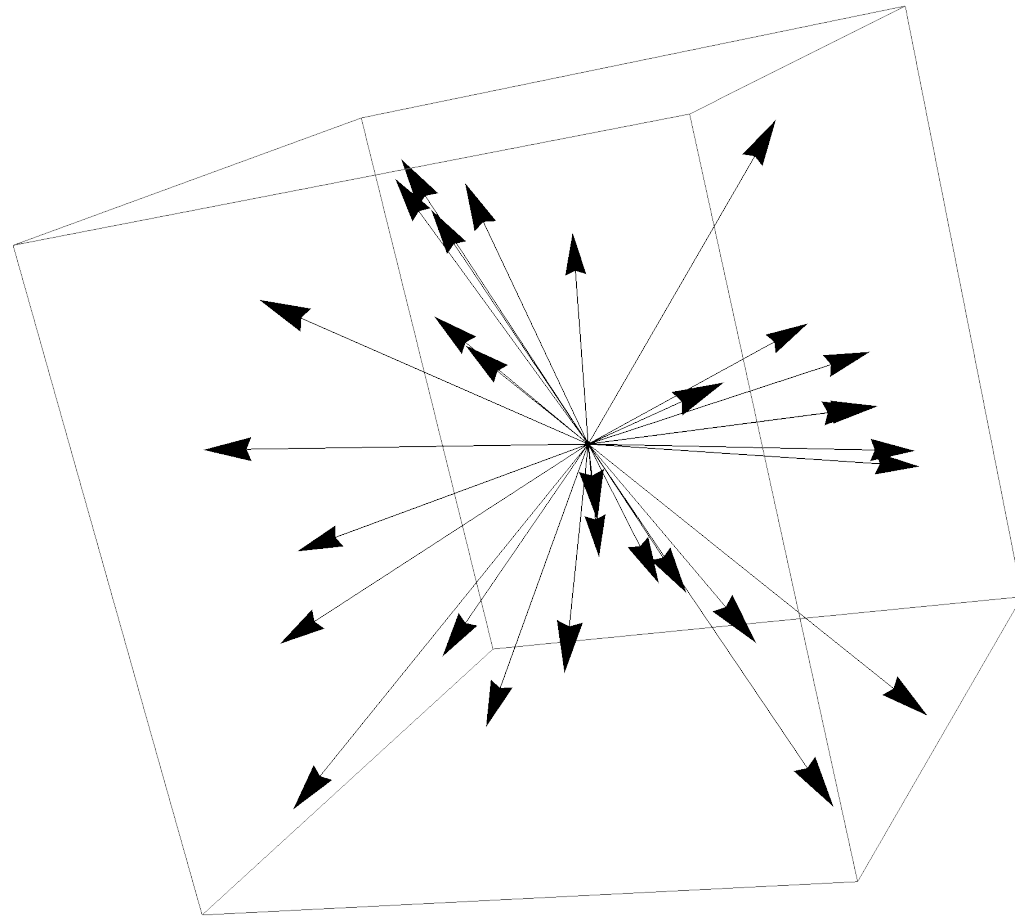
$$5x - 3y = 0$$

$$\left(\begin{array}{cc|c} 2 & 3 & 0 \\ 5 & -3 & 0 \end{array} \right)$$

Systems of equations
and matrices
and computations

CONCRETE

Abstract Linear Algebra



Proof. Suppose the system $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ is linearly dependent. Then there exist scalars α_k , $\sum_{k=1}^p |\alpha_k| \neq 0$ such that

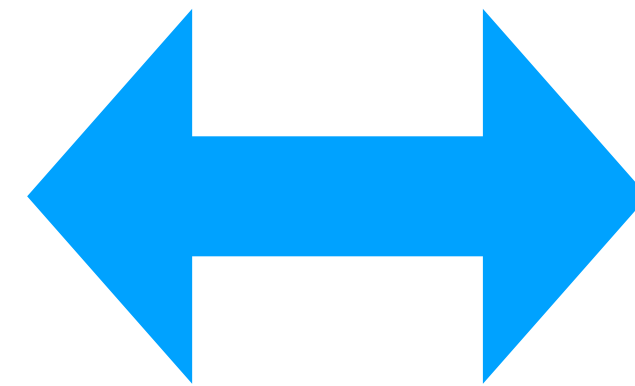
$$\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_p \mathbf{v}_p = \mathbf{0}.$$

Let k be the index such that $\alpha_k \neq 0$. Then, moving all terms except $\alpha_k \mathbf{v}_k$ to the right side we get

$$\alpha_k \mathbf{v}_k = - \sum_{\substack{j=1 \\ j \neq k}}^p \alpha_j \mathbf{v}_j.$$

Vector spaces
and linear transformations
and proofs

ABSTRACT



$$\begin{aligned} 2x + 3y &= 0 \\ 5x - 3y &= 0 \end{aligned}$$

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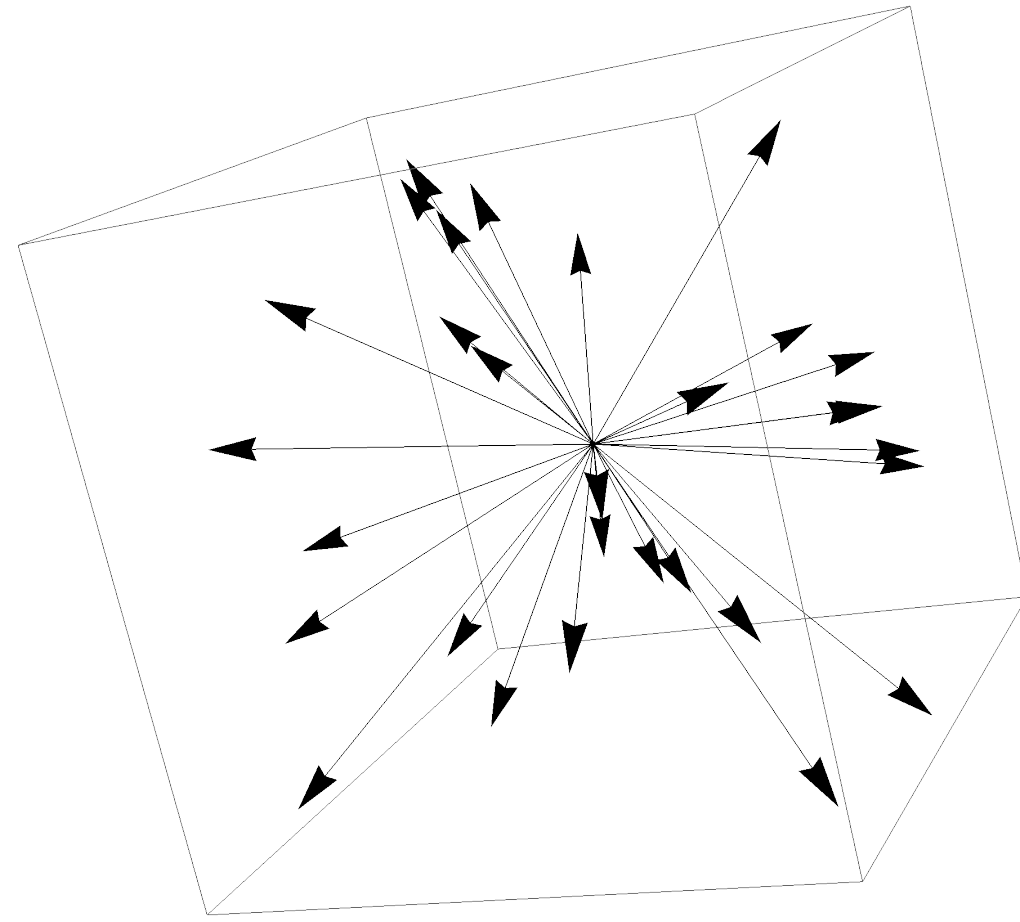
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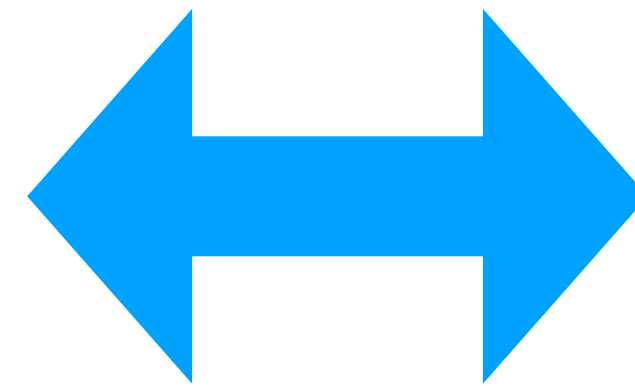
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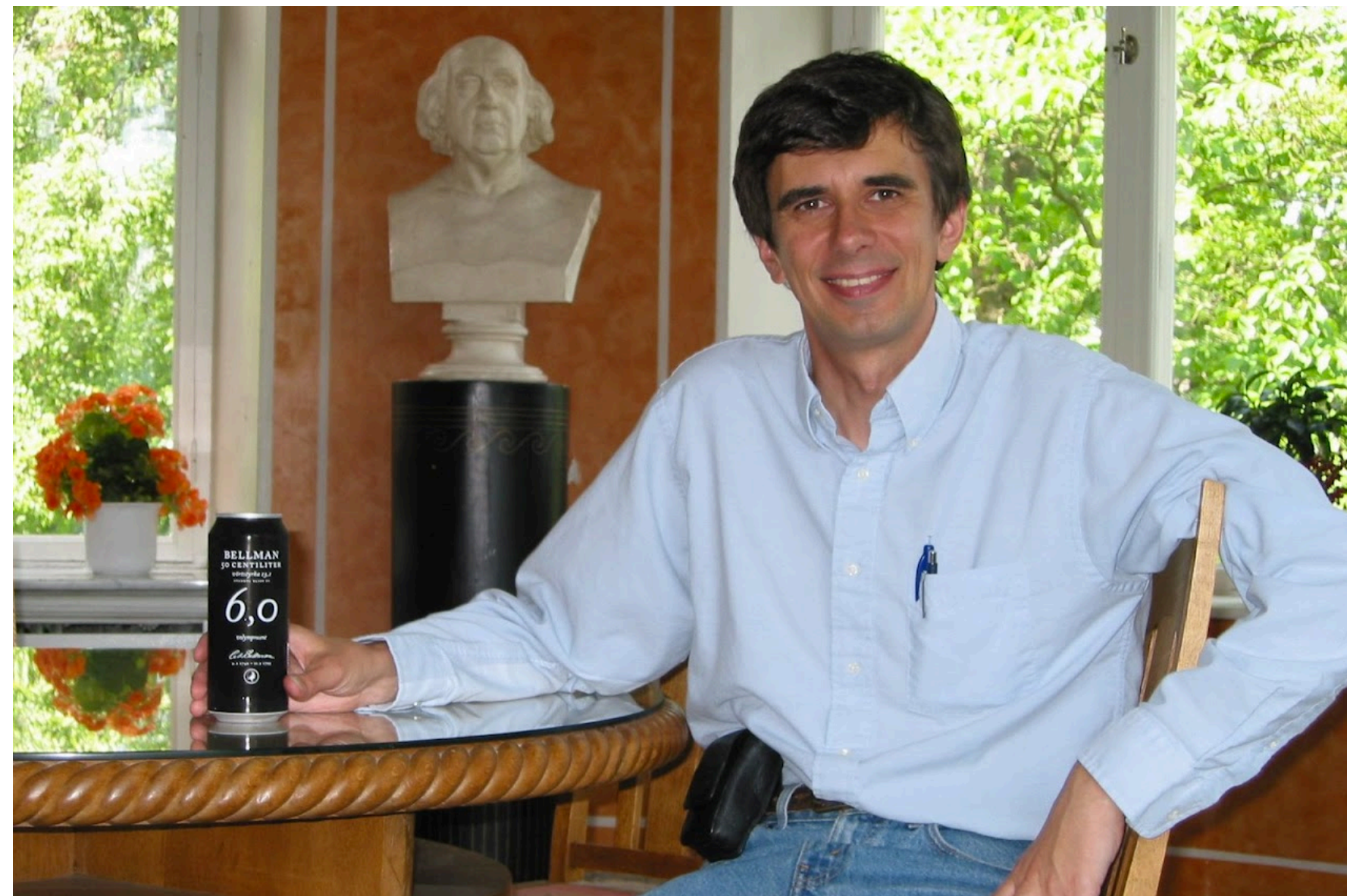
Systems of equations
and matrices
and computations

CONCRETE

Textbook:

Linear Algebra Done Wrong

Sergei Treil



Undergraduate Texts in Mathematics

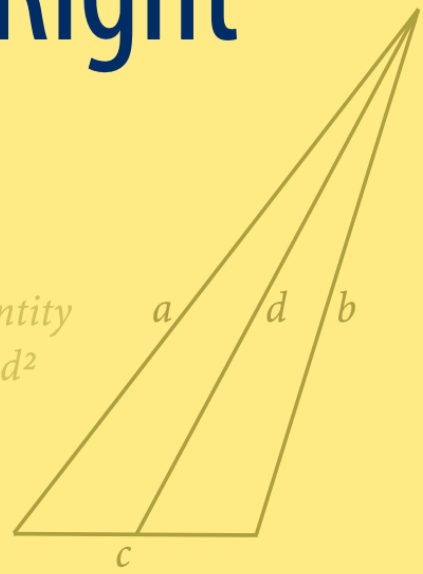
UTM

Sheldon Axler

Linear Algebra Done Right

Third Edition

Apollonius's Identity
 $a^2 + b^2 = \frac{1}{2}c^2 + 2d^2$



 Springer

Textbook:

Linear Algebra Done Wrong

Sergei Treil

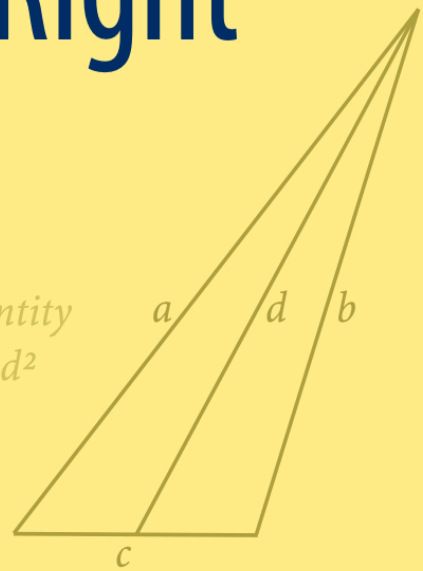


Sheldon Axler

Linear Algebra Done Right

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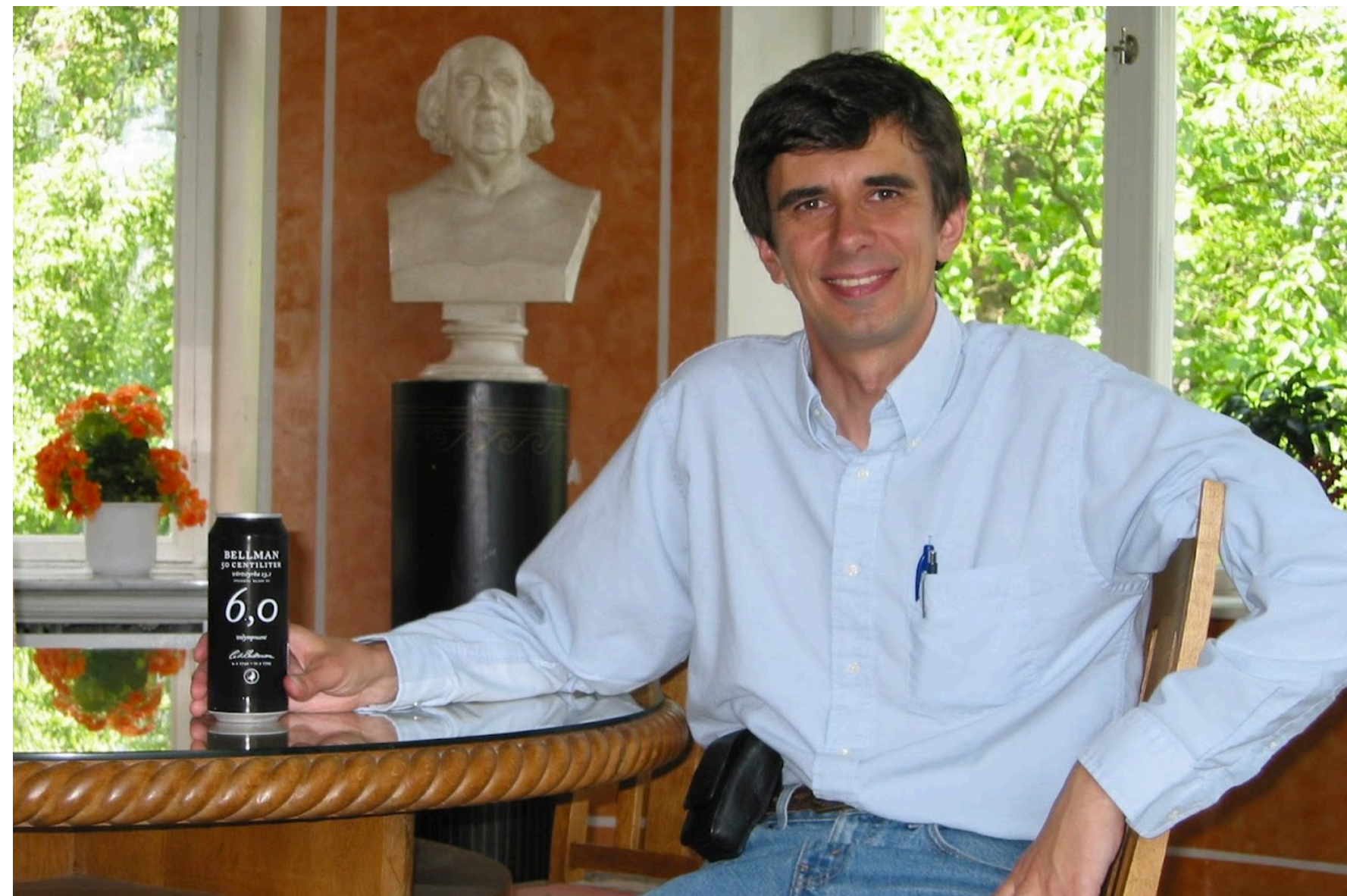
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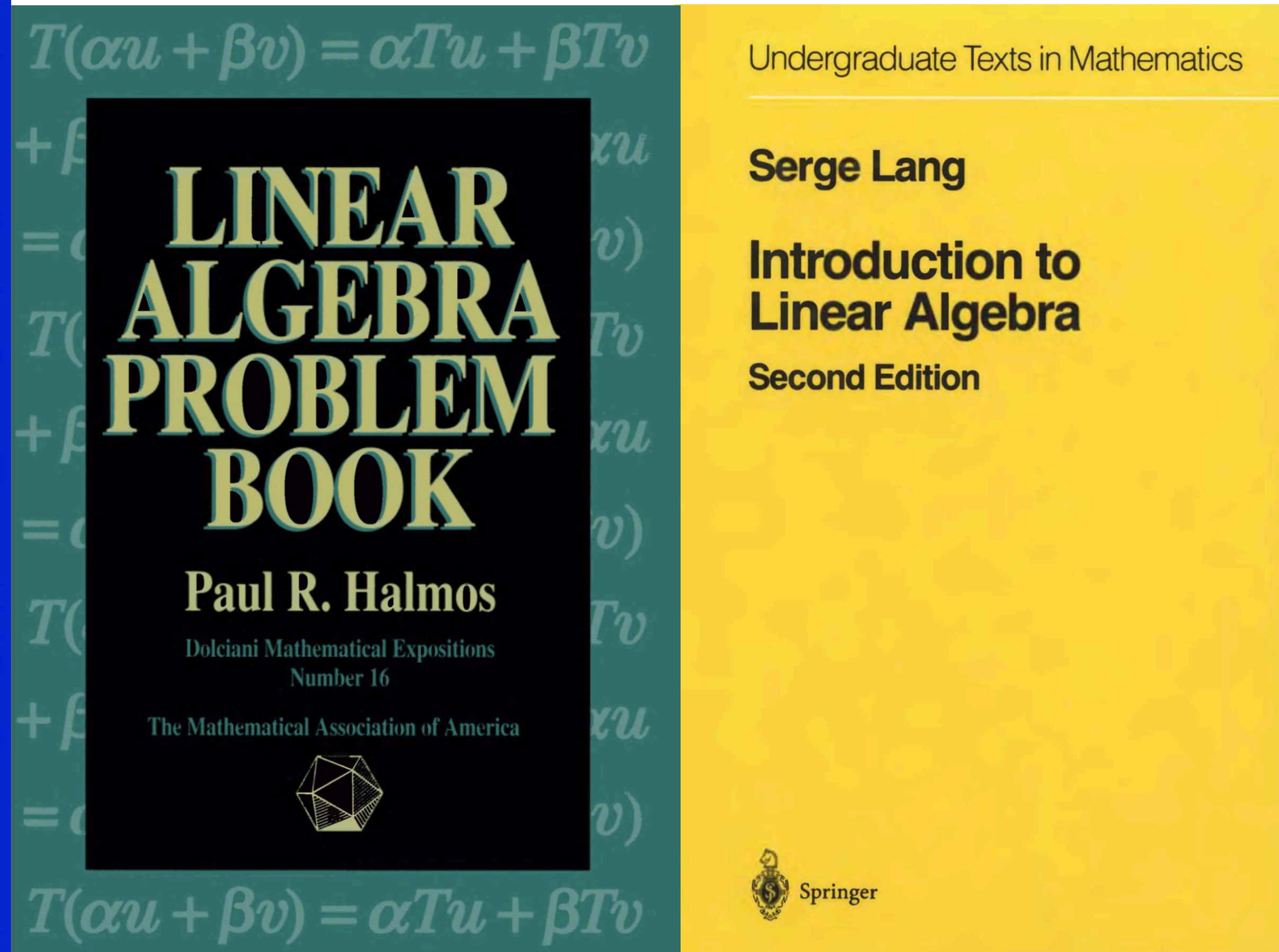
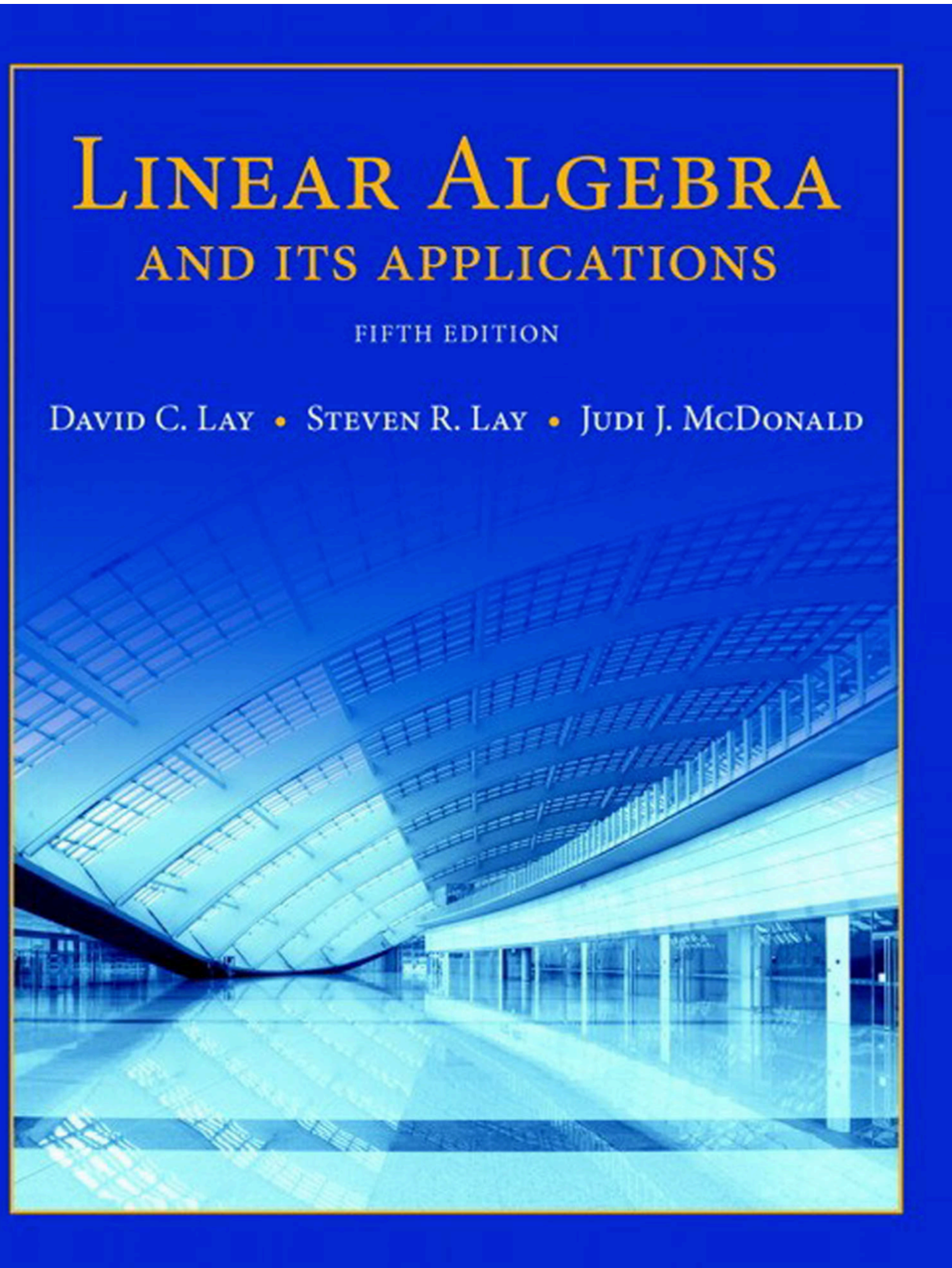
Sergei Treil



Emphasis on:

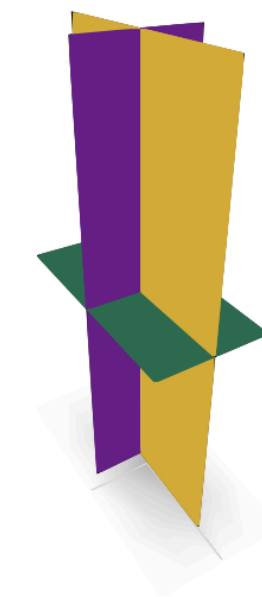
- finite dimensions
- the fields $\mathbb{R}$ and $\mathbb{C}$
- non-“algebraist” topics
- determinants

Other textbooks:



LINEAR ALGEBRA

Jim Hefferon
Fourth edition



<http://joshua.smcvt.edu/linearalgebra>

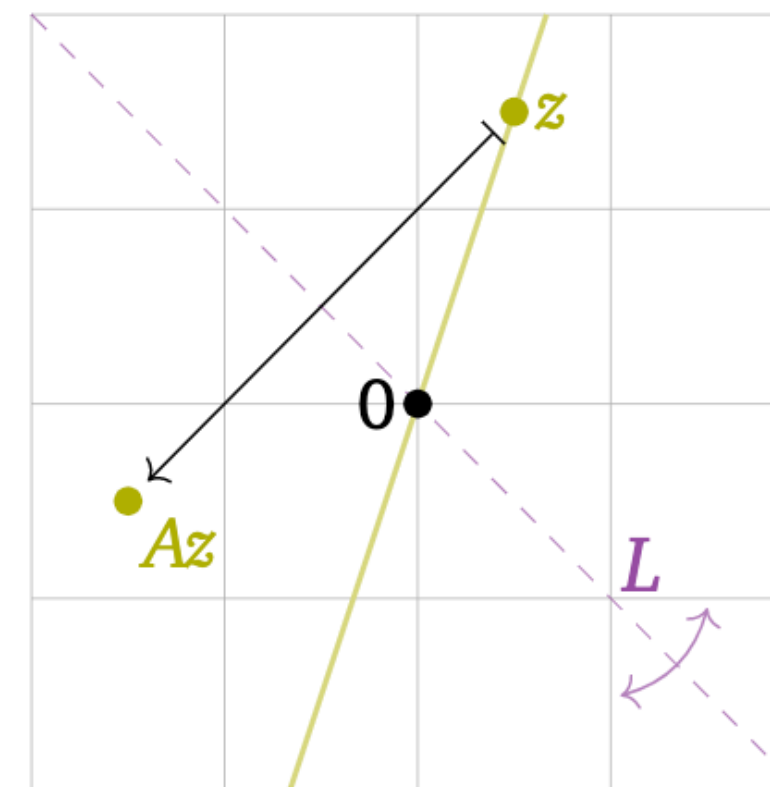
Interactive Linear Algebra

Dan Margalit
School of Mathematics
Georgia Institute of Technology

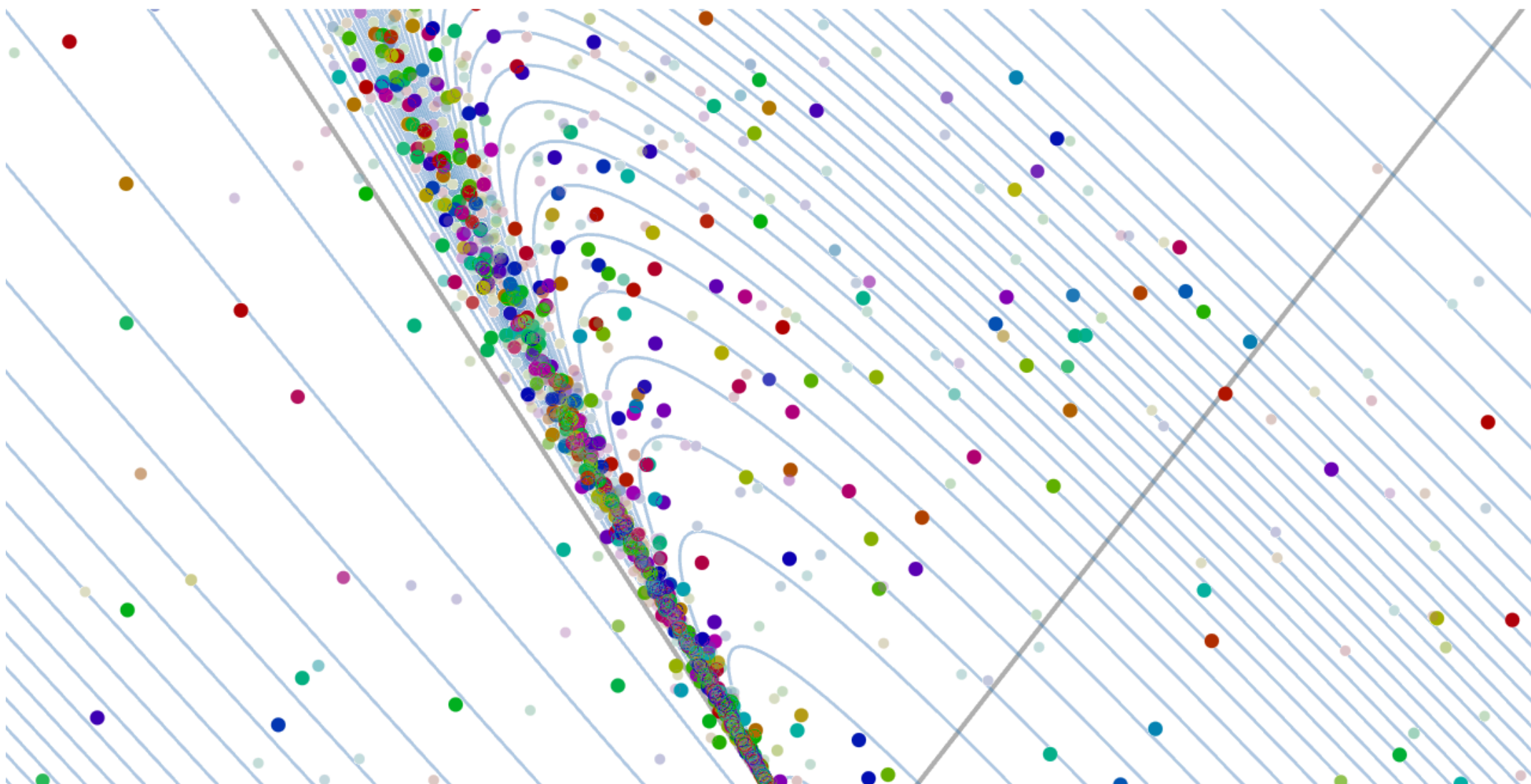
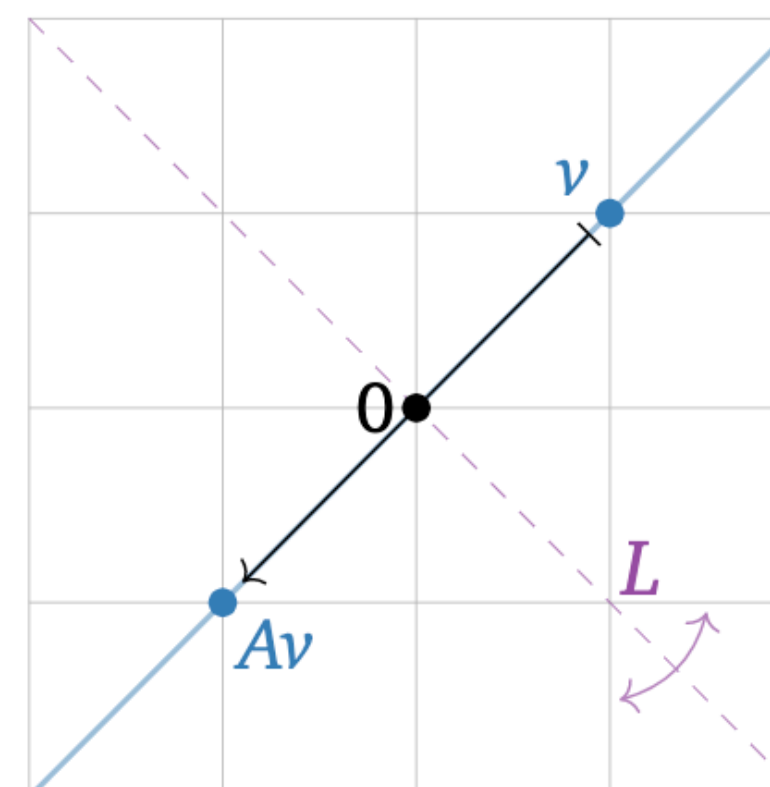
Joseph Rabinoff
School of Mathematics
Georgia Institute of Technology

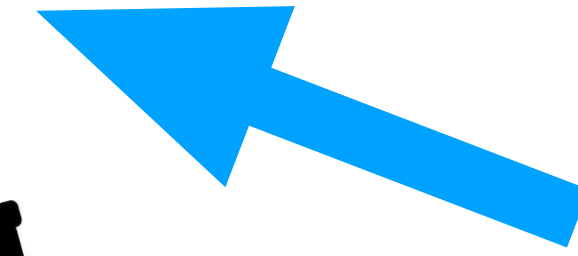
June 3, 2019

The vector u is not an eigenvector, because Au is not collinear with u and the origin.

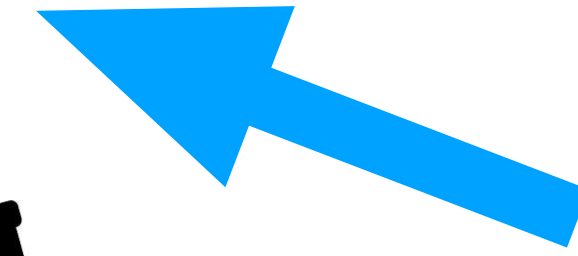


The vector z is not an eigenvector either.

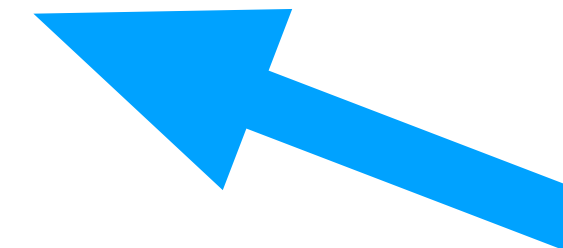




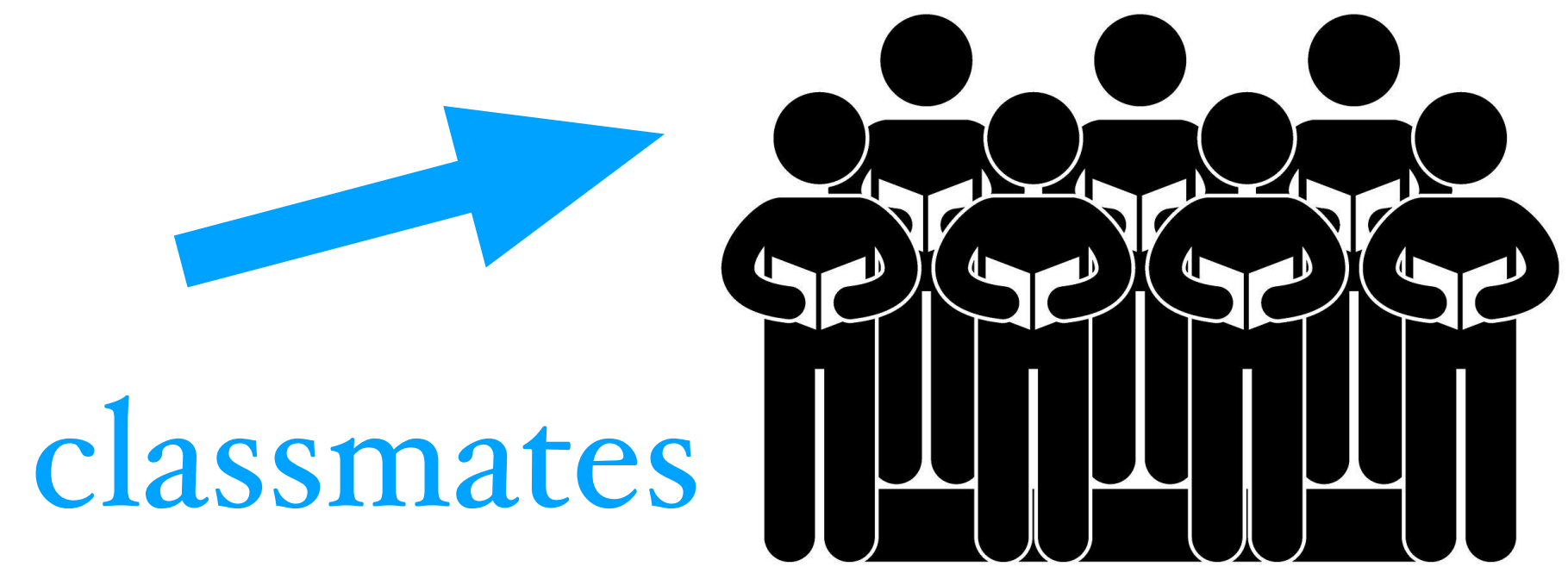
LADW



LADW



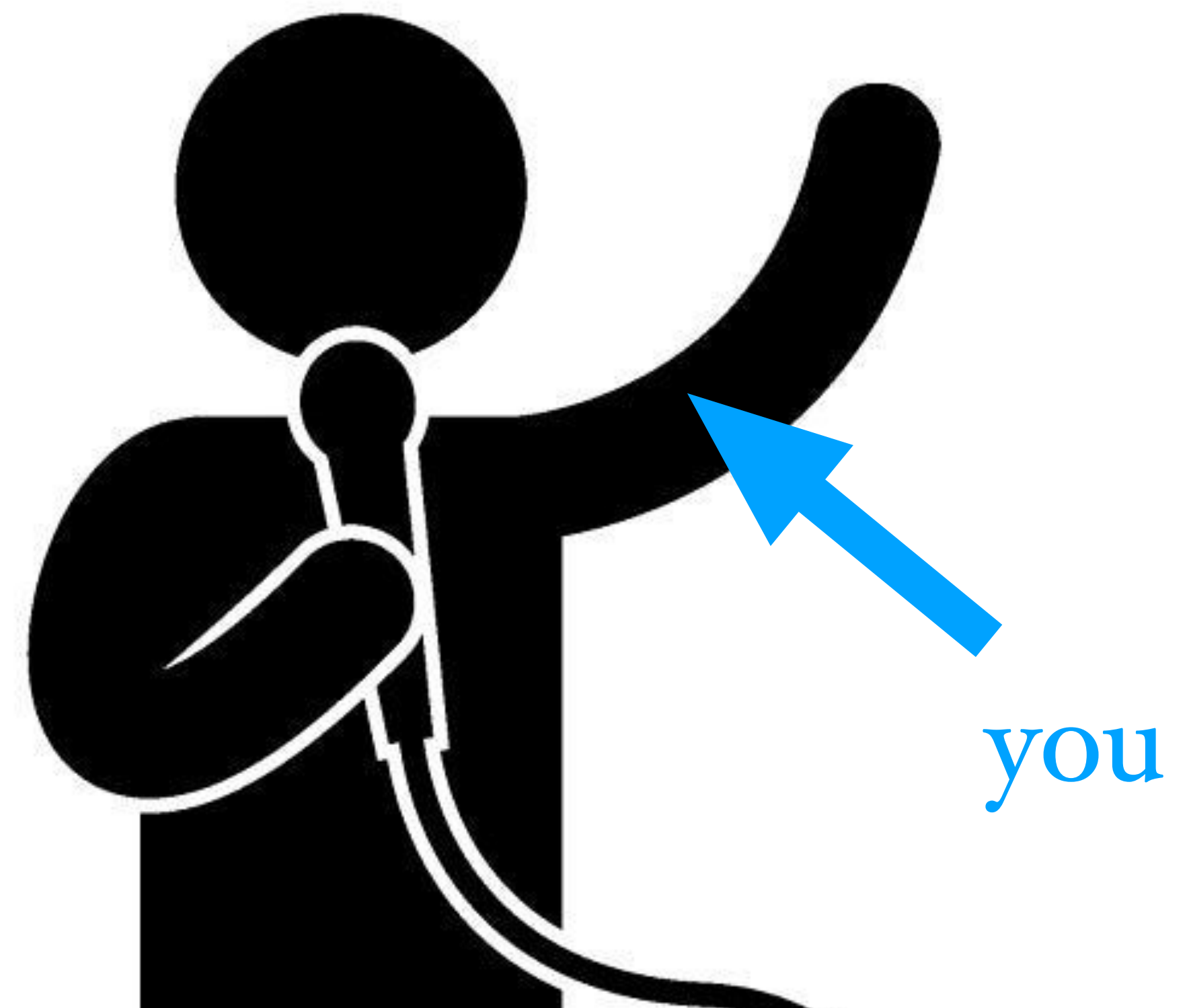
me



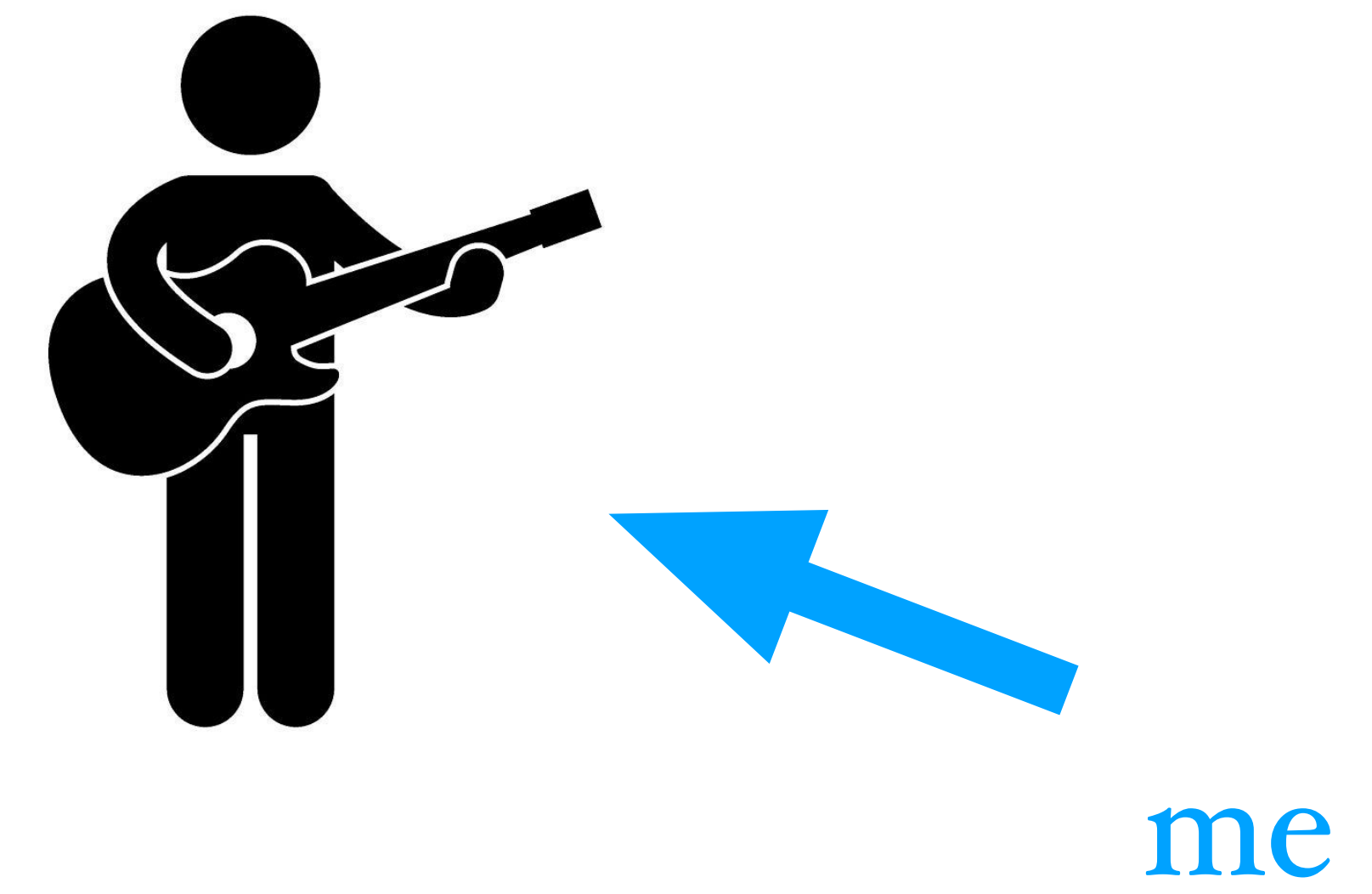
classmates



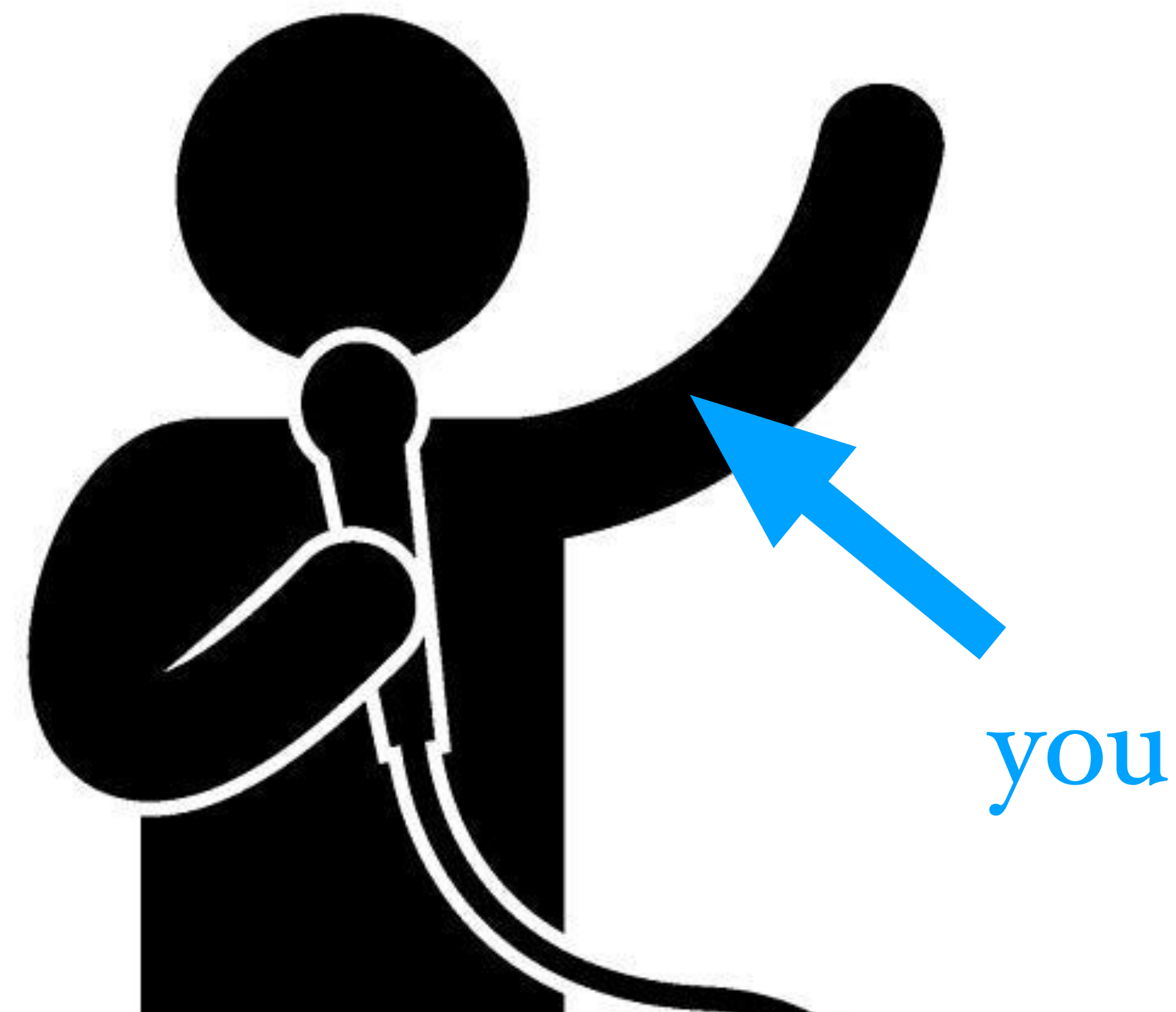
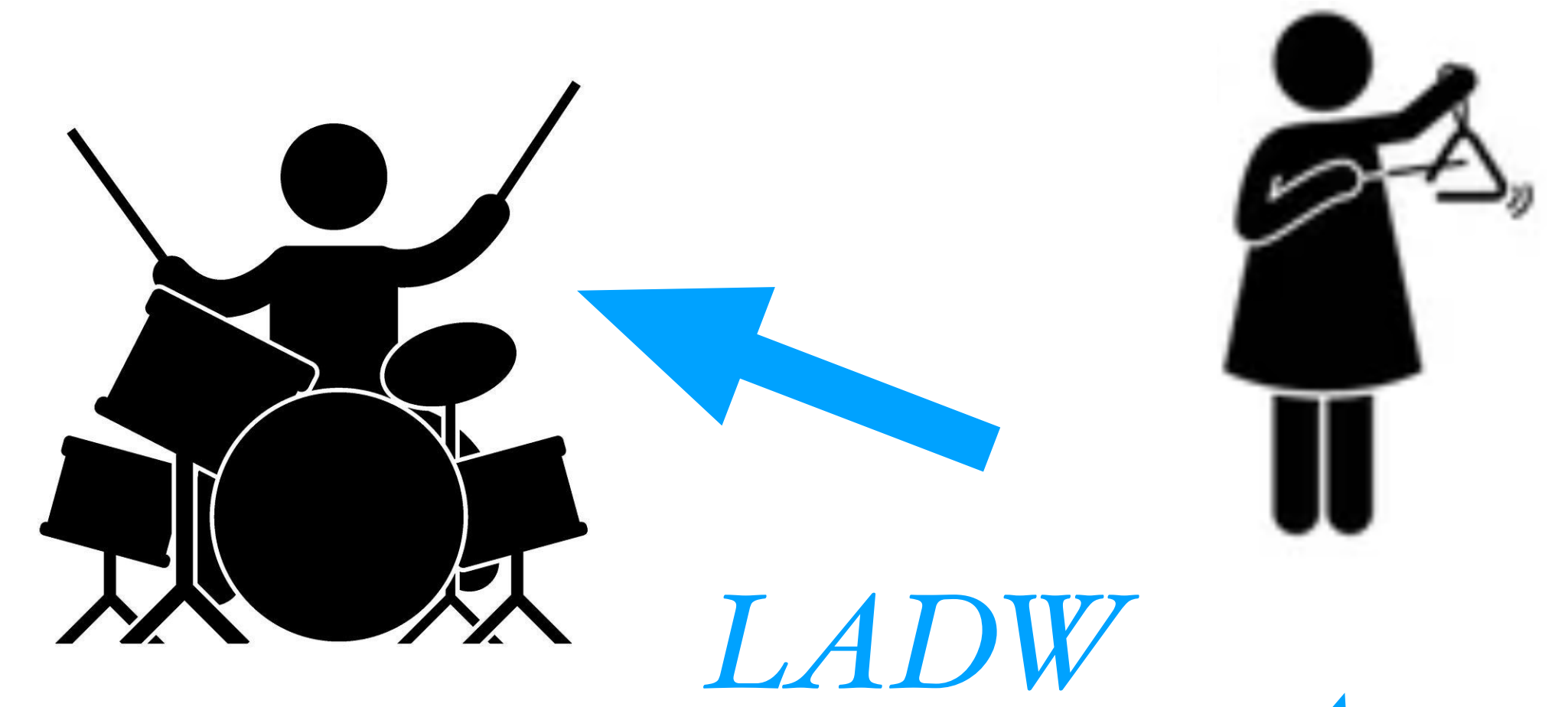
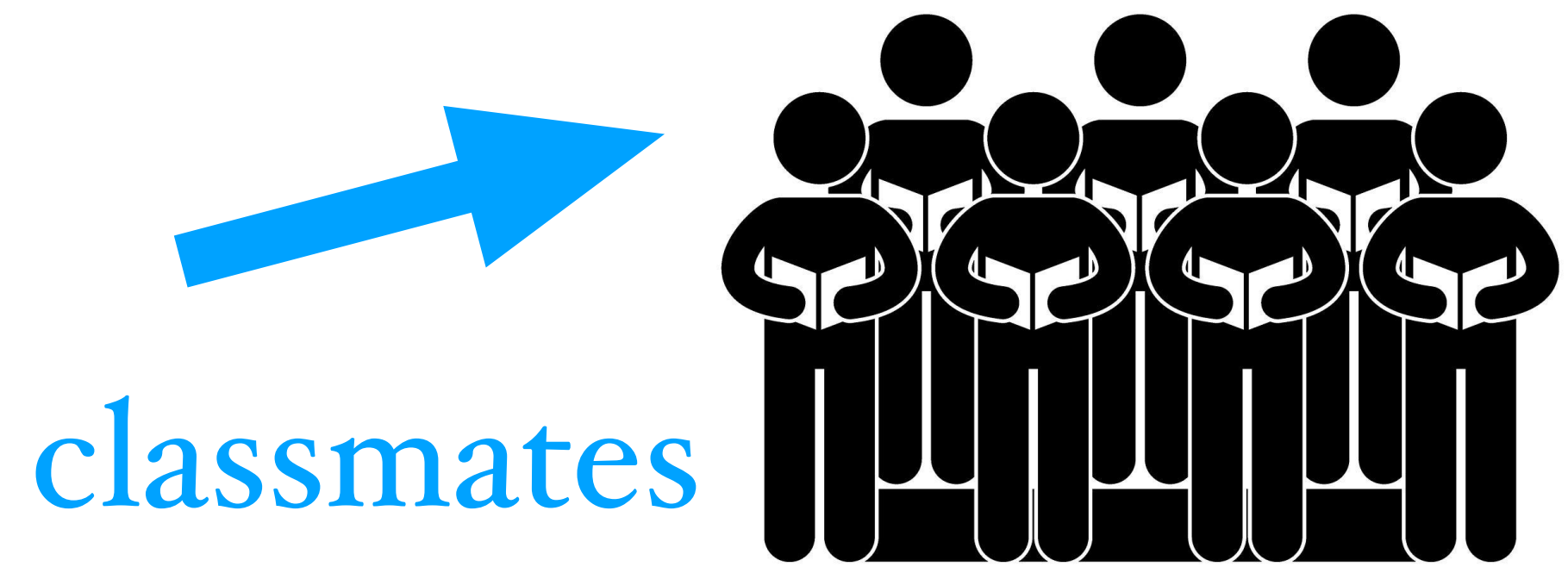
LADW



you



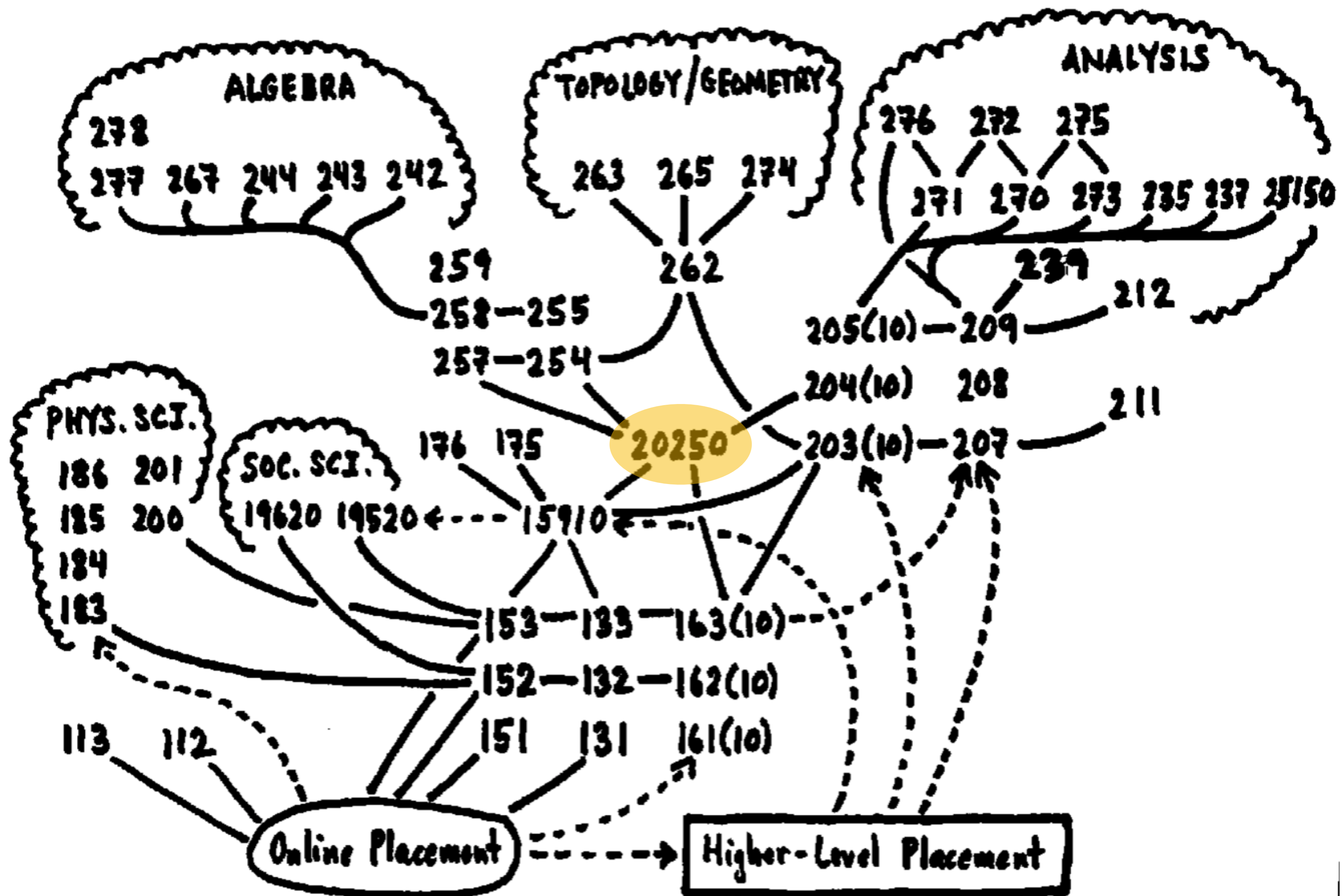
me



Goals of the course:

- To help you to learn the foundational ideas, skills, and results of linear algebra.
- To prepare you for upper-level math coursework, including the algebra and analysis sequences.
- To help you to grow as a mathematician.
- To make you feel both confident and stretched, during the course and once it is over.

The University of Chicago Mathematics Course Tree



This course is:

This course is *not*:

This course is:

ambitious

low-mid-high



```
public void processData()
{
    do
    {
        int data = getData();
        if(data < 0)
            performOperation1(data);
        else
            performOperation2(data);
    }
}
```



This course is *not*:

IBL / axiomatic

“hand hold-y” wrt skills

focused on applications

learning via programming

Assessments and grades

- Homework (20%): 7 assignments; written and computerized
- Quizzes (10%): end of Friday classes. Ten skills to master. Mastery based—you may re-quiz.
- Flex (10%): opt-in opportunities to engage with the course material further. Do even more, it raises your final grade.
- Midterm Exam (20%): Chapters 1-3. Mon April 24.
- Final Exam (40%): Whole course. Date/time TBA.

A: 90. B: 80. C: 70. D: 60.

Homework (20%)

Due on seven Fridays, 11:59pm

8 exercises from *LADW*
or my handout -
4 for correctness,
4 for completion
Submit online (15%)

10 exercises on WebWorK
via Canvas - many chances,
auto-graded (5%)

Plus a short homework reflection

Quizzes (10%)

On Fridays

TEN IMPORTANT SKILLS: SAMPLE QUIZZES

1. a) Let A be the following complex matrix and let v be the following vector in $\mathbb{C}^3$. Find Av .

$$A = \begin{pmatrix} 0 & i & 1 \\ 1+i & 1 & 0 \end{pmatrix}, \quad v = \begin{pmatrix} 1 \\ i \\ 1 \end{pmatrix}$$

b) Let B be the following real matrix:

$$B = \begin{pmatrix} 8 & 0 \\ 0 & 1 \\ 0 & 4 \end{pmatrix}$$

Complete the following sentence by filling in the four blanks:

The matrix B is a ____ $\times$ ____ matrix and it represents a linear transformation from $\mathbb{R}^-$ to $\mathbb{R}^-$.

mastery-based

Flex (10%)

Variable due dates

- I. Participate in group activities. (Join a study group.)
- II. Read and reflect. (Complete short reflections as you read the book.)
- III. Research and write. (Do a mini-project on an extension topic.)

Doing further flex work can raise your final grade.

More details Wednesday.

Fill out flex planning
form by next Sunday!

Midterm and Final (20%, 40%)

Questions about important definitions and major theorems (10%)

Problems applying important skills (10%)

Problems modeled closely on homework exercises (60%)

Instructor's choice (20%)

Review packets for both.

Midterm can be revised.

Final exam can be curved.

Office hours:

1. Half-hour “small group” sessions
2. “Open” drop-in hours
3. Homework debriefs
4. Individual meetings by request

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My office hours for the first week of the course are held in Eckhart 328 (my office):

Monday, 3:30-4:30pm

Tuesday, 12:00-1:00pm

Wednesday, 7:00-7:30pm on [Zoom](#)  (not in my office)

Friday, 3:30-4:30pm

and by appointment.

Drop by!

Accommodations & Religious Accommodation & Health Pact

Accommodations & Religious Accommodation & Health Pact

I encourage communication. Just ask!

Course Calendar

Course Calendar
 MATH 20250 - Abstract Linear Algebra
 Spring 2023, Sections 45 and 55, Prof. Lanier

Week	Sections covered in <i>LADW</i>	Monday	Wednesday	Friday
Week 1 March 20	Ch 1.1-4 18 pages			HW1 assigned
Week 2 March 27	Ch 1.5-7 Ch 2.1-3 28 pages			Quiz 1 & 2 HW1 due, HW2 assigned
Week 3 April 3	Ch 2.4-8 28 pages			Quiz 3 & 4 HW2 due, HW3 assigned
Week 4 April 10	Ch 3.1-6 21 pages		Problem swap 1: problems submit	Quiz 5 HW3 due, HW4 assigned Midterm review distributed
Week 5 April 17	Ch 4.1-2 17 pages		Problem swap 1: solutions submit	Prof. Lanier away Quiz 6 & 7 HW4 due, HW5 assigned
Week 6 April 24	finishing Ch 4, starting Ch 5	Midterm on Ch 1-3		Quiz 8 HW5 due, HW6 assigned
Week 7 May 1	Ch 5.1-3 Ch 6.1-2 27 pages		Problem swap 2: problems submit	Quiz 9 HW6 due, HW7 assigned Midterm revisions due
Week 8 May 8	SVD (see Ch 6.3-4) Ch 9.1-2 6+ pages		Problem swap 2: solutions submit	Quiz 10 HW7 due
Week 9 May 15	Ch 9.3-5 15 pages			Final review distributed Last requiz session Due date for most flex

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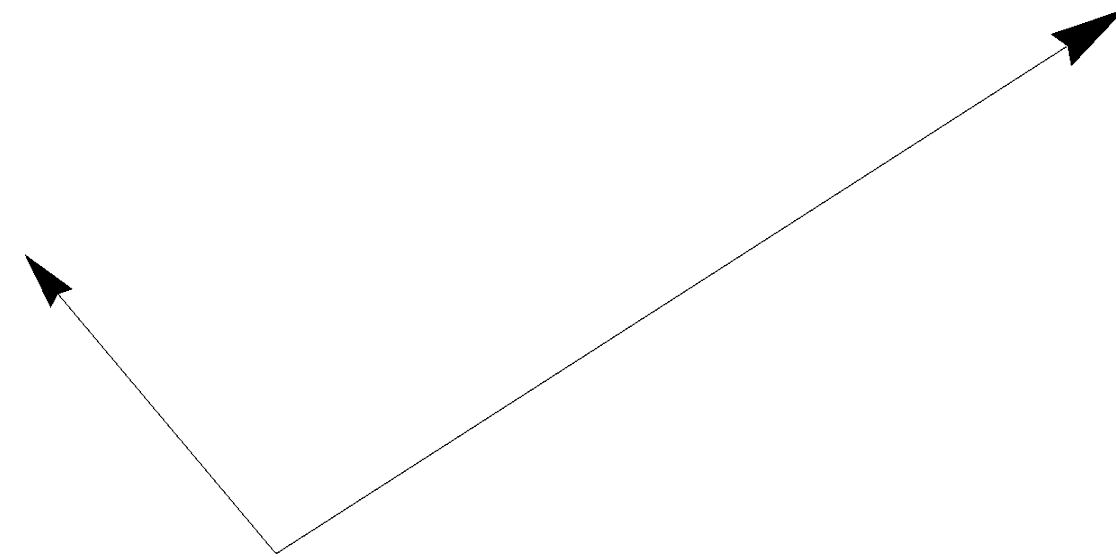
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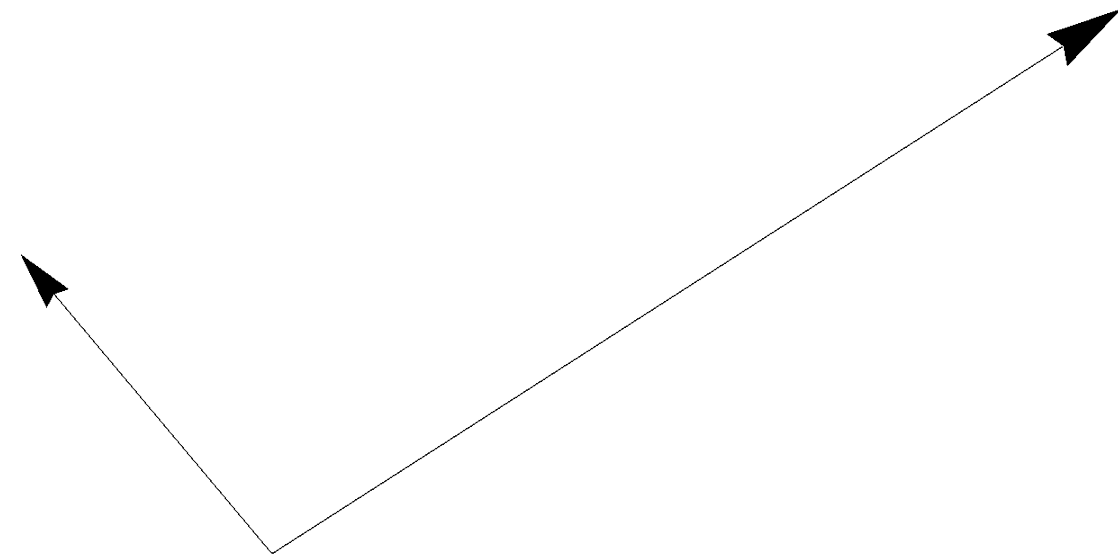
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- Begin to give an overview of the main ideas of the course (Ch 1-2)

What is Linear Algebra?

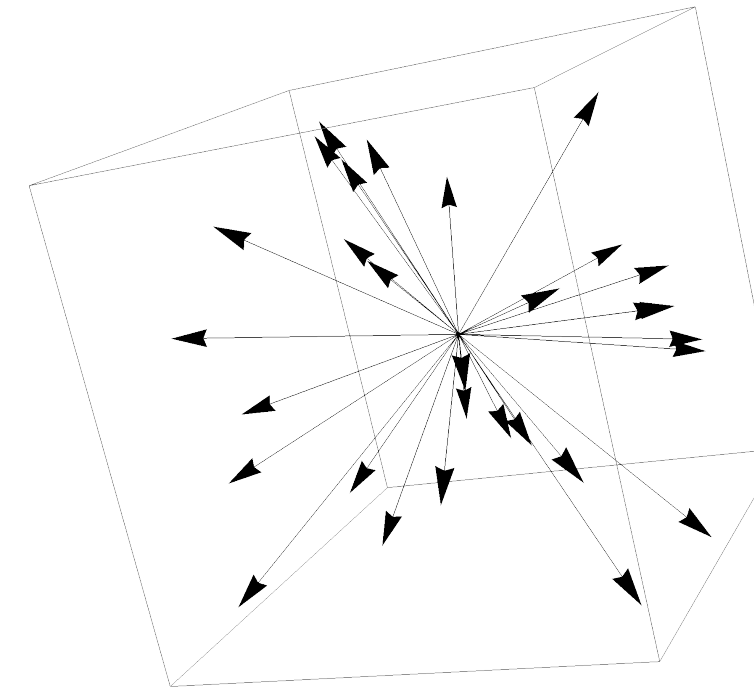


vectors

What is Linear Algebra?

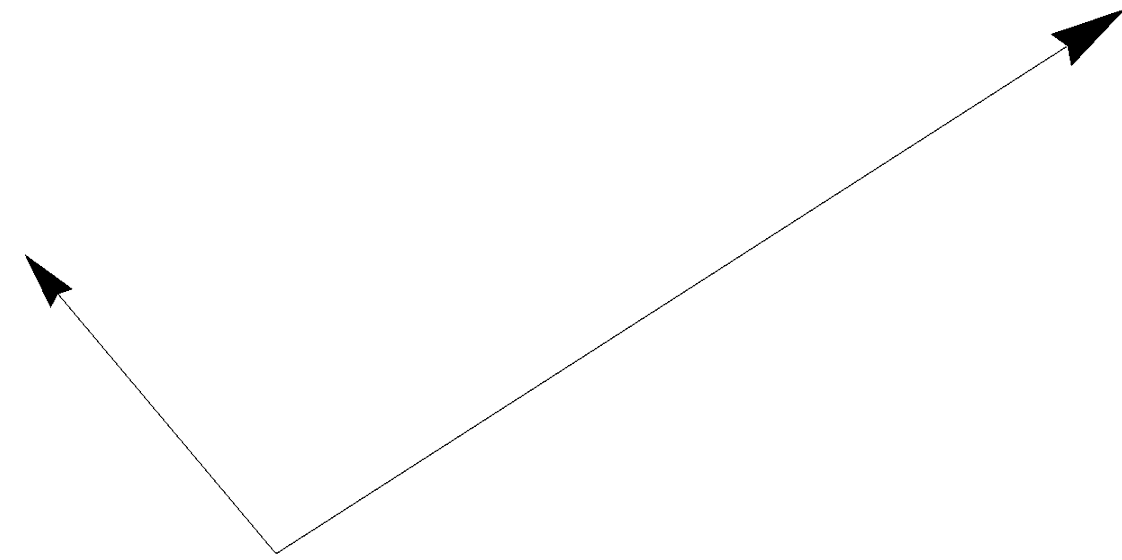


vectors

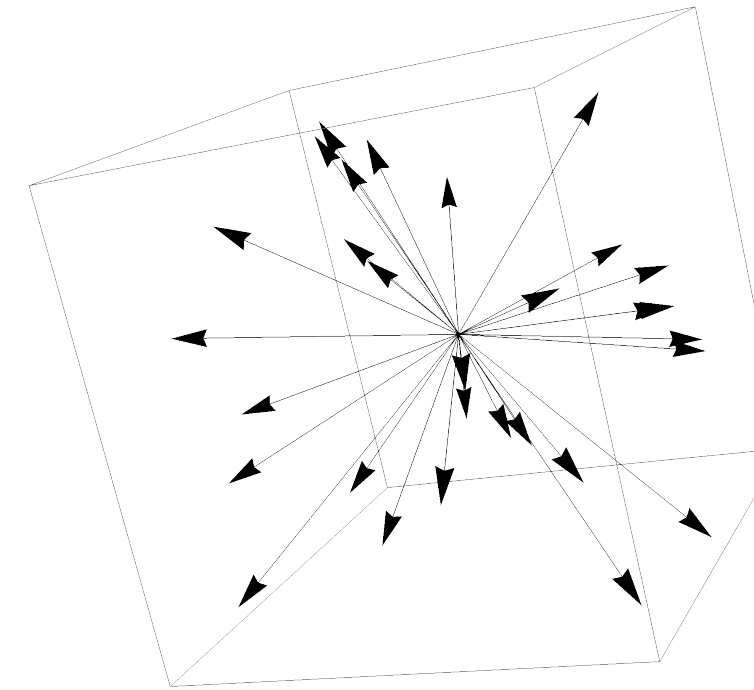


vector spaces

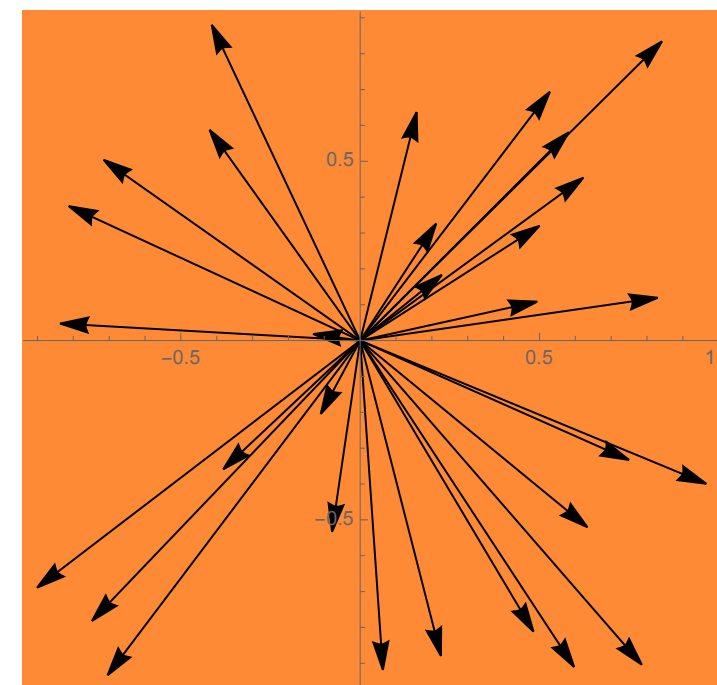
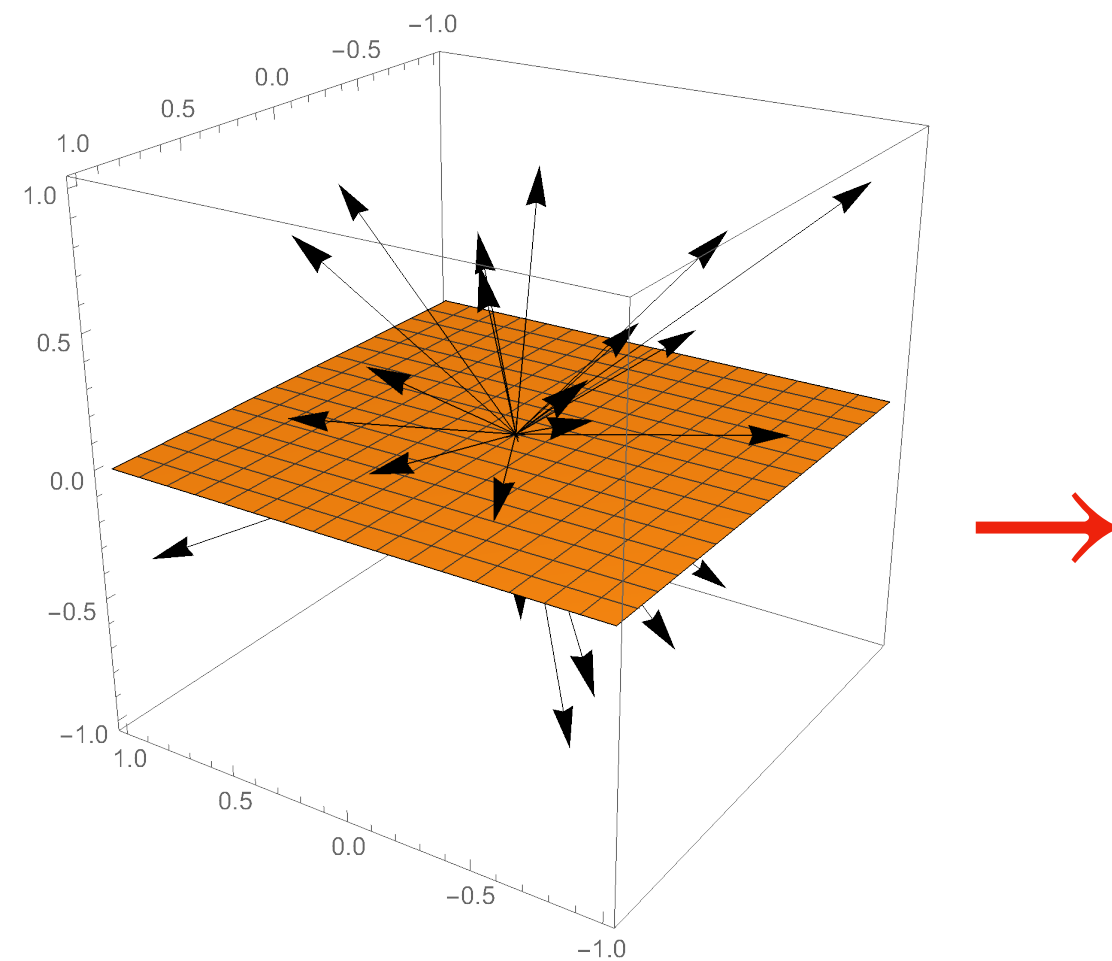
What is Linear Algebra?



vectors

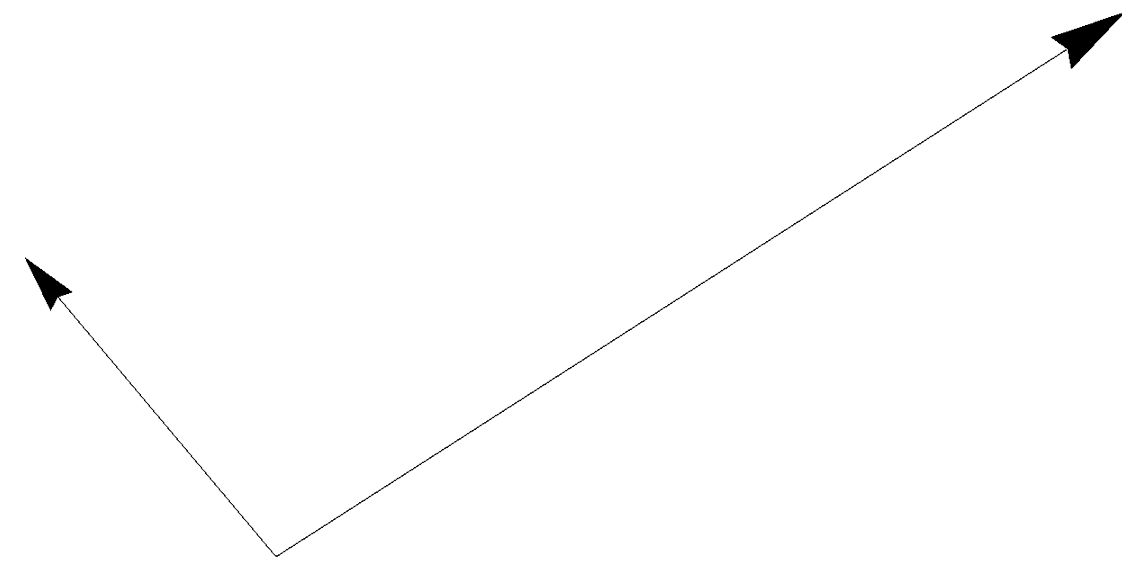


vector spaces

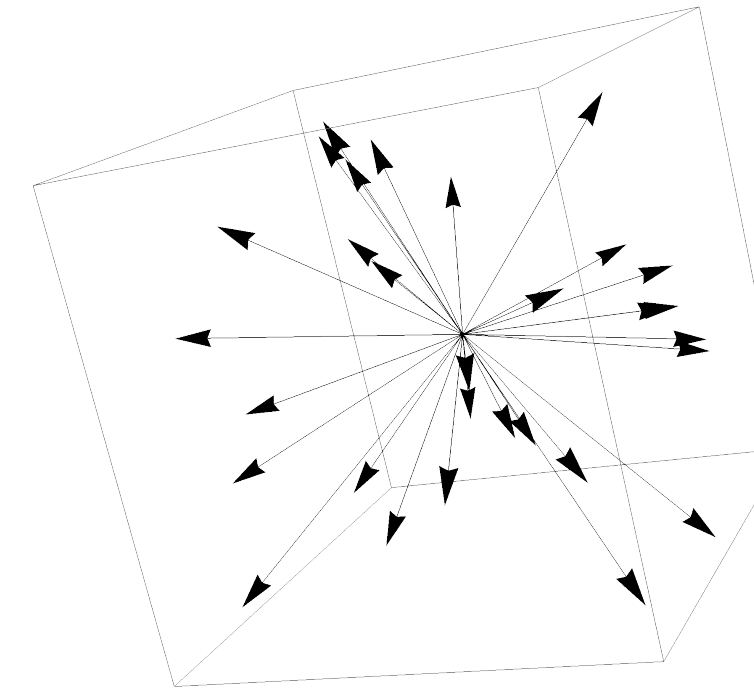


linear maps between
vector spaces

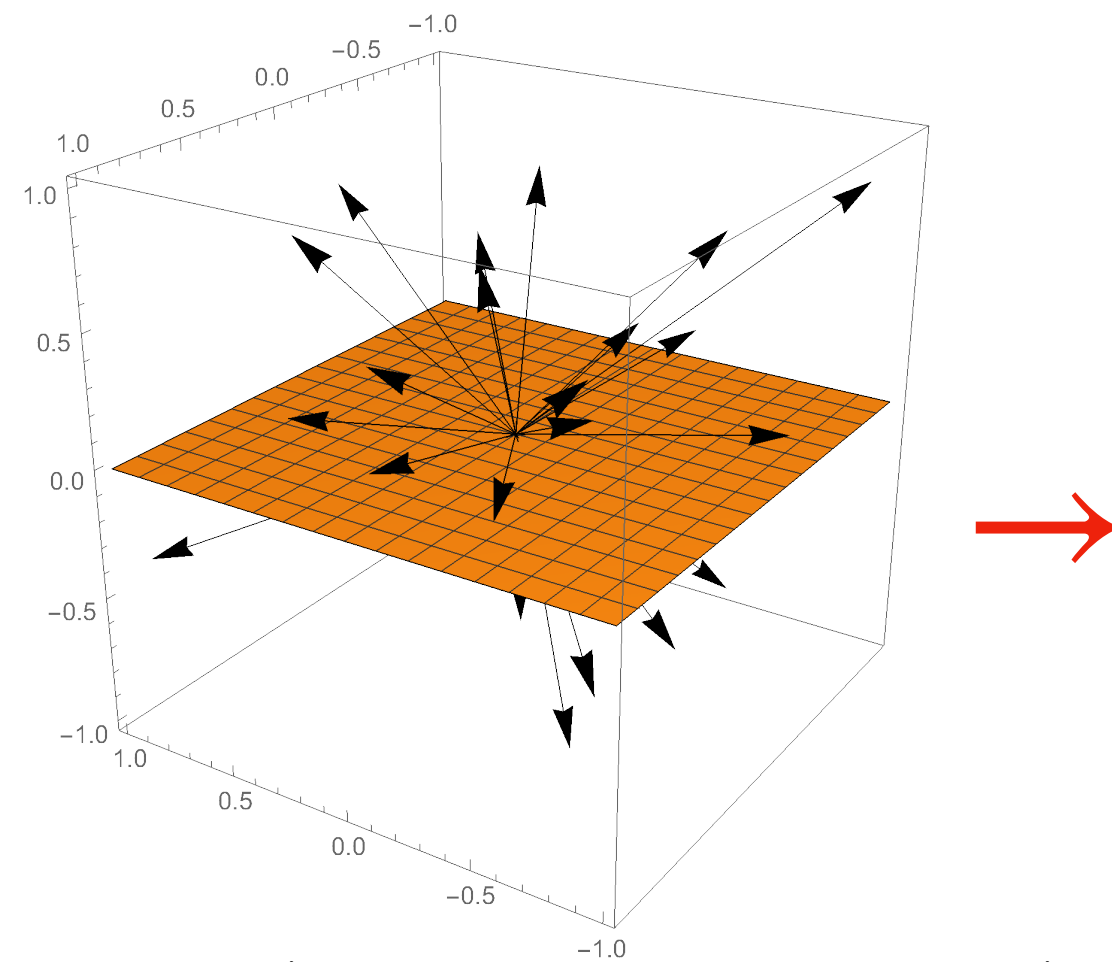
What is Linear Algebra?



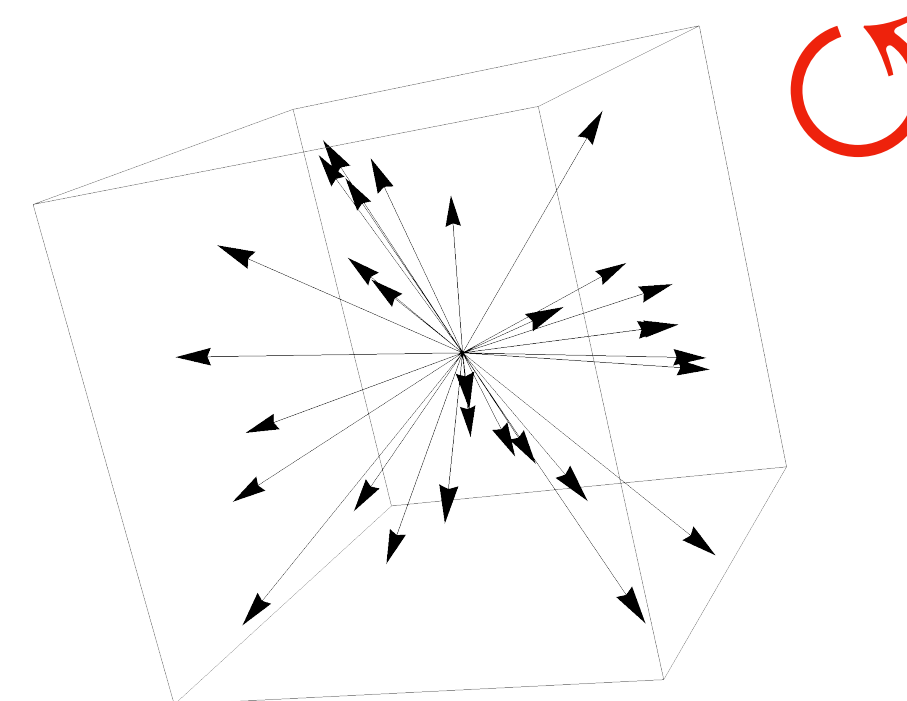
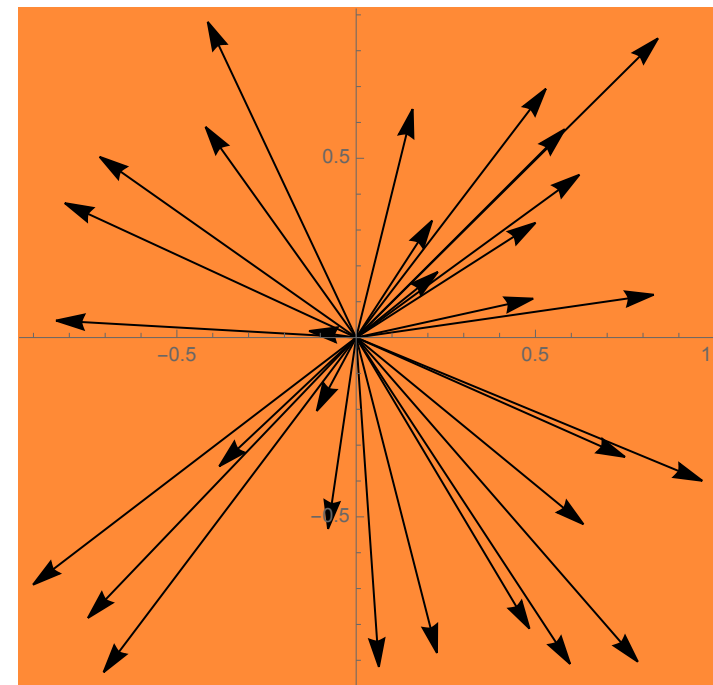
vectors



vector spaces

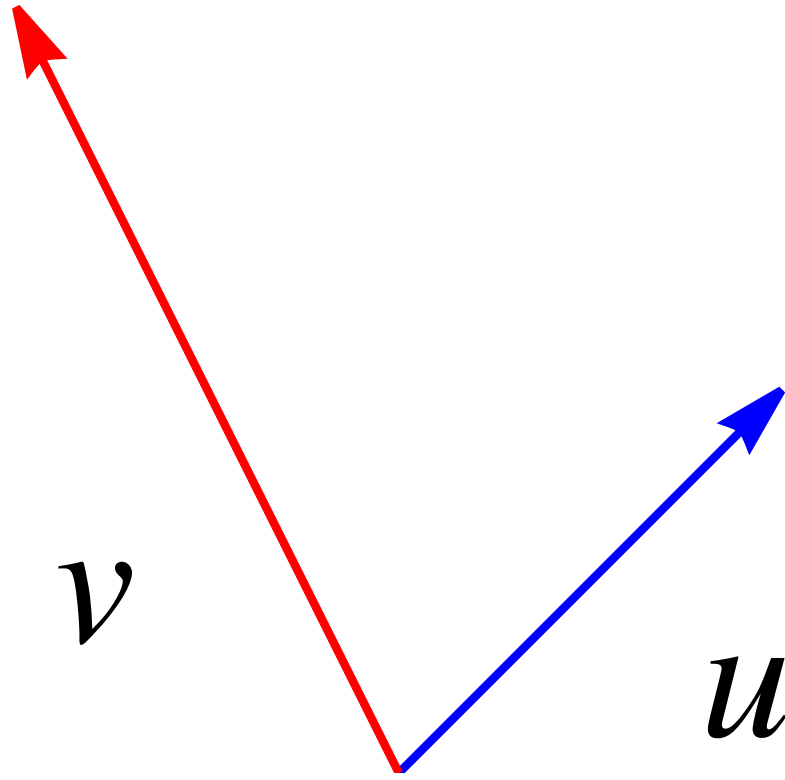


linear maps between
vector spaces

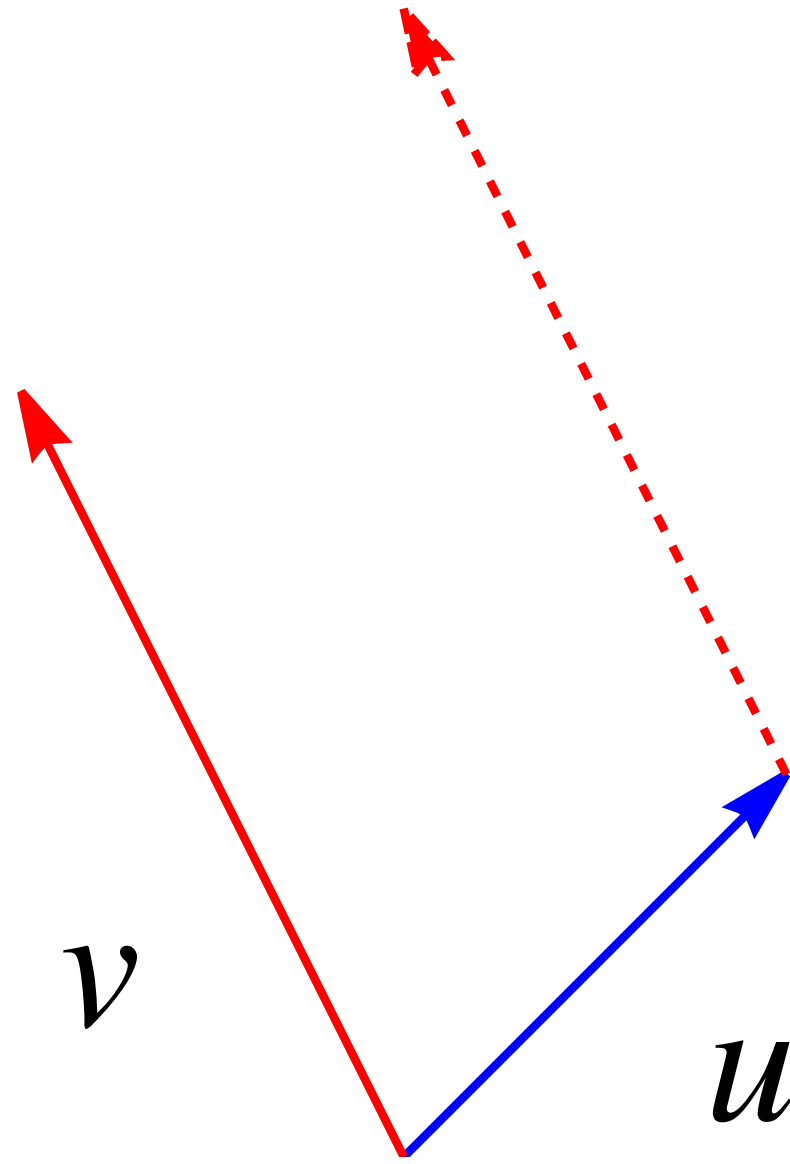


linear self-maps of
vector spaces

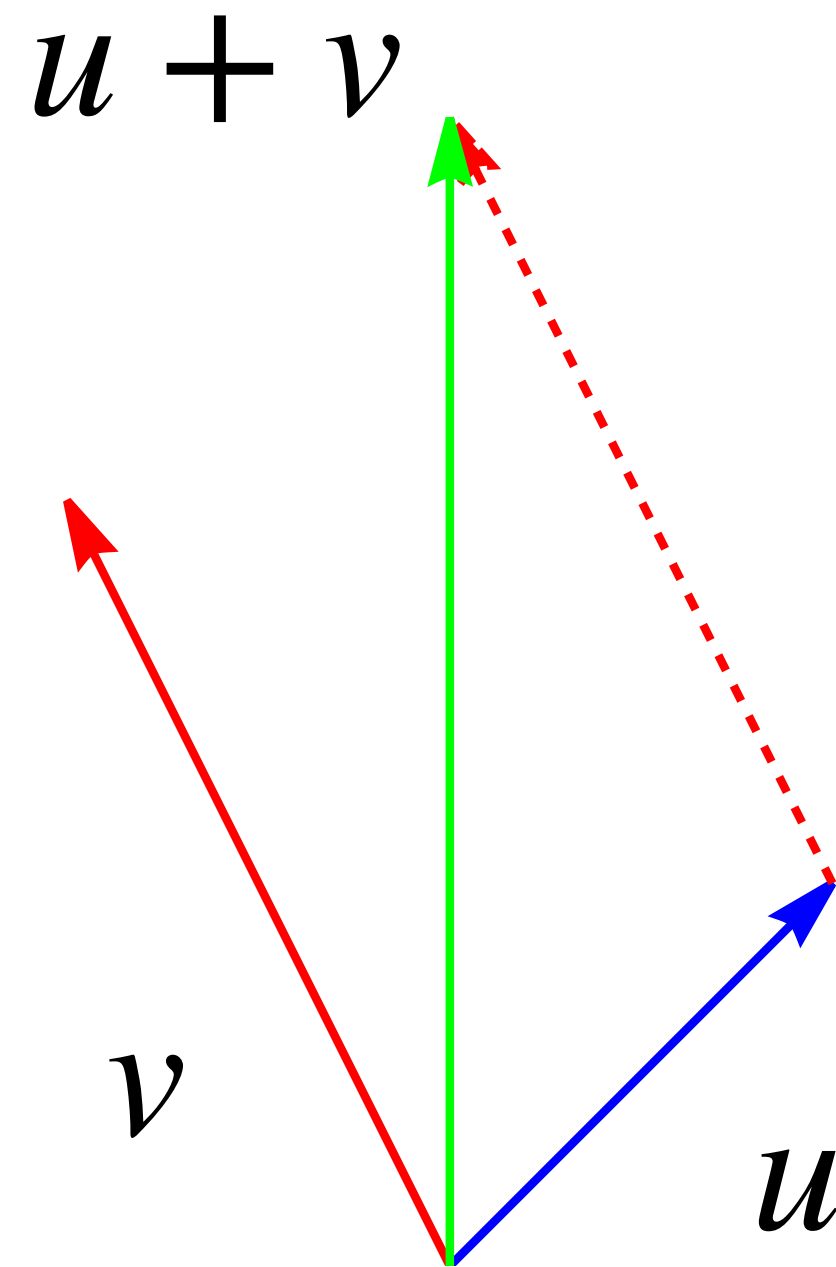
vectors



vectors

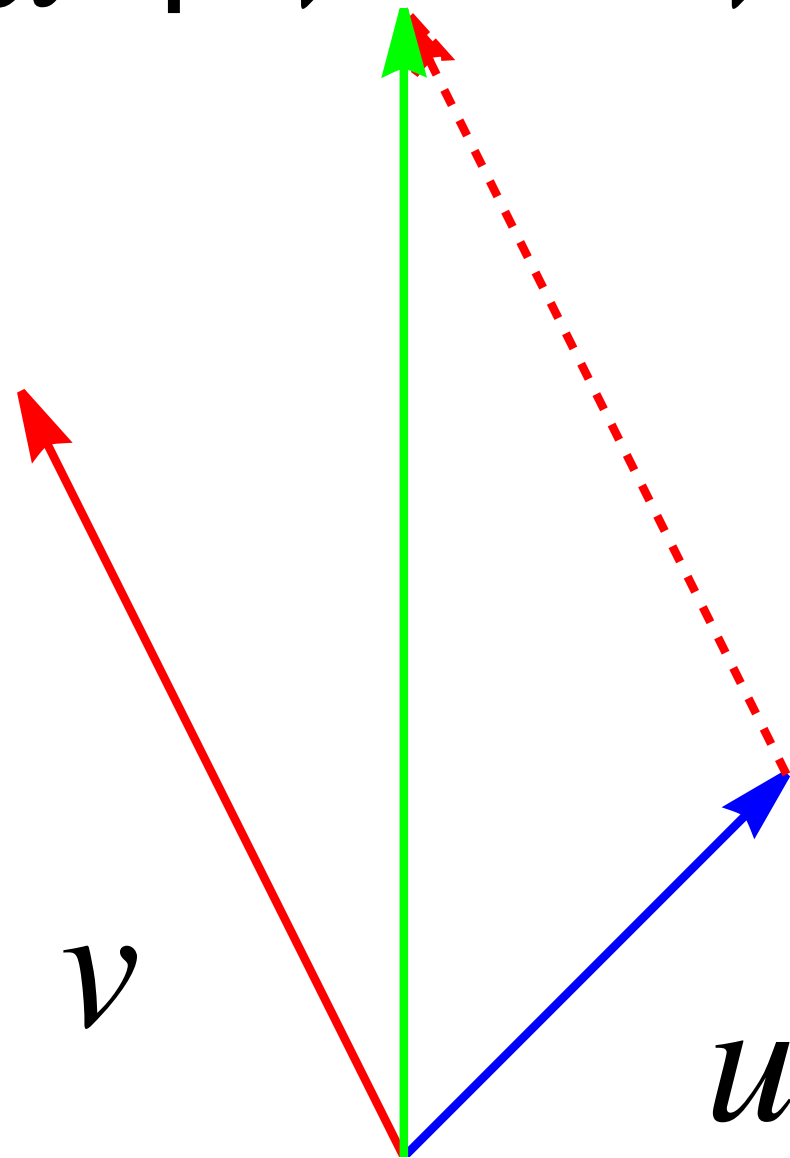


vectors

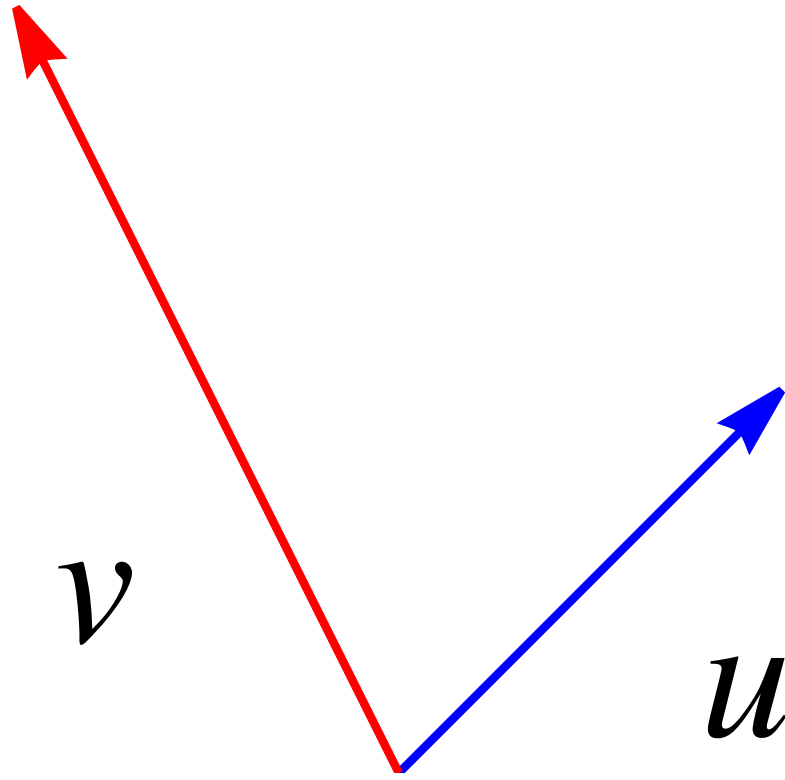


vectors

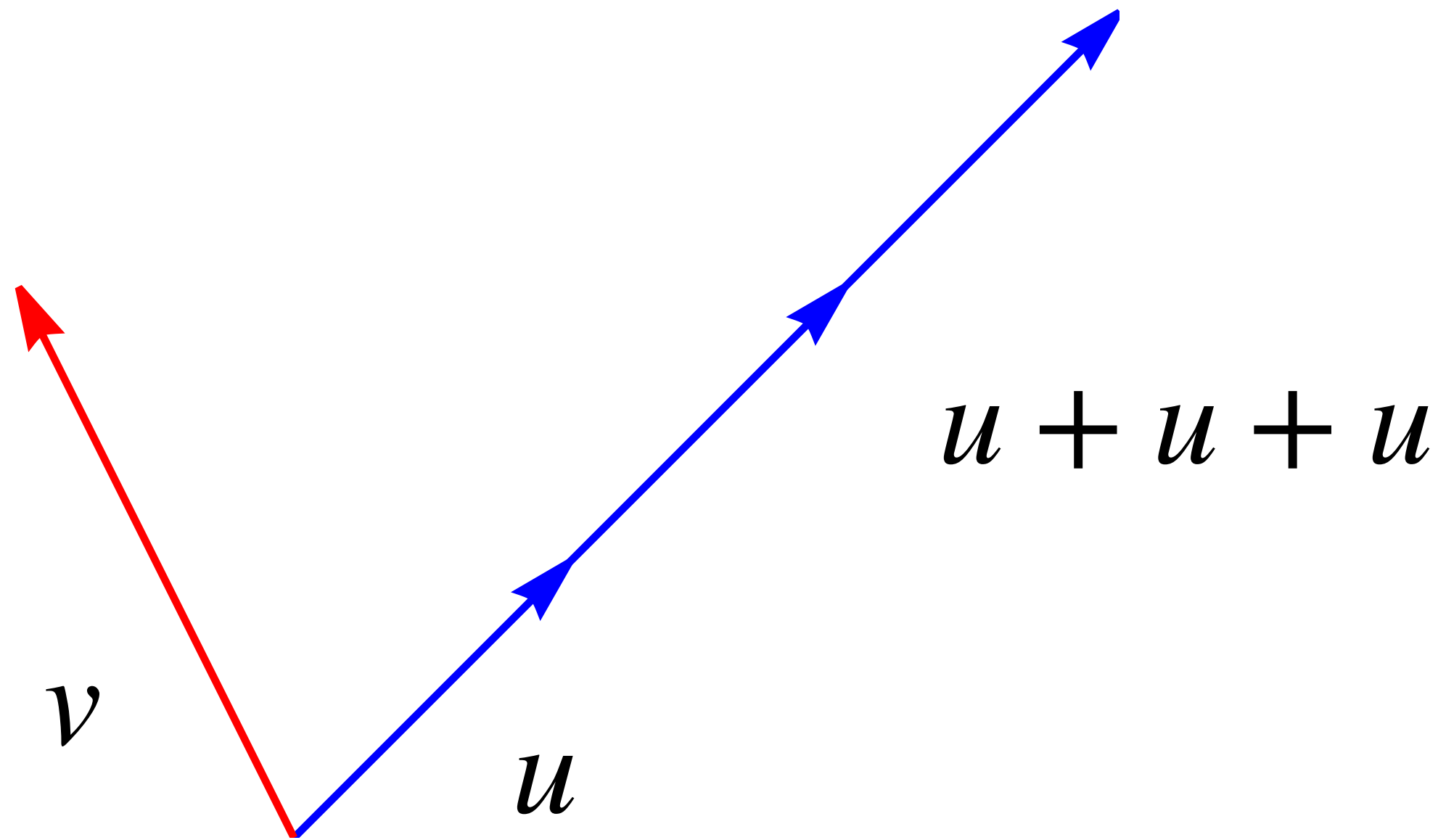
$$u + v = v + u$$



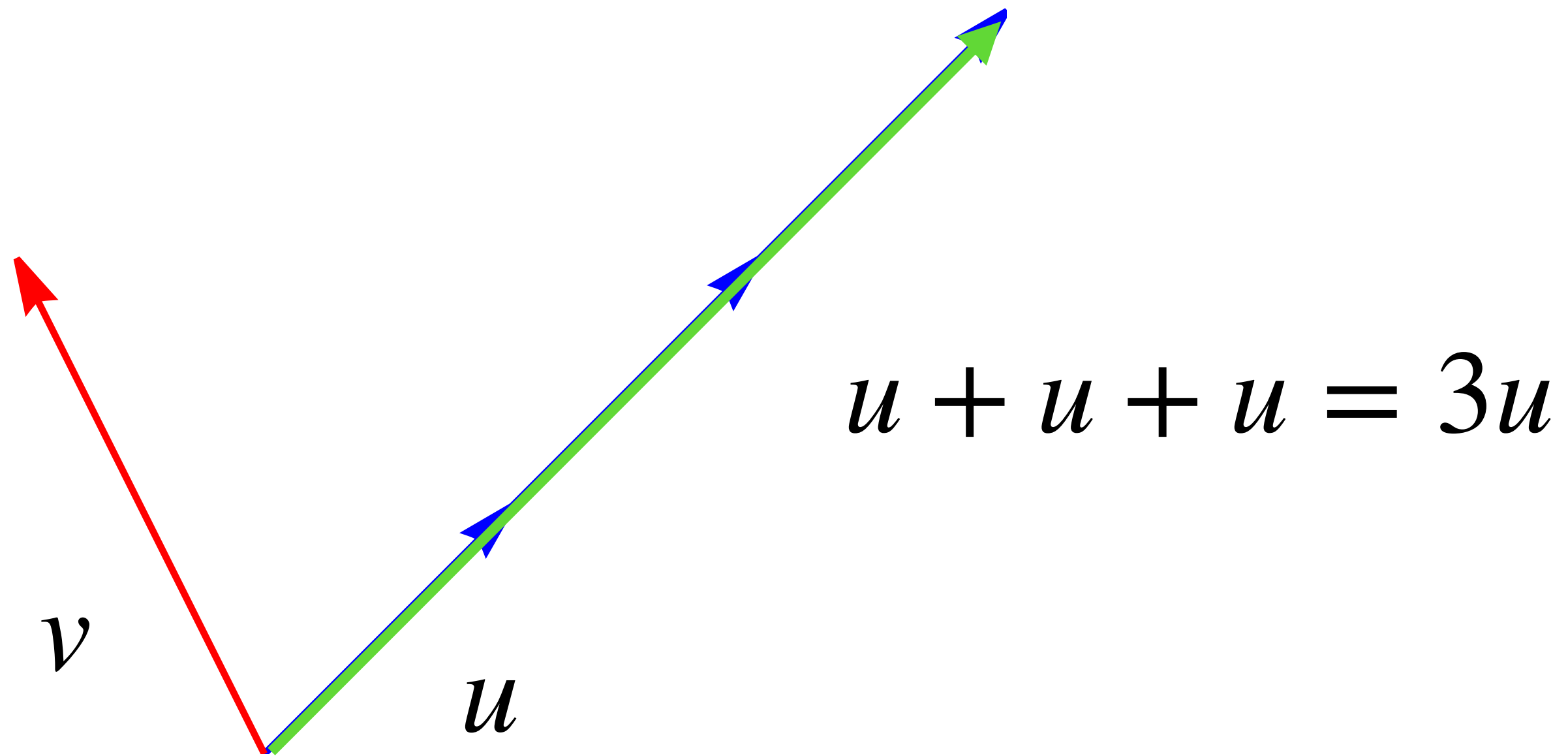
vectors



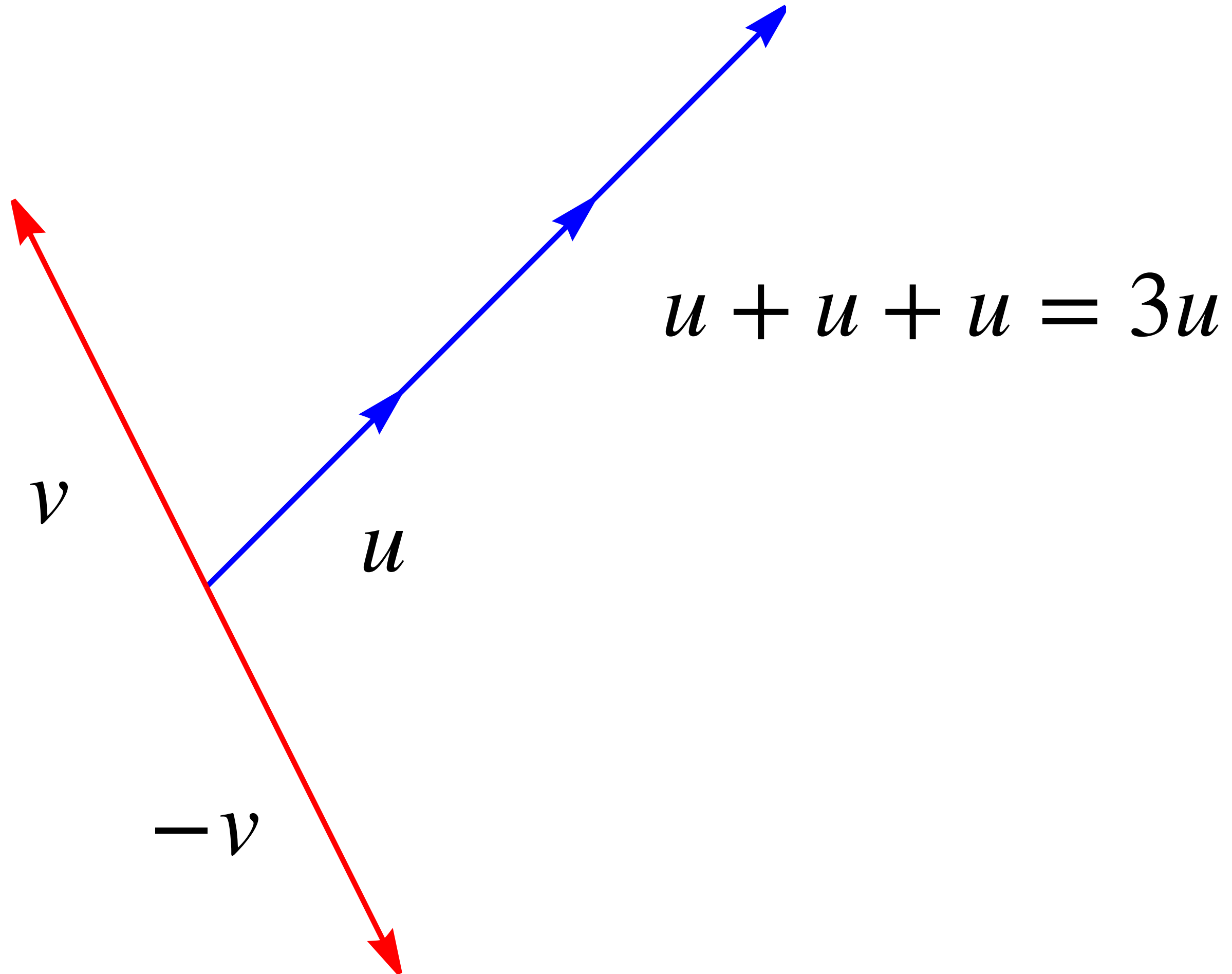
vectors



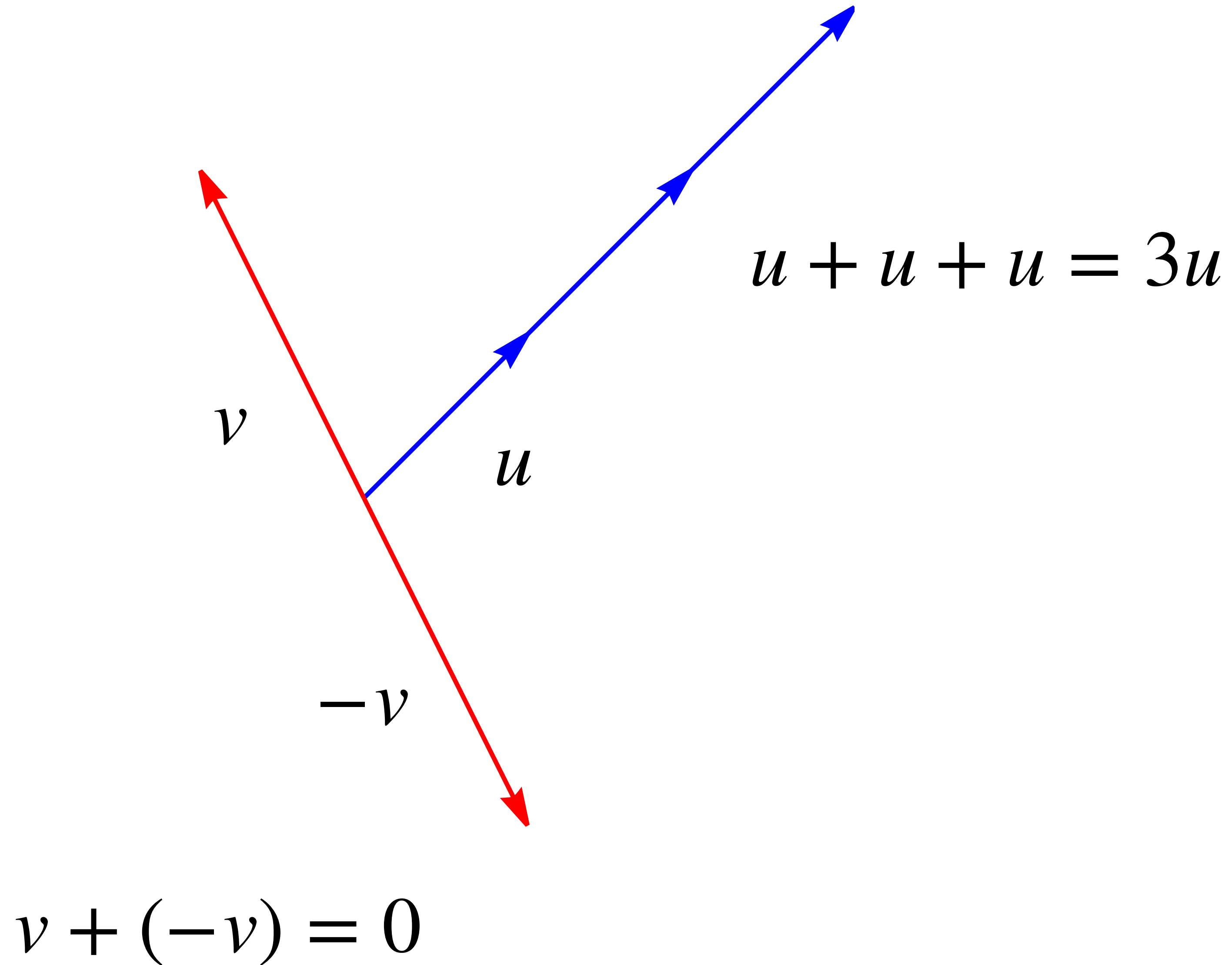
vectors



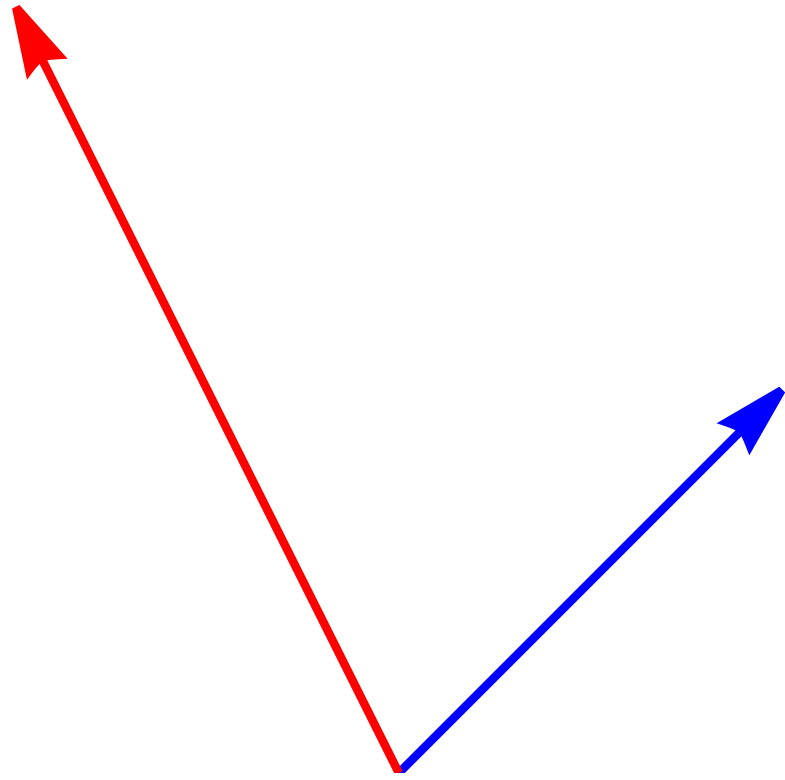
vectors



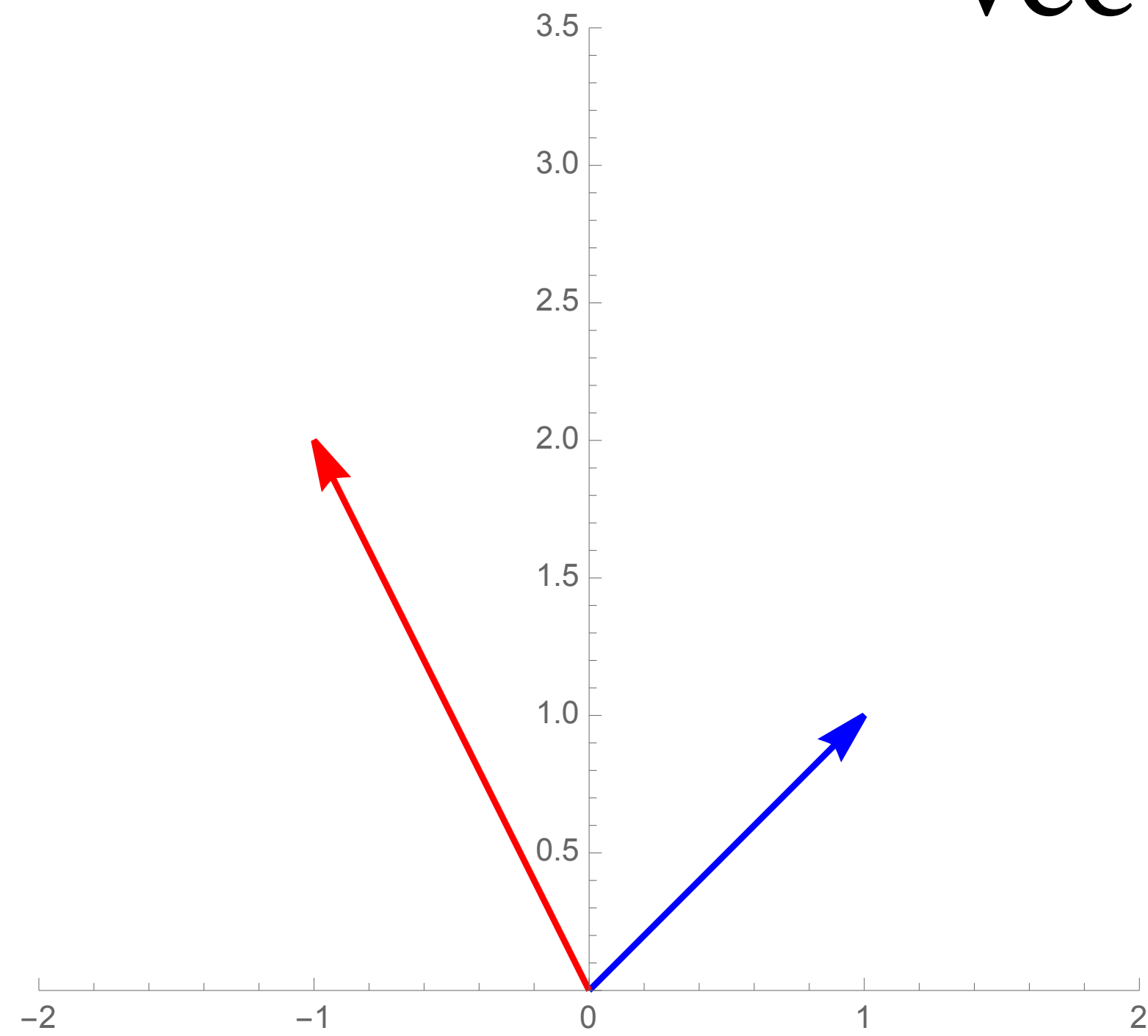
vectors



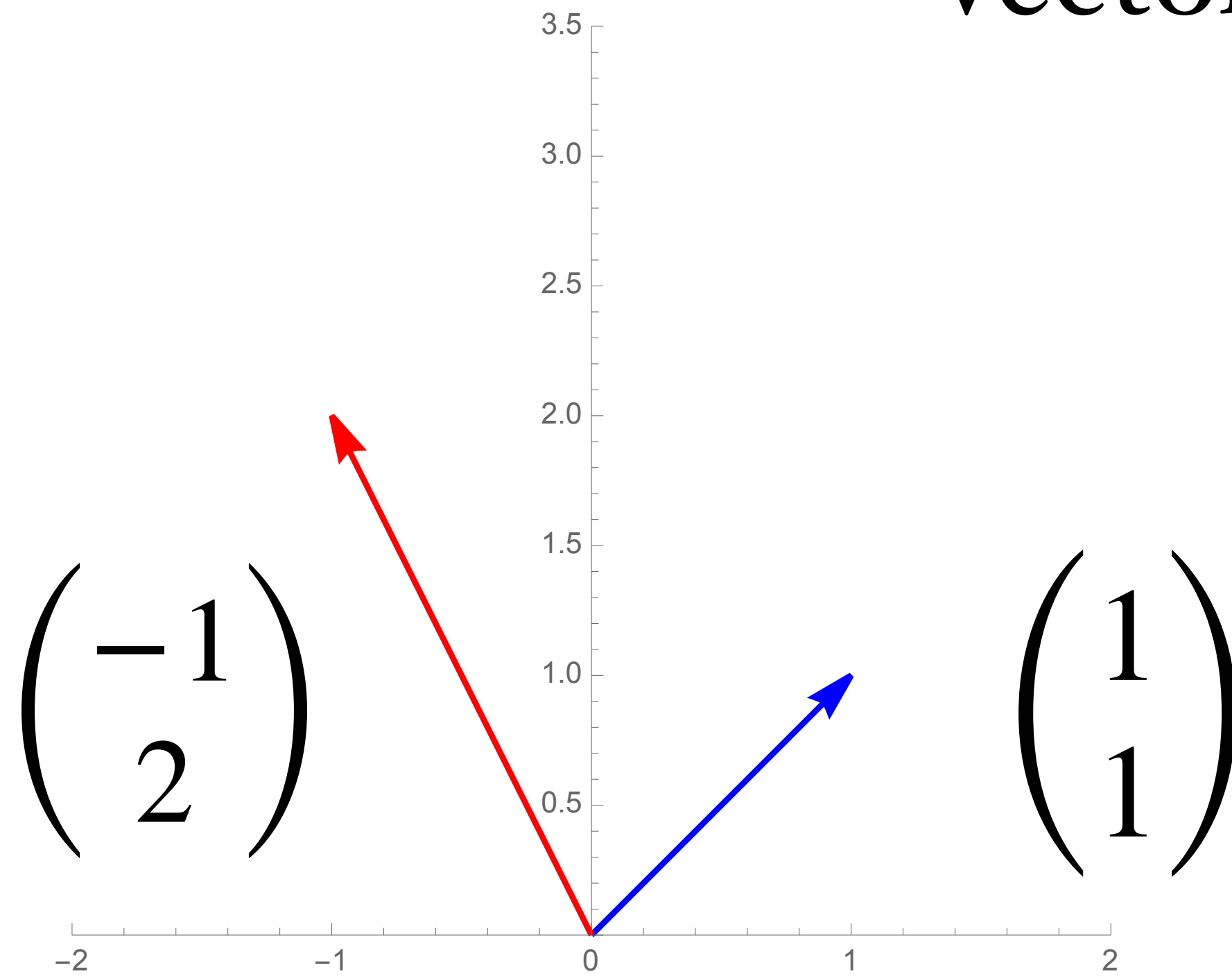
vectors in $\mathbb{R}^2$



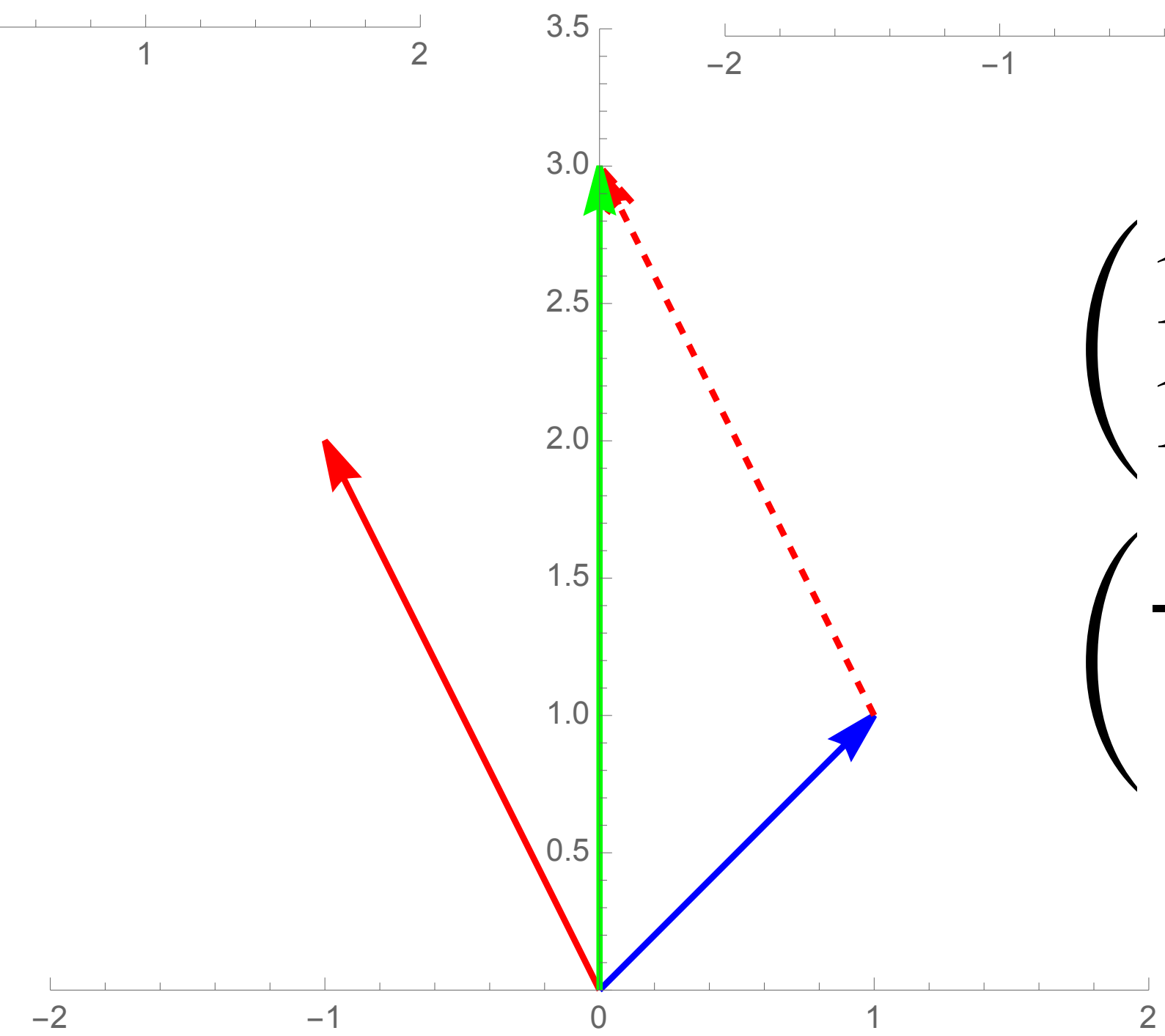
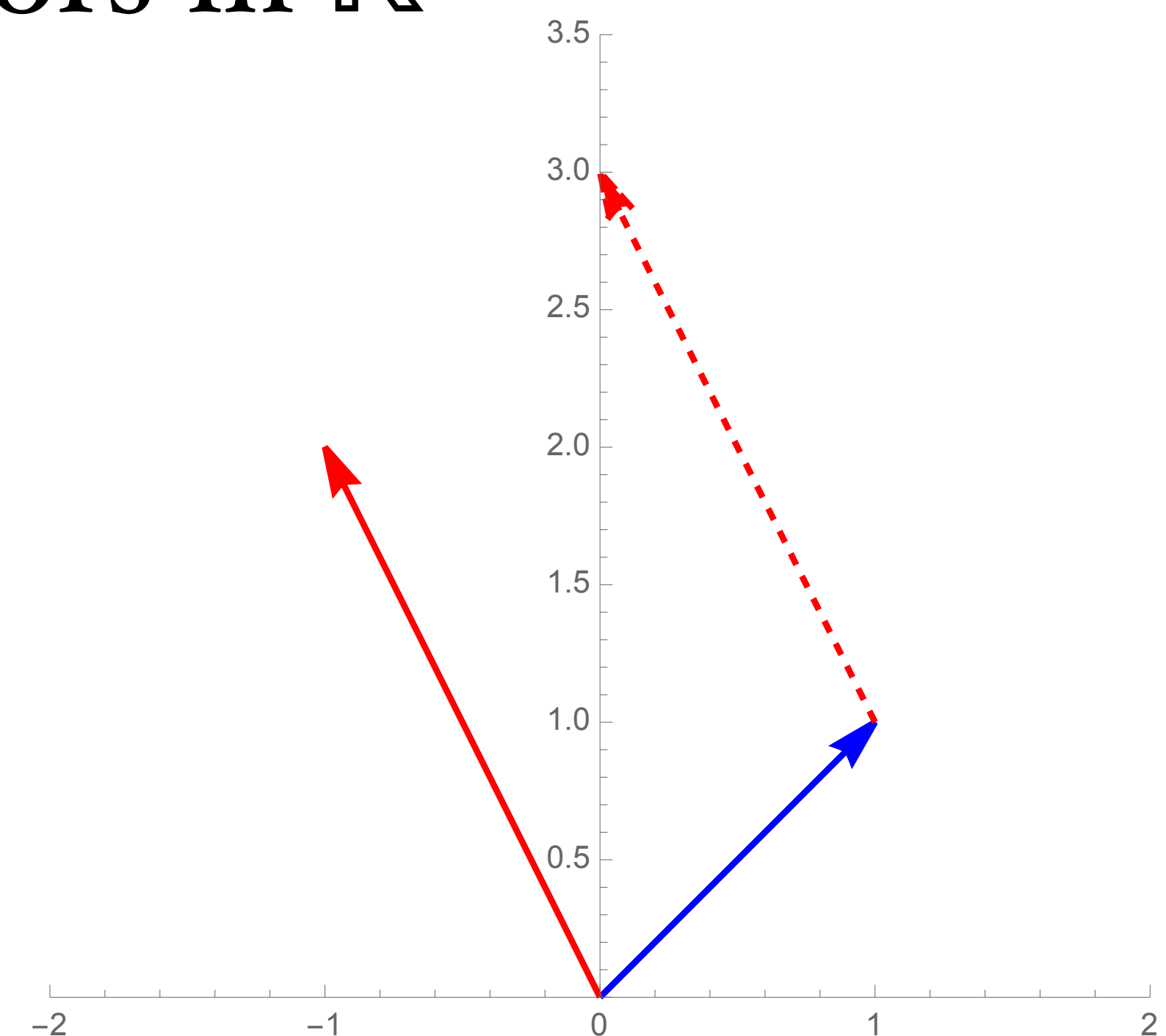
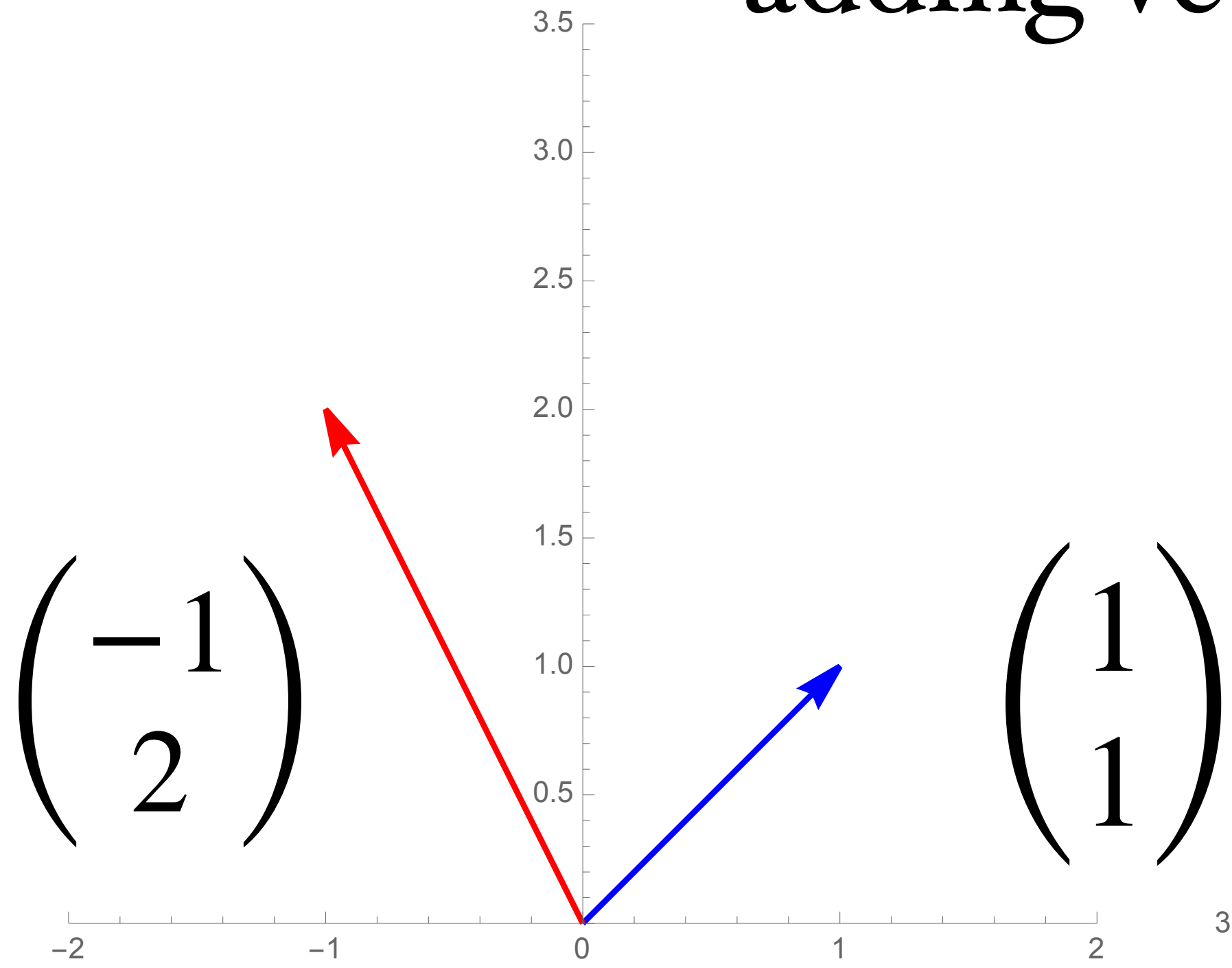
vectors in $\mathbb{R}^2$



vectors in $\mathbb{R}^2$

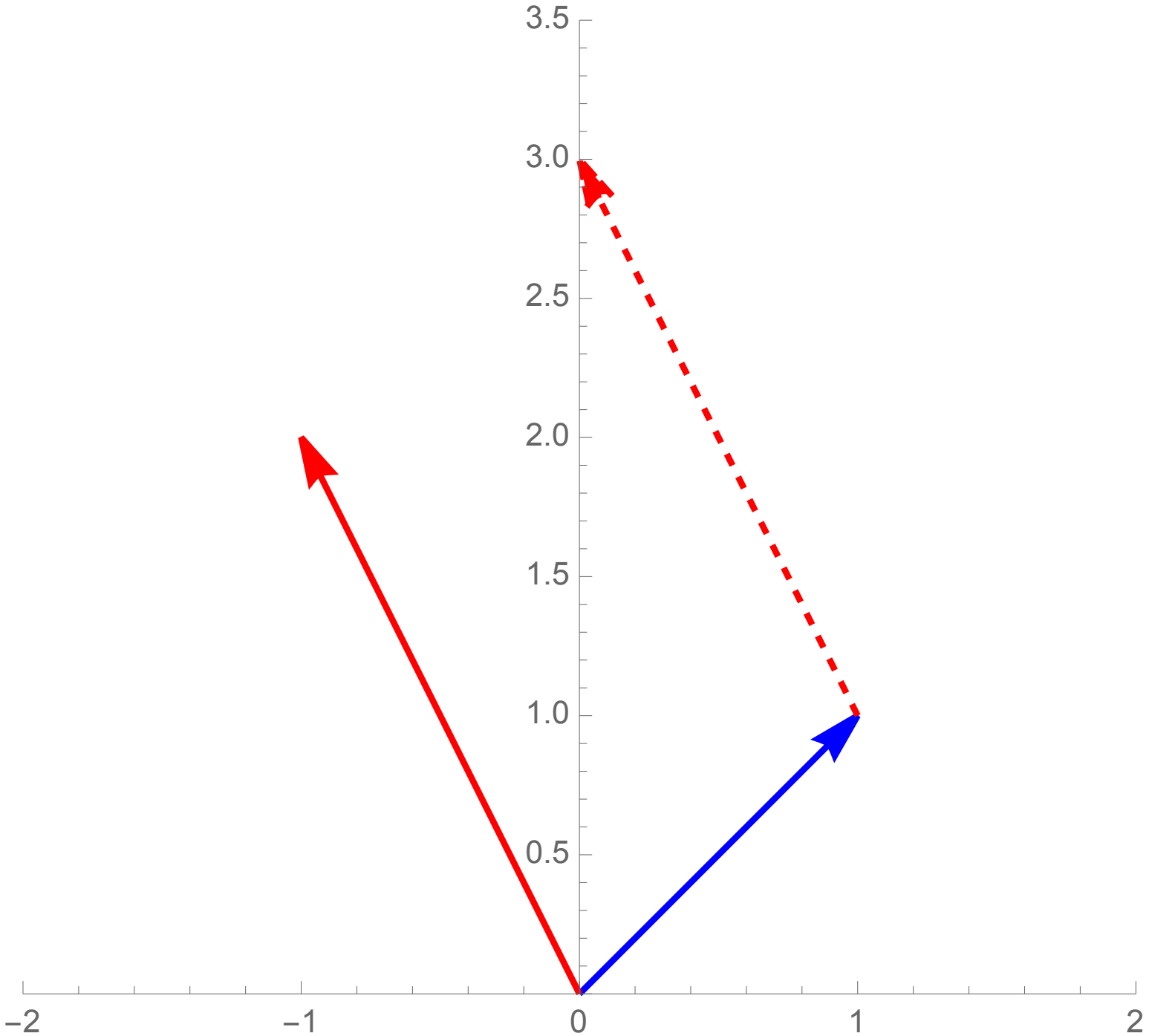
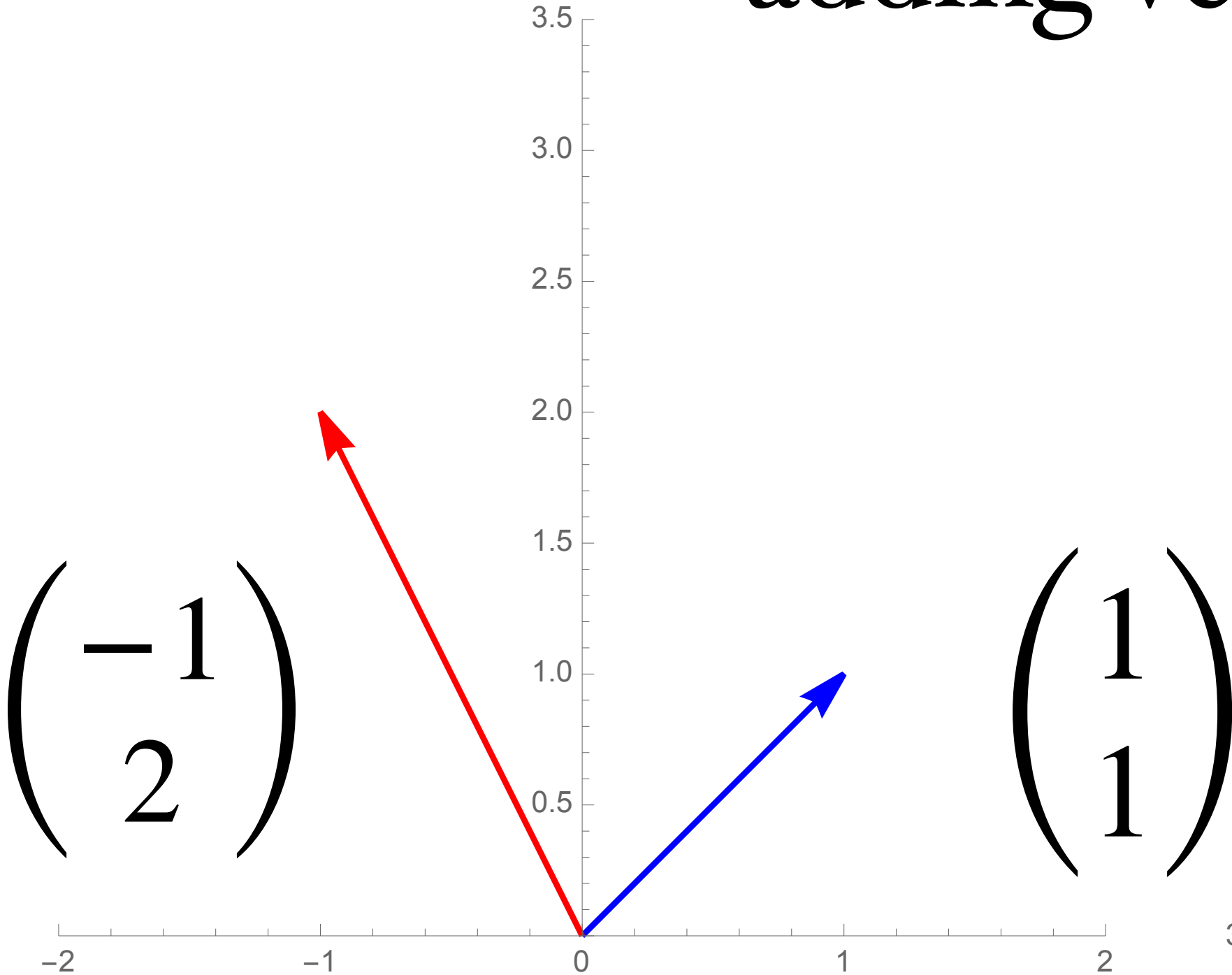


adding vectors in $\mathbb{R}^2$

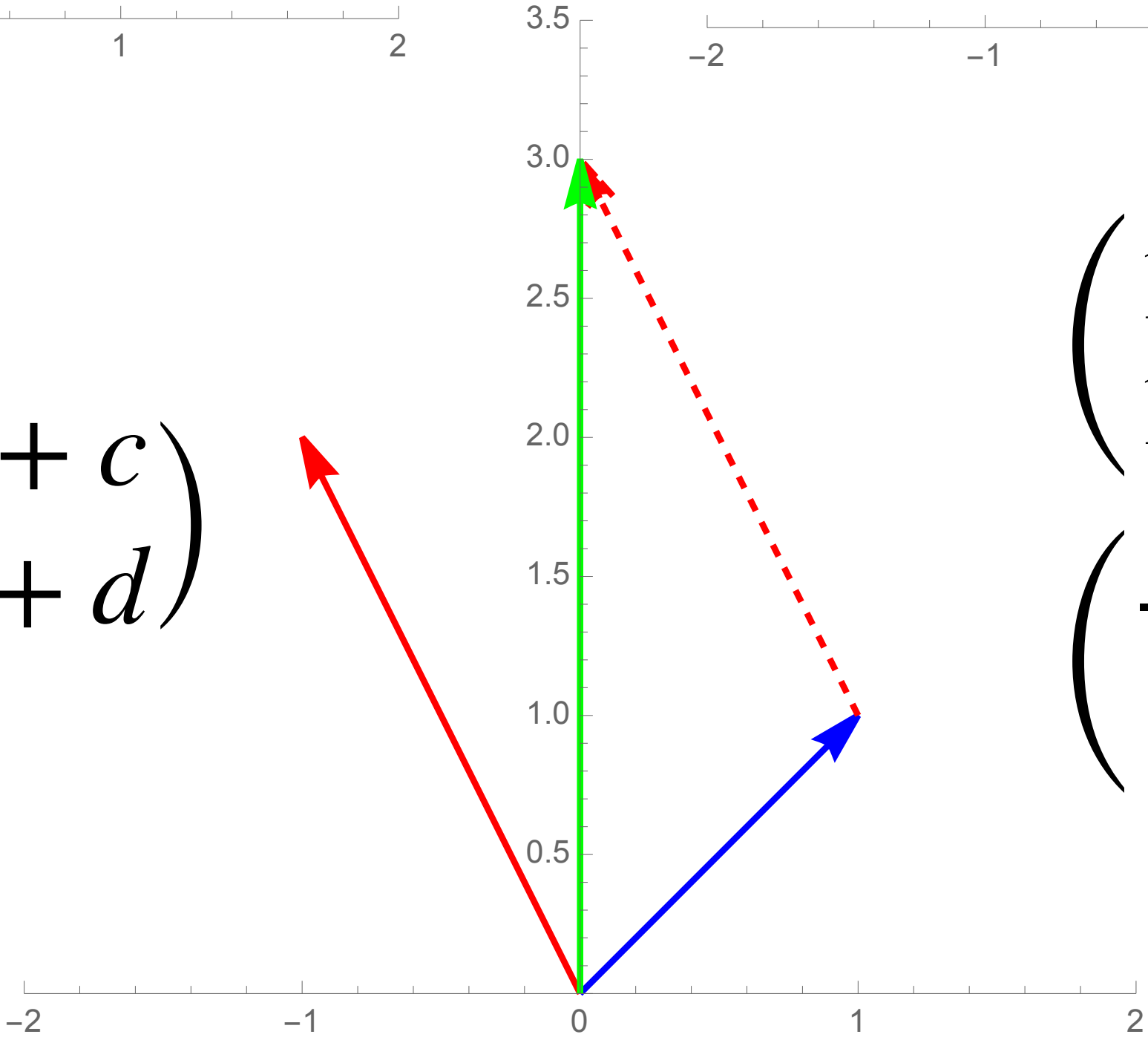


$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 2 \end{pmatrix} =$$
$$\begin{pmatrix} -1 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}$$

adding vectors in $\mathbb{R}^2$

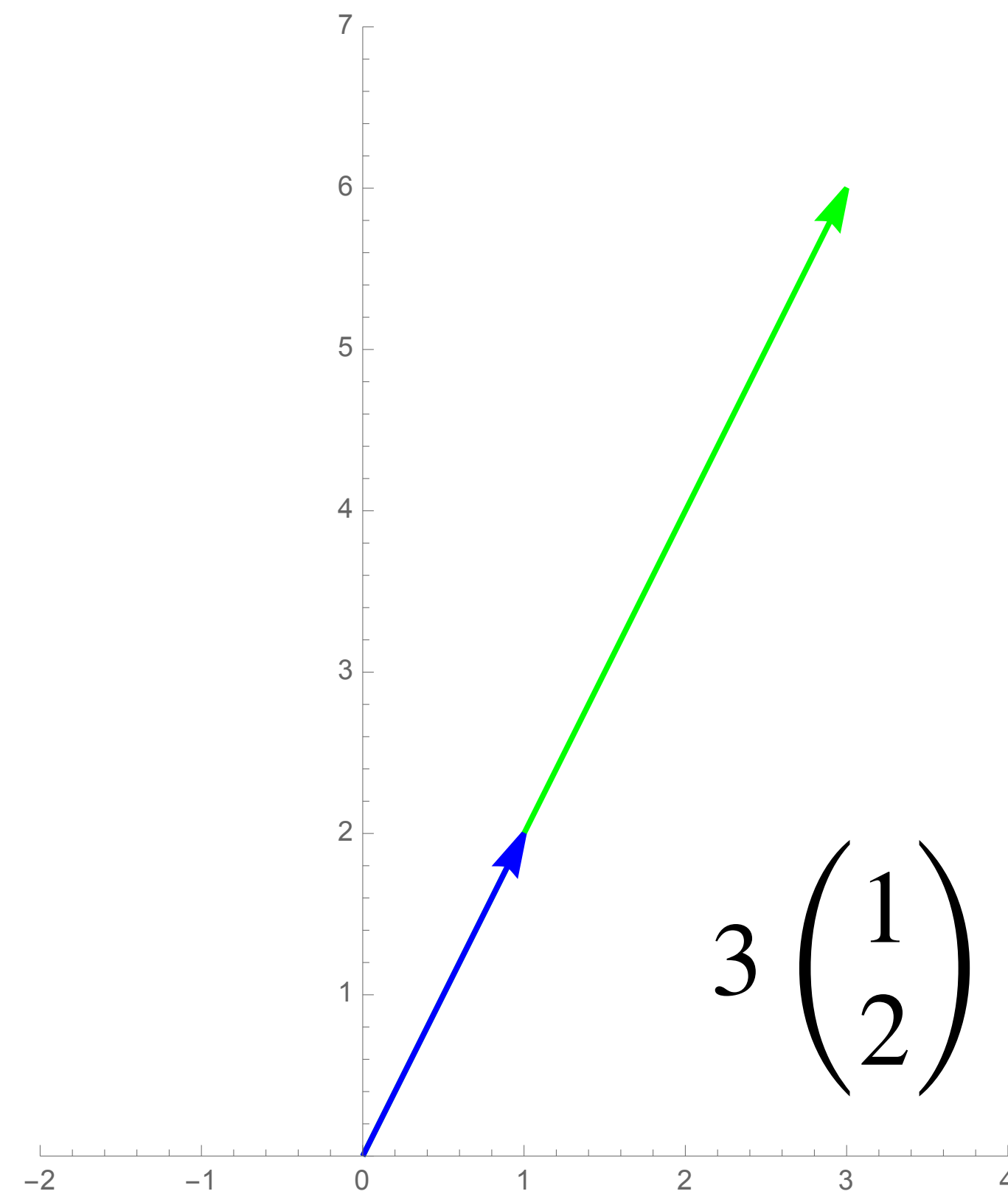
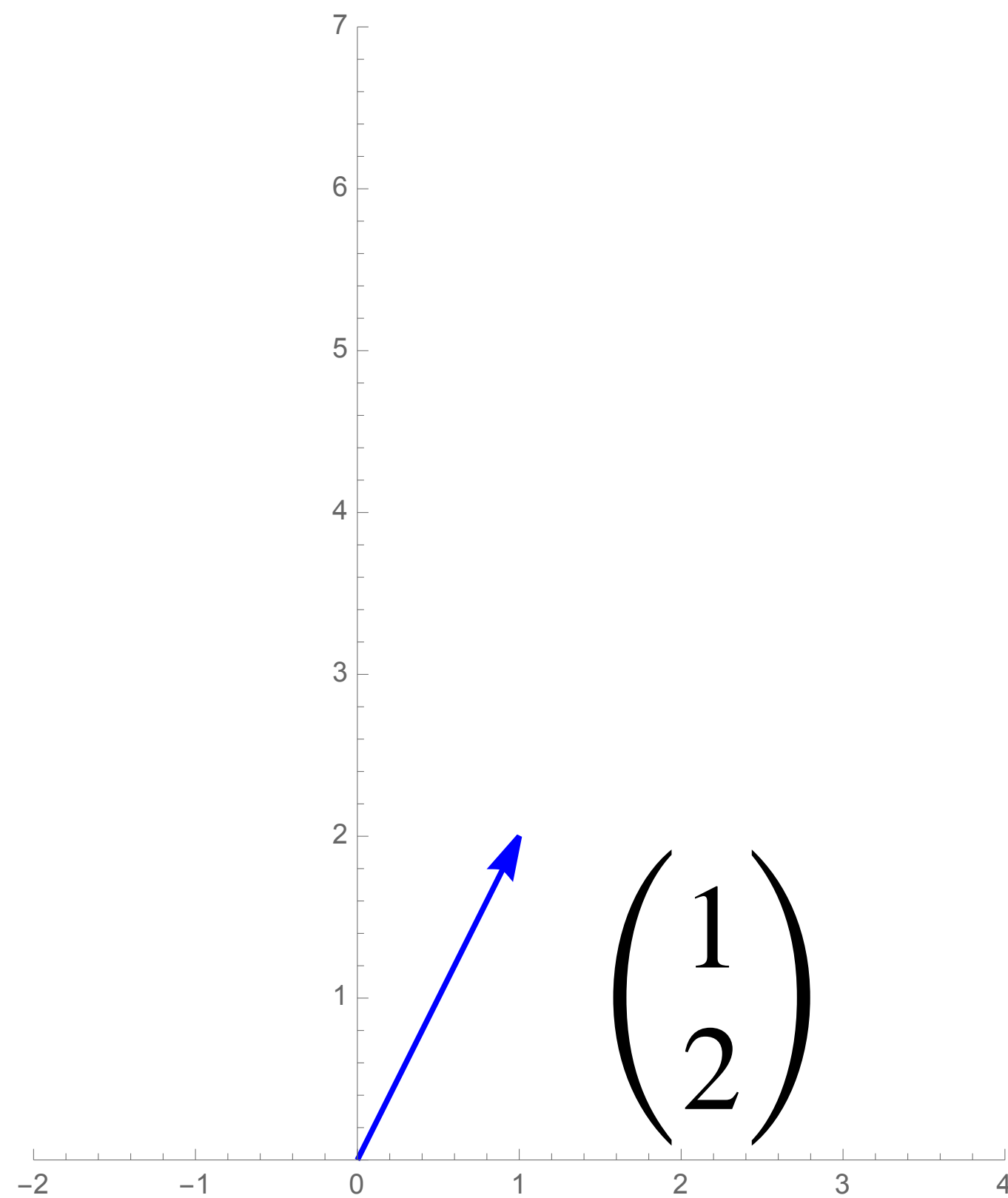


$$\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a + c \\ b + d \end{pmatrix}$$



$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 2 \end{pmatrix} =$$
$$\begin{pmatrix} -1 \\ 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}$$

scaling vectors in $\mathbb{R}^2$



$$k \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ka \\ kb \end{pmatrix}$$

distribution

$$k\left(\binom{a}{b} + \binom{c}{d}\right) \stackrel{?}{=} k\binom{a}{b} + k\binom{c}{d}$$

distribution

$$k\left(\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix}\right) \stackrel{?}{=} k\begin{pmatrix} a \\ b \end{pmatrix} + k\begin{pmatrix} c \\ d \end{pmatrix}$$

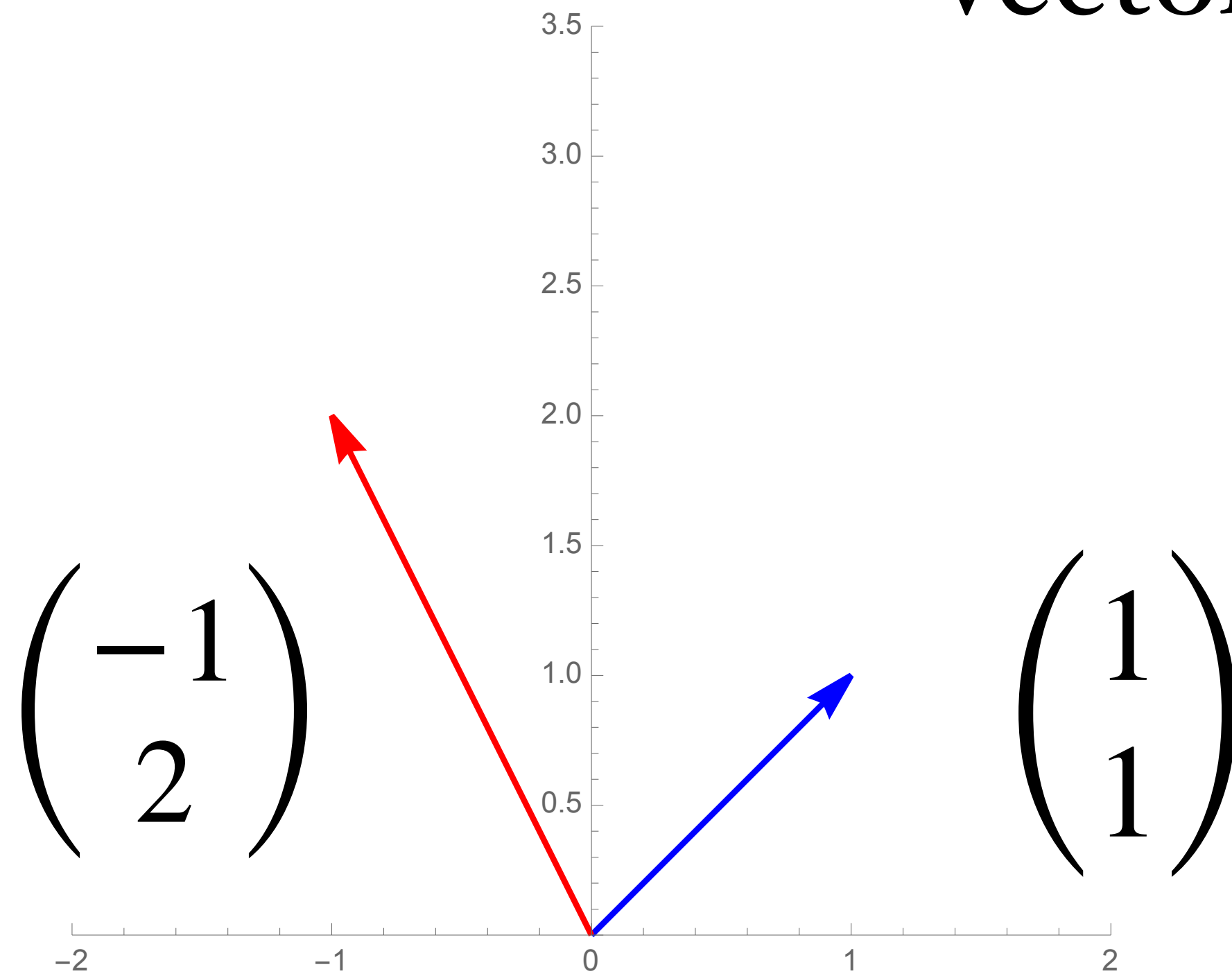
$$\begin{pmatrix} ka + kc \\ kb + kd \end{pmatrix}$$



linear combination

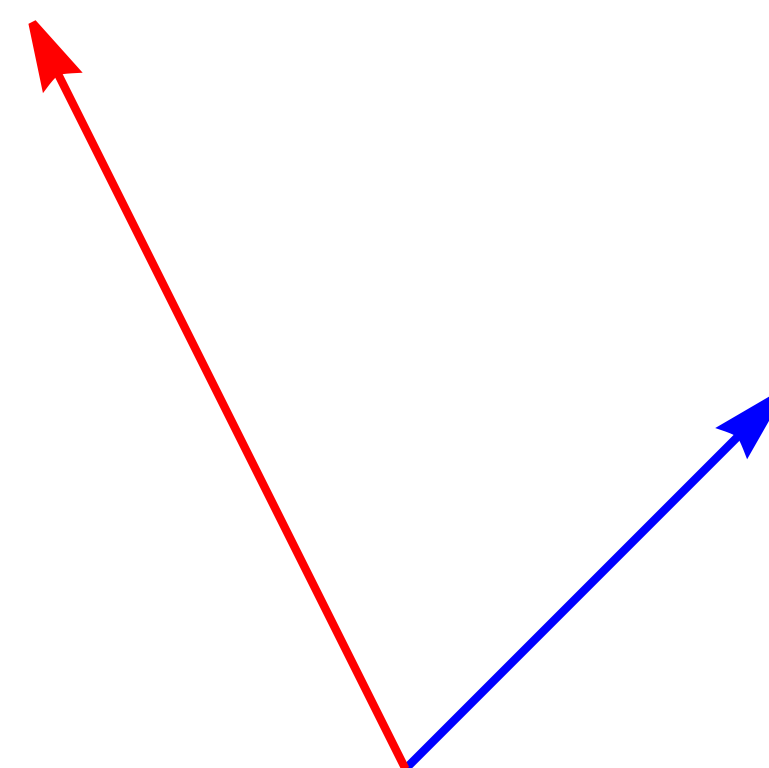
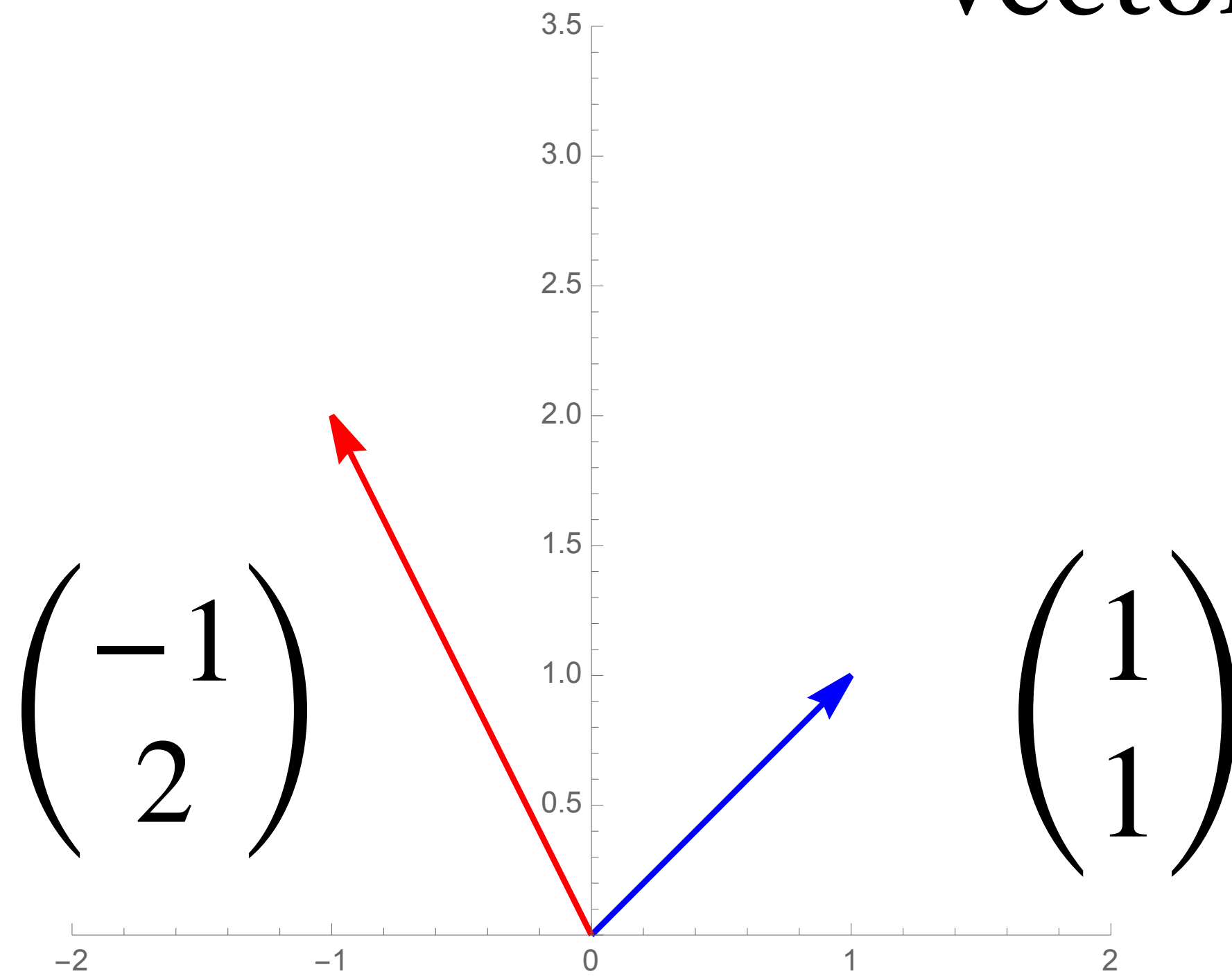
$$u, v \in V \quad \rightsquigarrow \quad au + bv$$

vectors in $\mathbb{R}^2$



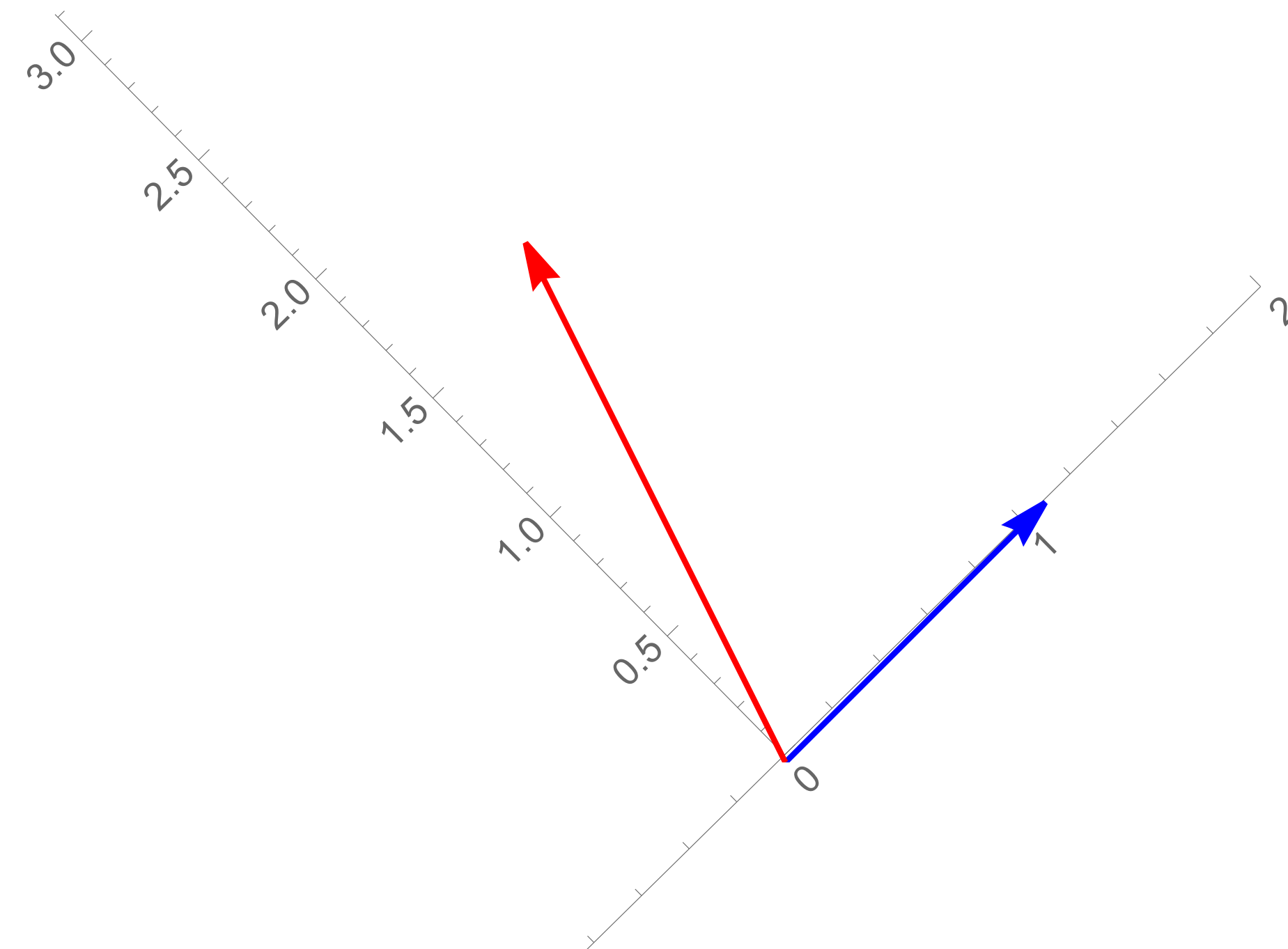
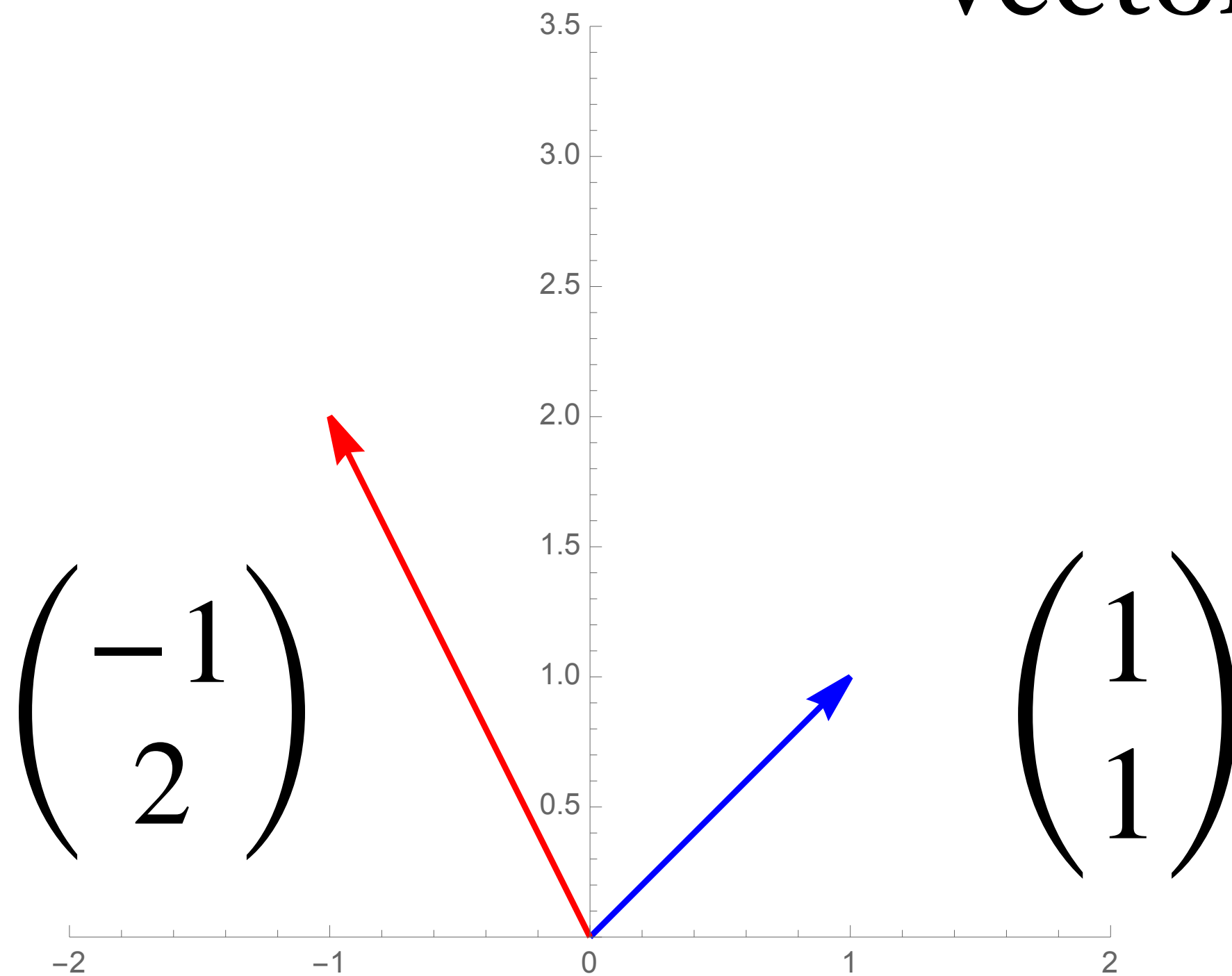
coordinates vs. coordinate-free

vectors in $\mathbb{R}^2$



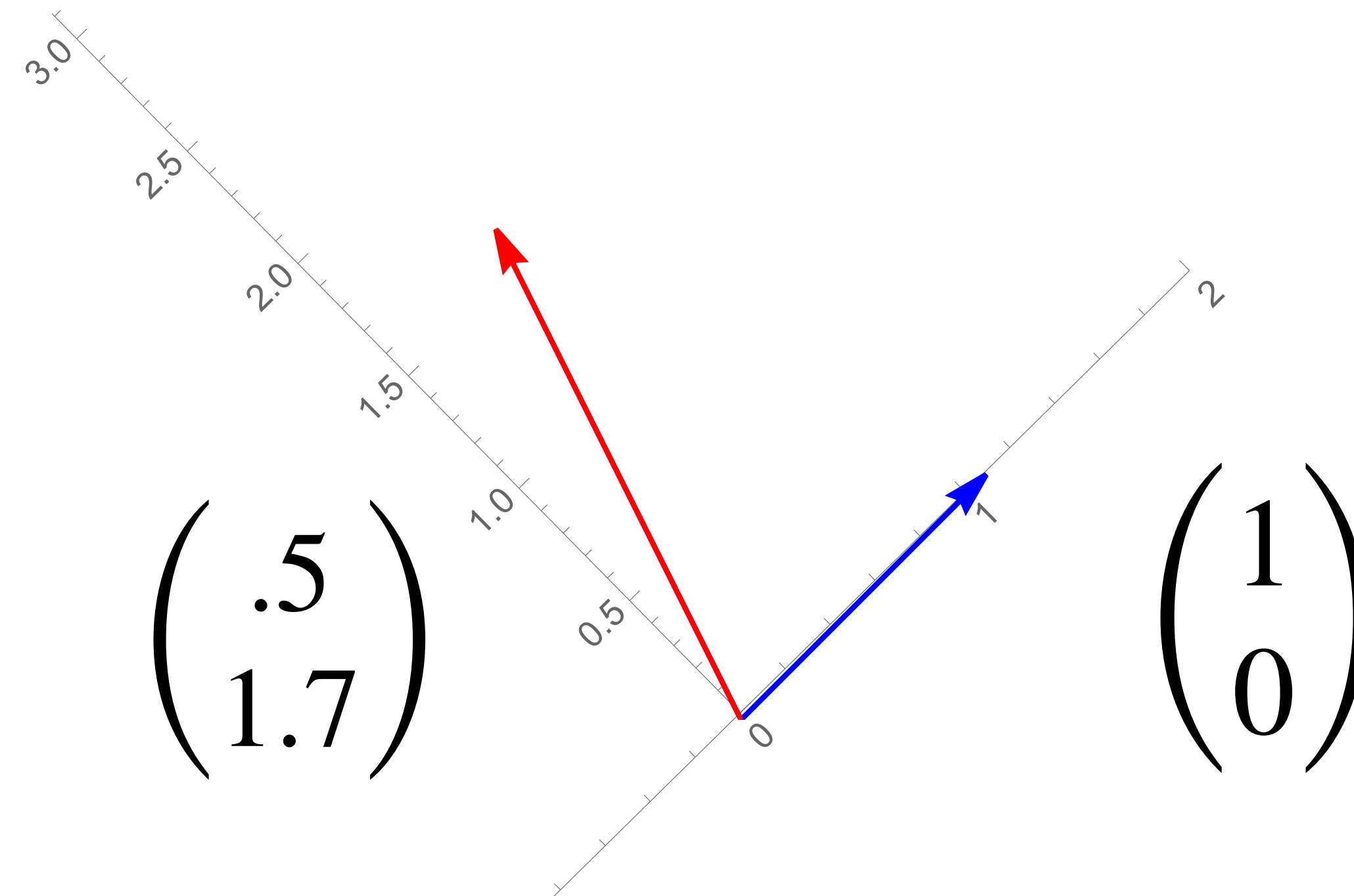
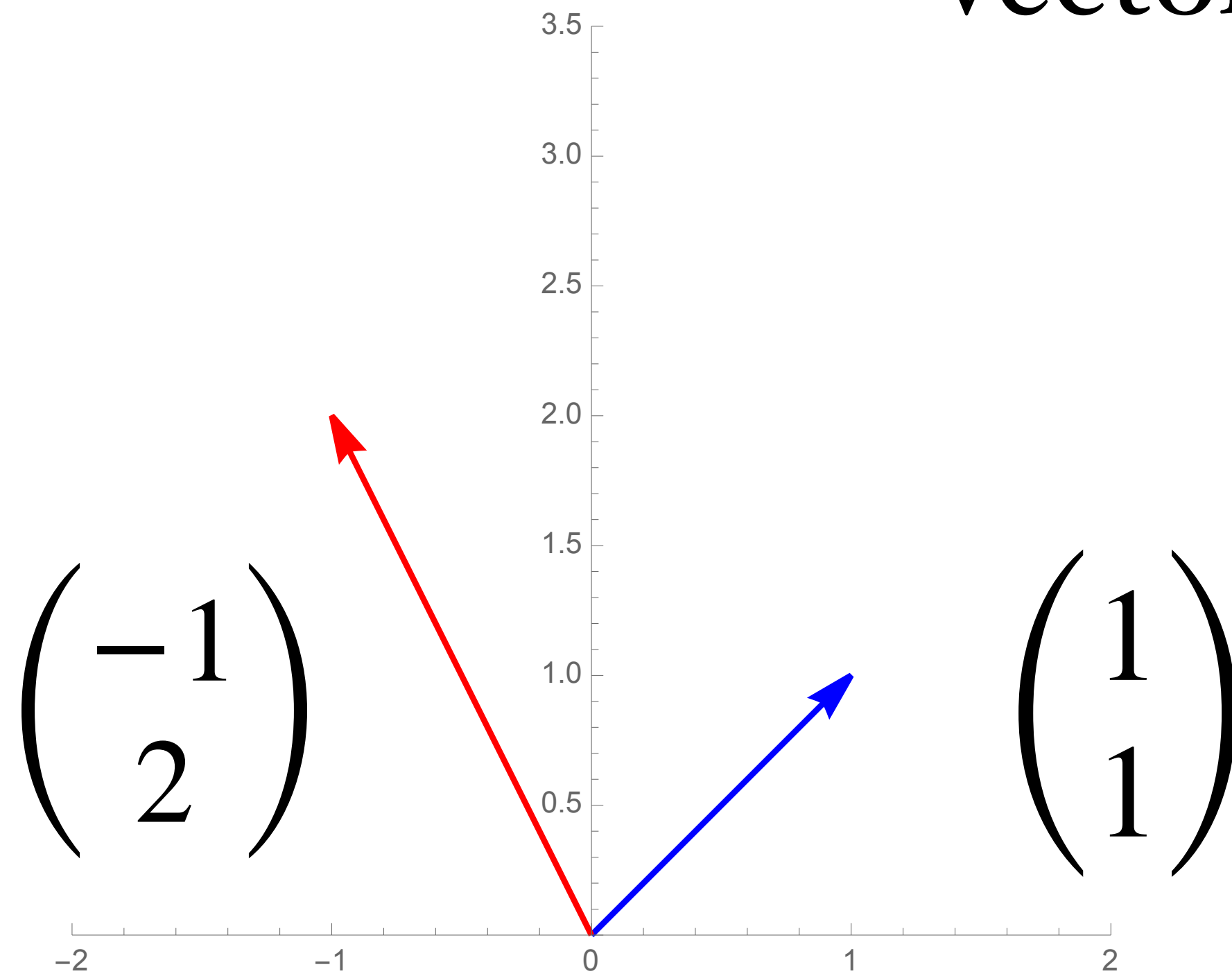
coordinates vs. coordinate-free

vectors in $\mathbb{R}^2$



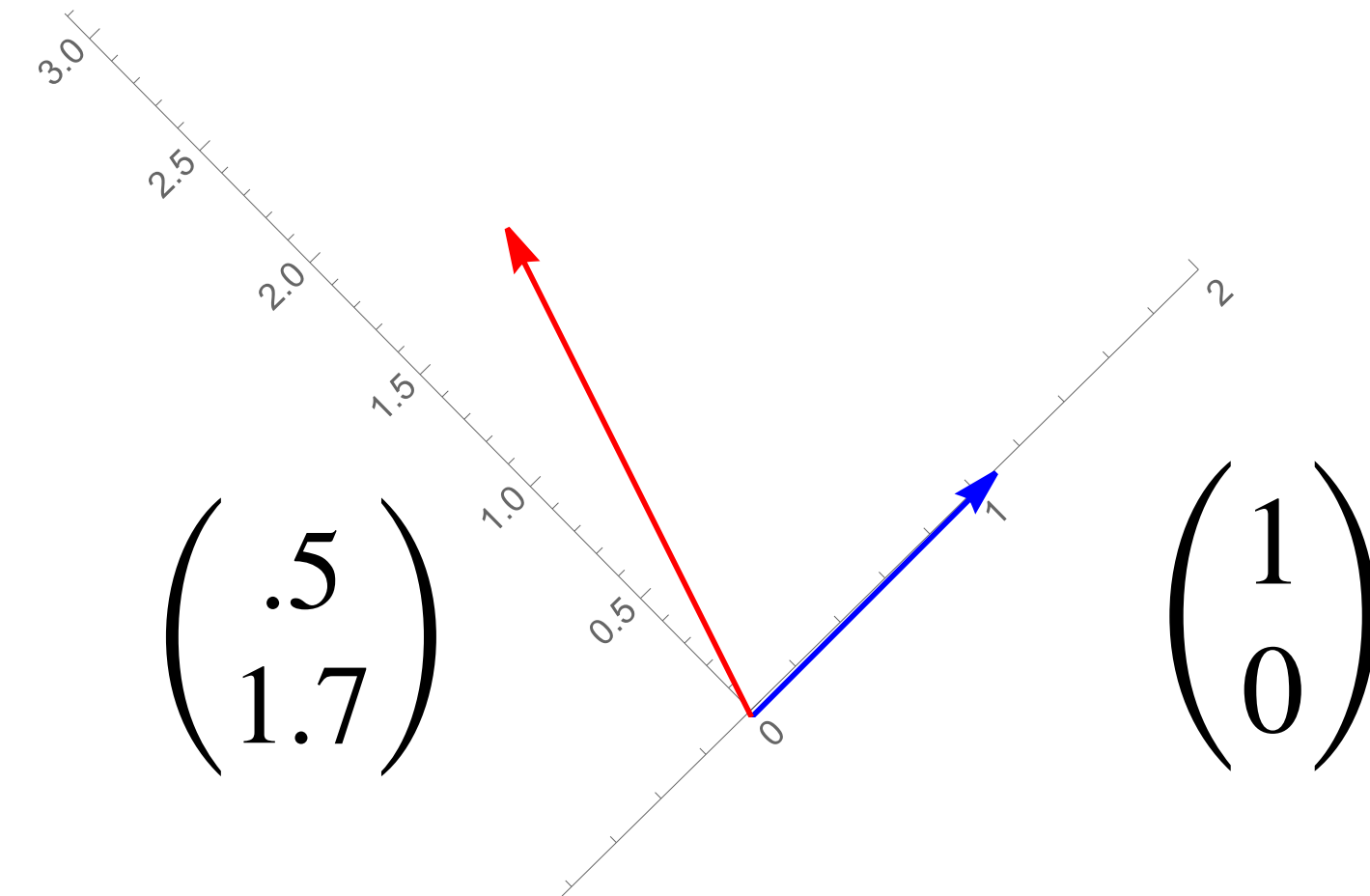
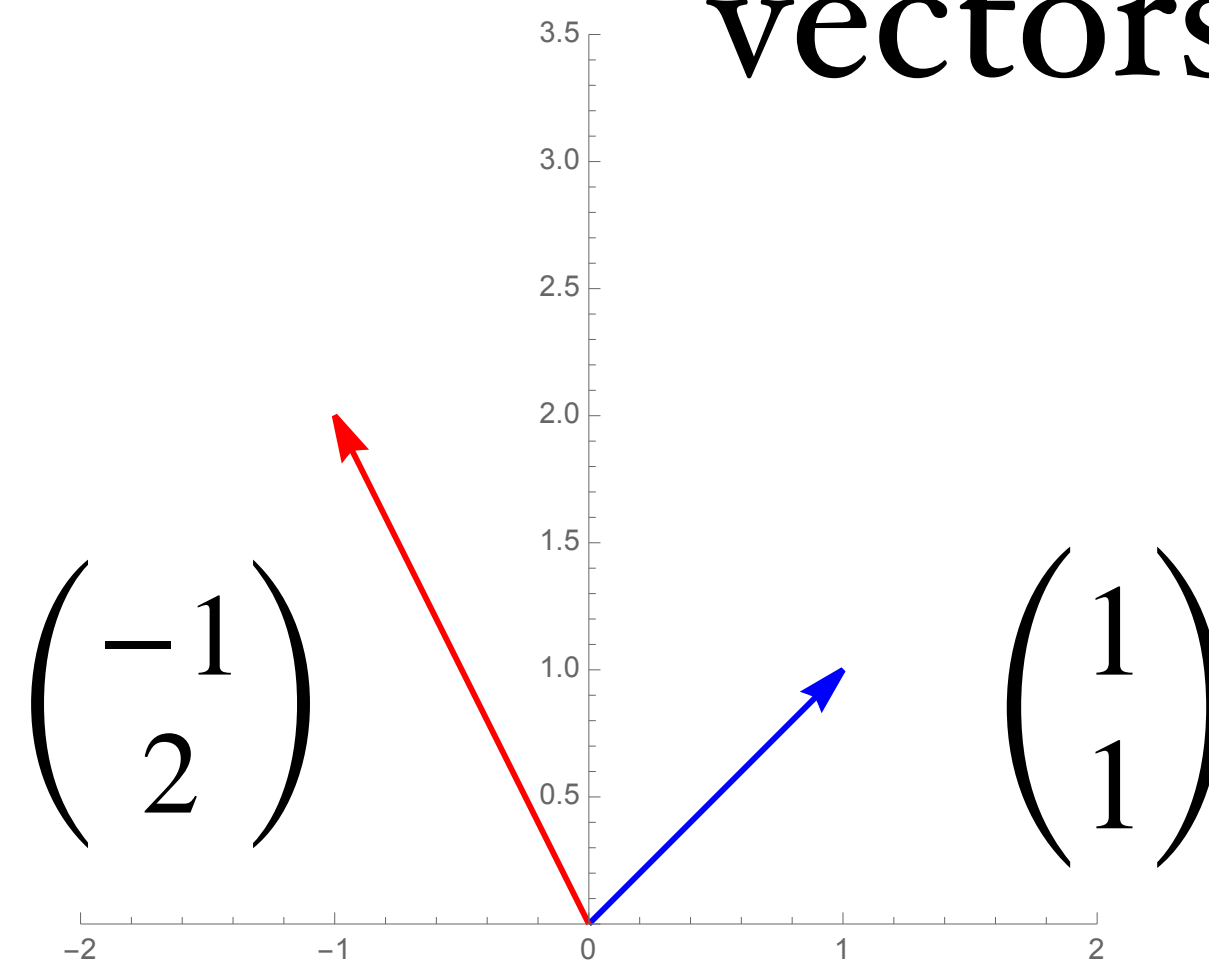
coordinates vs. coordinate-free

vectors in $\mathbb{R}^2$



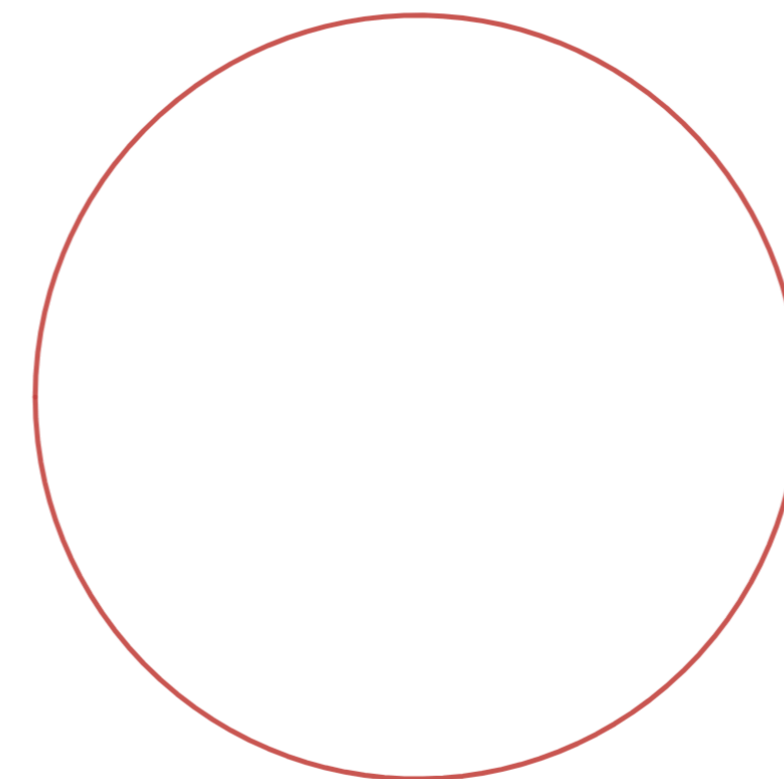
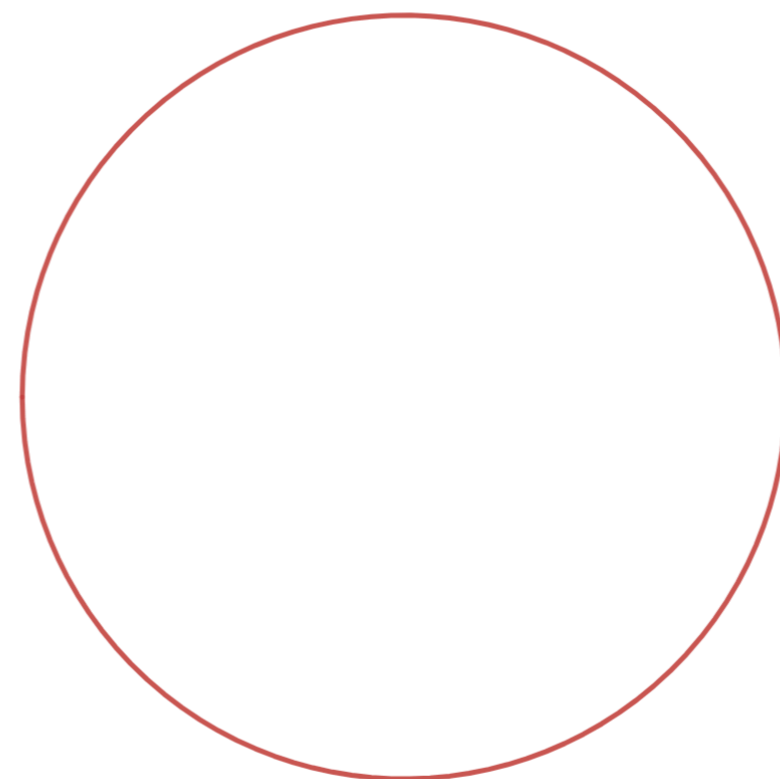
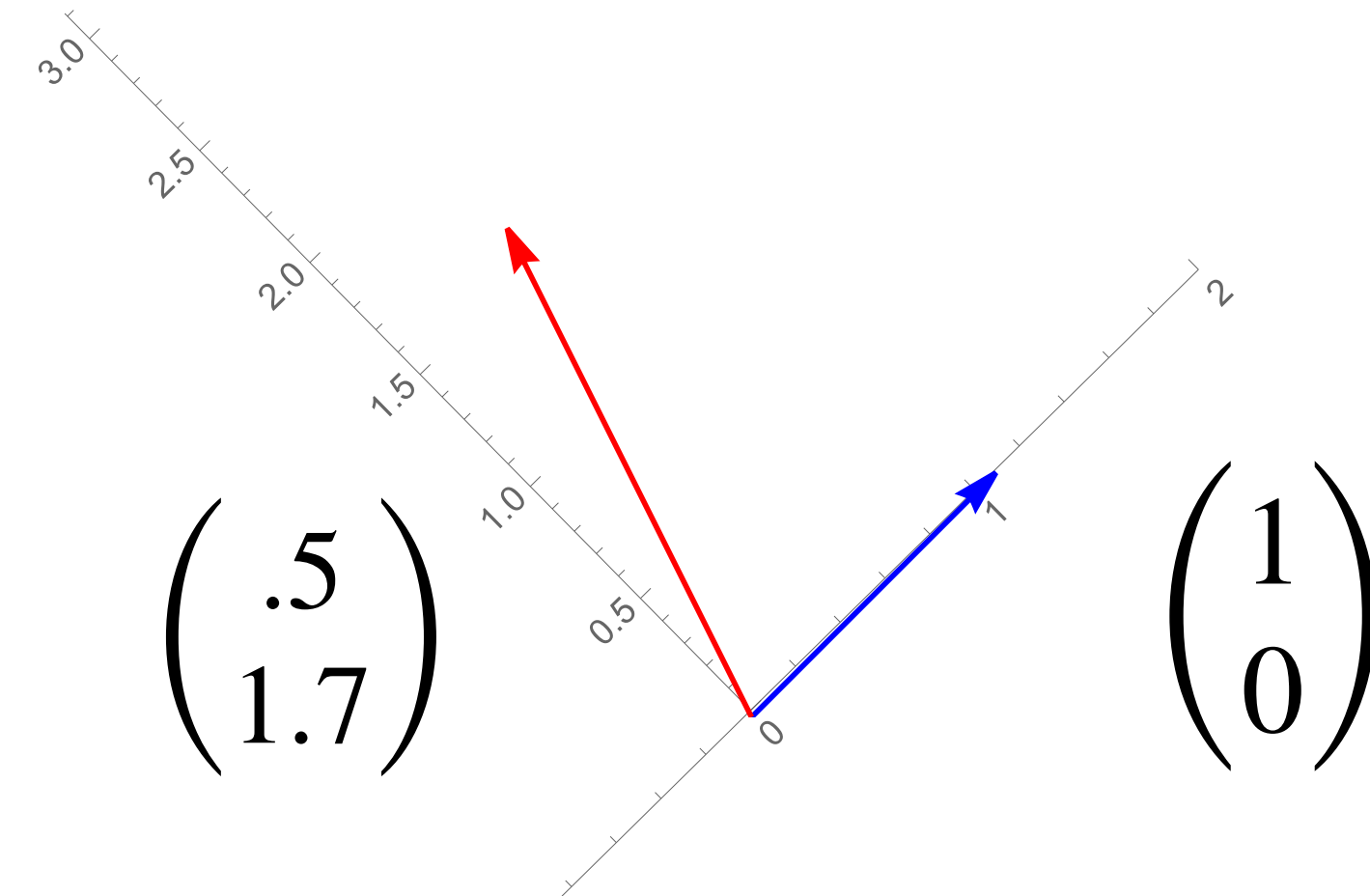
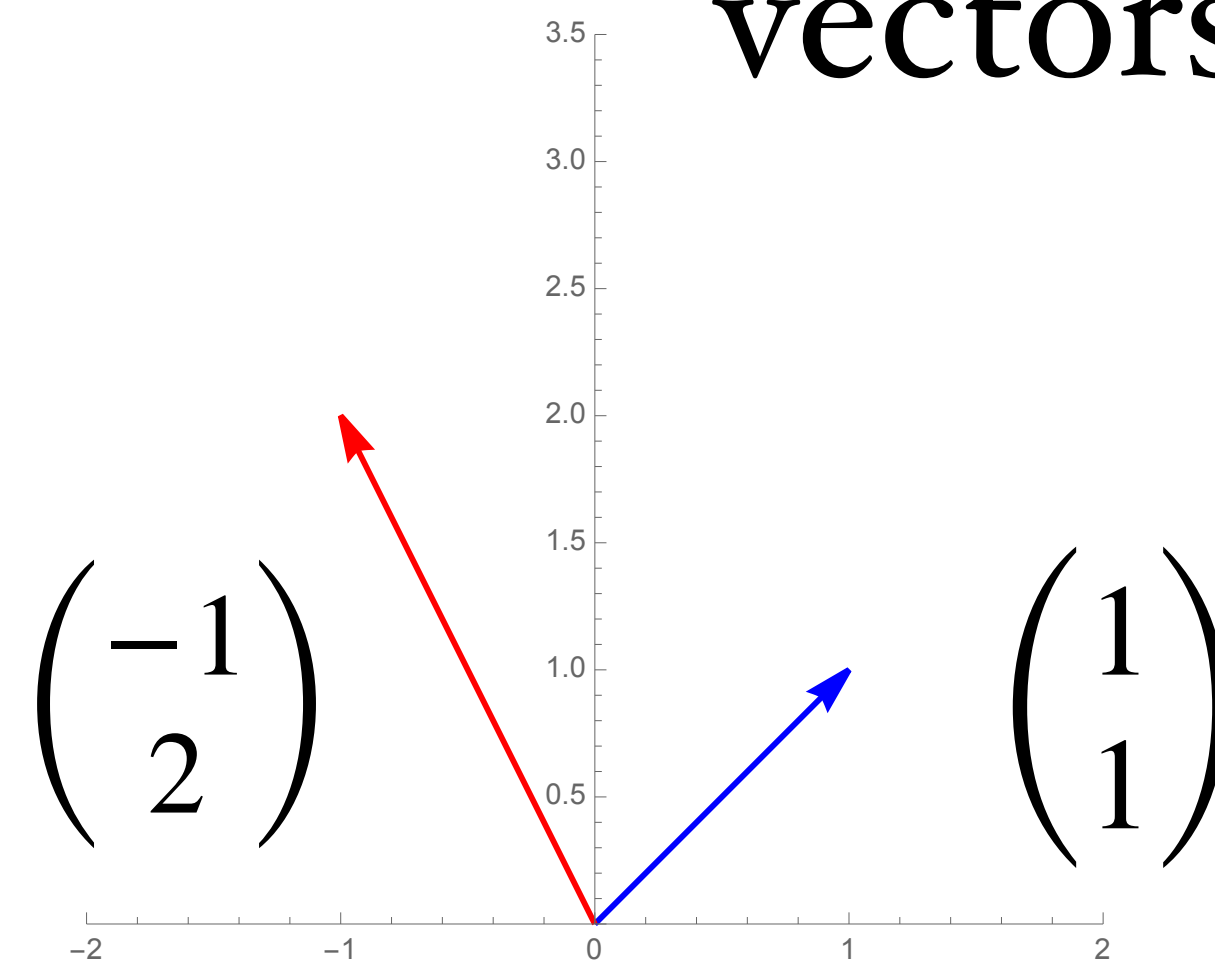
coordinates vs. coordinate-free

vectors in $\mathbb{R}^2$



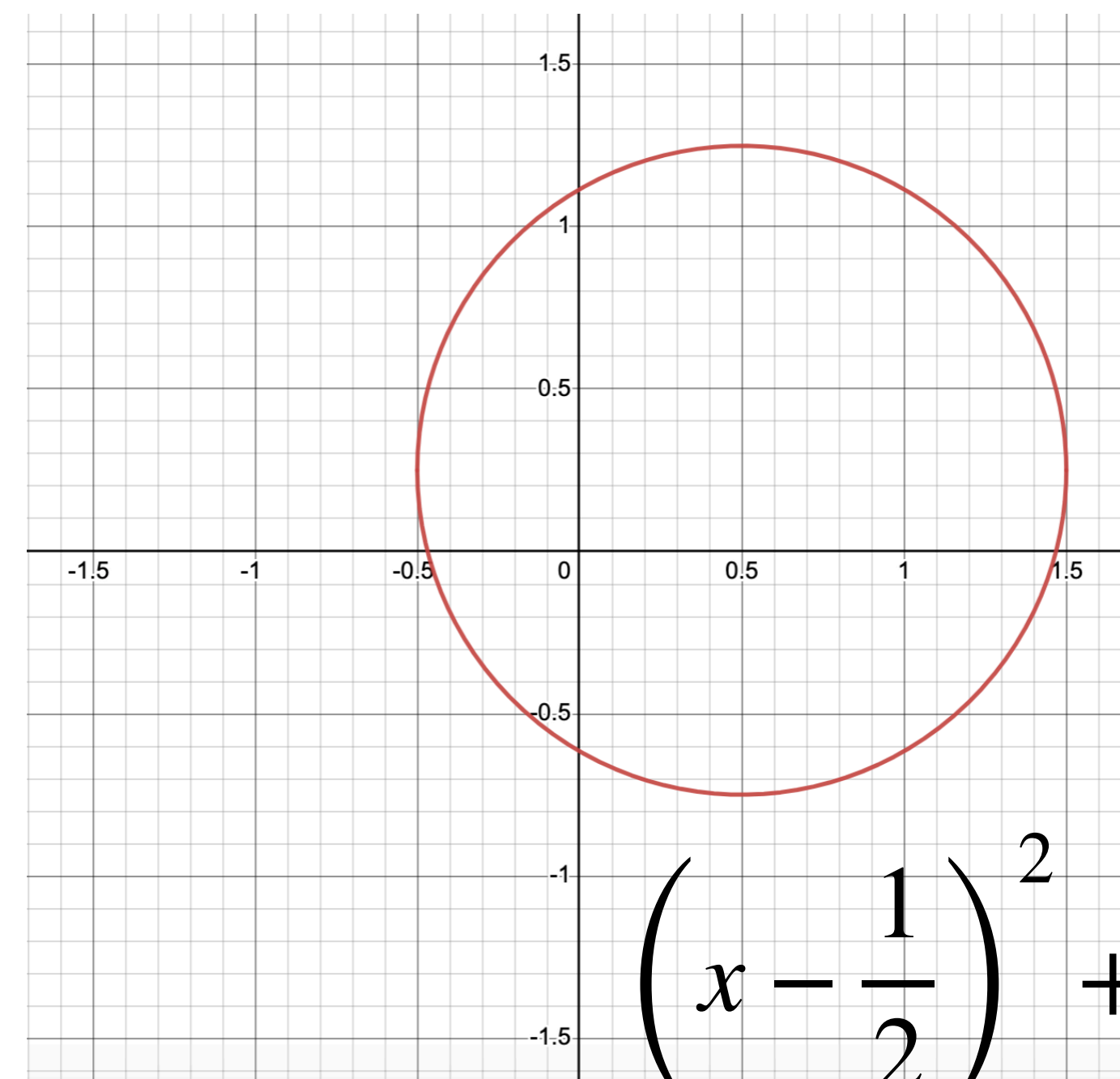
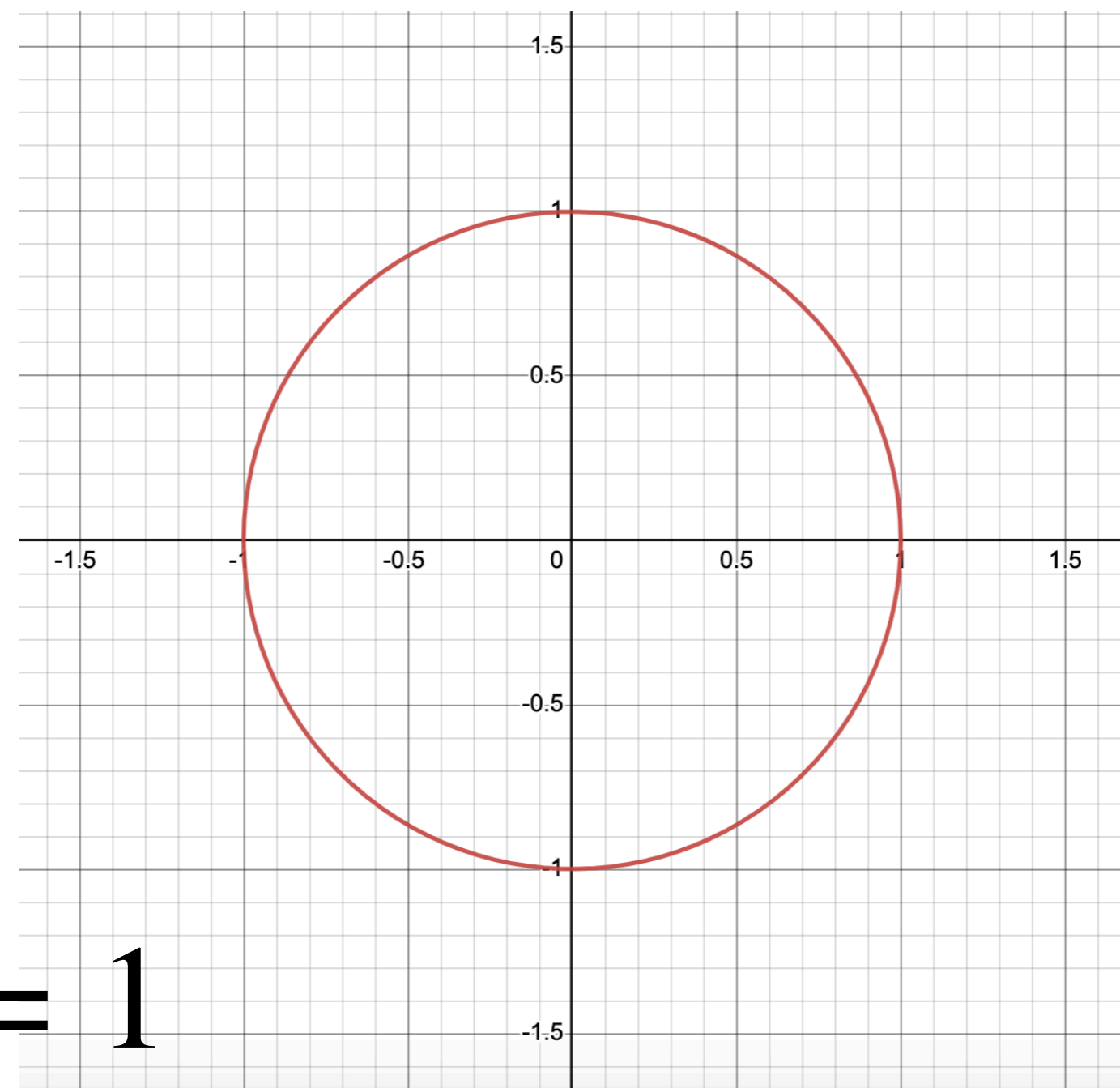
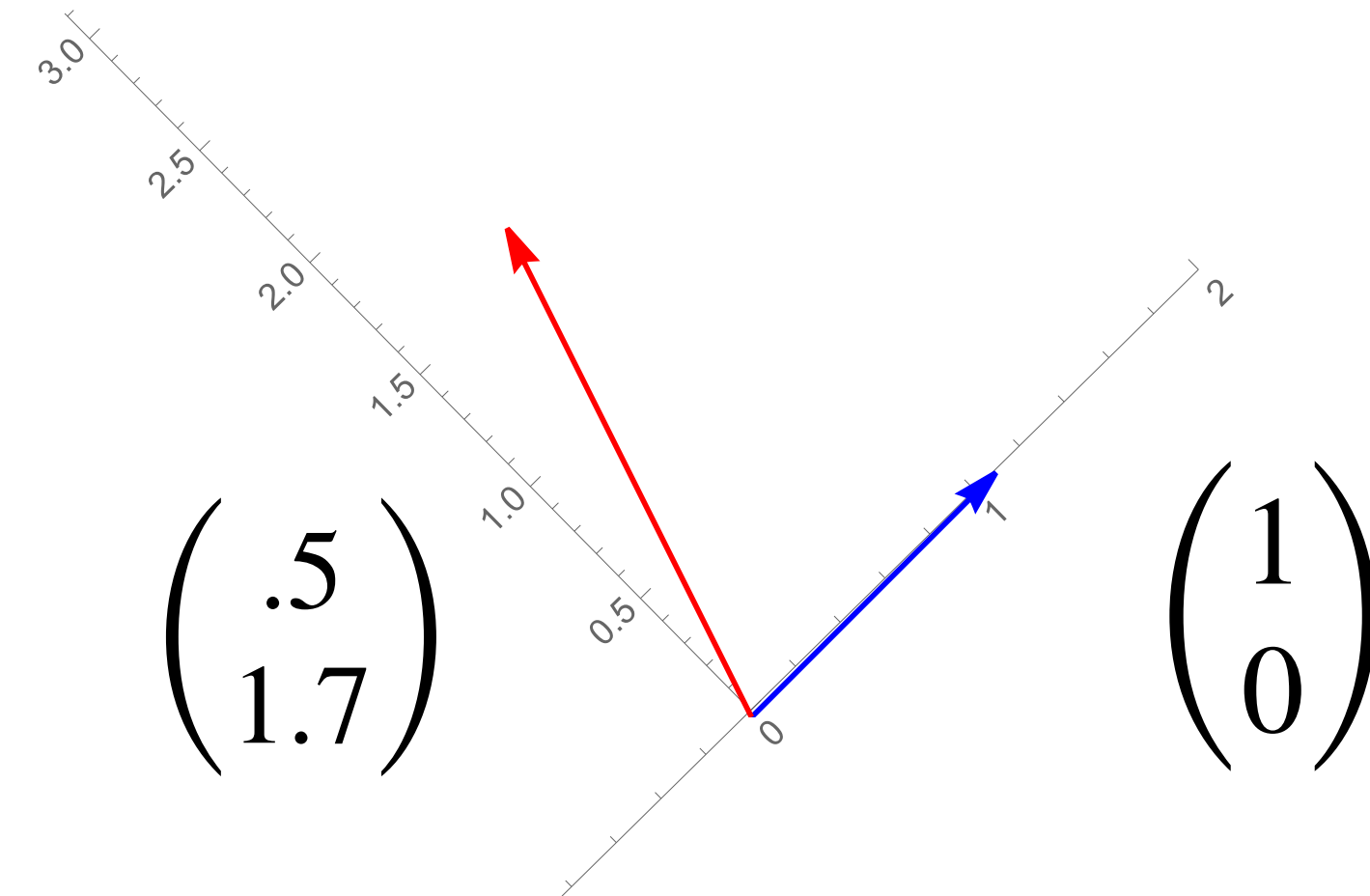
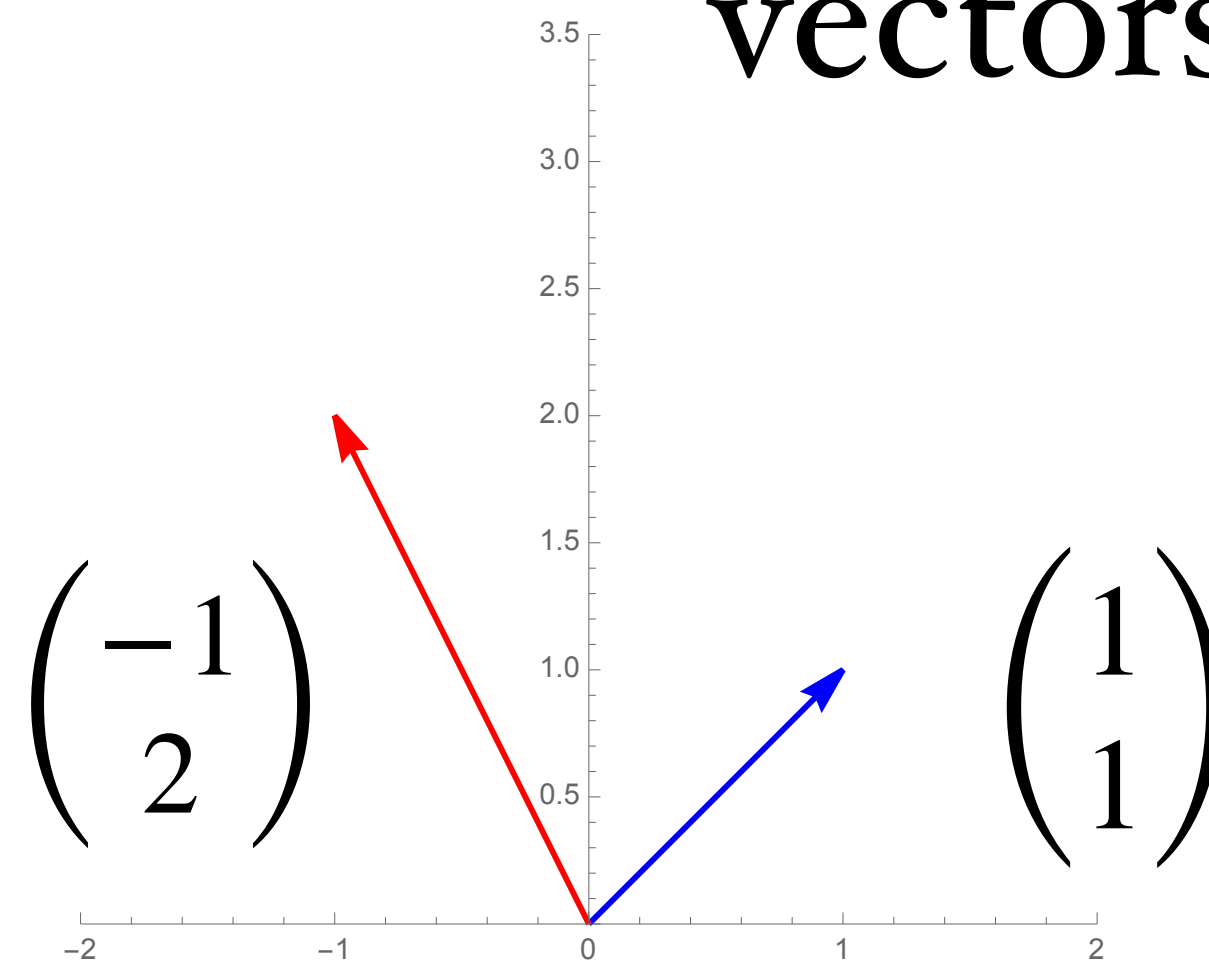
coordinates vs. coordinate-free

vectors in $\mathbb{R}^2$



coordinates vs. coordinate-free

vectors in $\mathbb{R}^2$



$$x^2 + y^2 = 1$$

$$\left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{4}\right)^2 = 1$$

coordinates vs. coordinate-free

vector space:

vectors you can add and subtract with each other,

a zero vector, inverse (opposite) vectors;

and a way to scale the vectors that “respects” the addition

vector space:

vectors you can add and subtract with each other,
a zero vector, inverse (opposite) vectors;
and a way to scale the vectors that “respects” the addition

a group of vectors and a field of scalars

For you to do ASAP:

- Read through the syllabus and course docs
- Fill out the start-of-course survey
- Read sections 1-4 of Chapter 1 of *LADW*
- Form a study group
- Explore the Canvas site some
- Come to office hours!

Decide whether this course
is right for you.